```
import { createClientFromRequest } from 'npm:@base44/sdk@0.5.0';

const isNum = (v) => v !== null && v !== undefined && Number.isFinite(v);
const safeDiv = (a, b) => (isNum(a) && isNum(b) && Math.abs(b) > 1e-9) ? a / b :
null;

const stdev = (arr) => {
    const filtered = arr.filter(isNum);
    if (filtered.length < 2) return null;
    const n = filtered.length;
    const mean = filtered.reduce((a, b) => a + b) / n;
    const variance = filtered.map(x => Math.pow(x - mean, 2)).reduce((a, b) => a
+ b) / (n - 1);
    return Math.sqrt(variance);
};

const cagr = (series, years) => {
    if (!series || series.length < 2) return null;
    const end = series[series.length - 1];
    const endDate = new Date(end.date);
    const startCutoff = new Date(endDate);
    startCutoff.setFullYear(startCutoff.getFullYear() - years);

    const start = series.reduce((acc, s) => {
        const sDate = new Date(s.date);
        return (sDate <= startCutoff && (!acc || sDate > new Date(acc.date))) ?
s : acc;
    }, null);

    if (!start || !isNum(start.value) || start.value <= 0) return null;

    const actualYears = (endDate.getTime() - new Date(start.date).getTime()) /
(365.25 * 24 * 3600 * 1000);
    if (actualYears < years - 0.5) return null;

    return (Math.pow(end.value / start.value, 1 / actualYears) - 1) * 100;
};

Deno.serve(async (req) => {
  const base44 = createClientFromRequest(req);
  const logs = [];
  const log = (msg) => logs.push(`[${new Date().toISOString().split('T')
[1].slice(0, -1)}] ${msg}`);

  try {
    if (!(await base44.auth.isAuthenticated())) {
      return new Response(JSON.stringify({ ok: false, error: 'Unauthorized' }),
{ status: 401 });
    }

    const { ticker } = await req.json();
    const symbol = String(ticker || '').trim().toUpperCase();
    if (!symbol) {
      return new Response(JSON.stringify({ ok: false, error: 'ticker
required' }), { status: 400 });
    }

    log(`[FundMetrics] Starting computation for ${symbol}`);

    // Fetch all necessary data
    const [metricsRow, financials] = await Promise.all([
      (async () => (await
base44.asServiceRole.entities.Metrics.filter({ ticker_symbol: symbol }))[0] || {
ticker_symbol: symbol, ticker: symbol })(),
```

```
      base44.asServiceRole.entities.FinancialsHistory.filter({ ticker: symbol },
'-period_end', 40)
    ]);

    if (!financials || financials.length === 0) {
      log(`[FundMetrics] No financial history found for ${symbol}`);
      return new Response(JSON.stringify({
        ok: false,
        error: 'No financial history available',
        logs
      }), { status: 400 });
    }

    const quarterly = financials.filter(f => f.freq === 'quarterly').slice(0,
8);
    const annual = financials.filter(f => f.freq === 'annual').slice(0, 6);

    log(`[FundMetrics] Found ${quarterly.length} quarterly and ${annual.length}
annual records`);

    // Build TTM aggregates
    const ttmSum = (field) => {
      const values = quarterly.slice(0, 4).map(q => q[field]).filter(isNum);
      return values.length === 4 ? values.reduce((a, b) => a + b, 0) : null;
    };

    const TTM = {
      revenue: ttmSum('revenue'),
      net_income: ttmSum('net_income'),
      cfo: ttmSum('cfo'),
      capex: ttmSum('capex'),
      operating_income: ttmSum('operating_income'),
      ebit: ttmSum('ebit'),
      interest_expense: ttmSum('interest_expense'),
      depreciation: ttmSum('depreciation'),
      sbc: ttmSum('stock_based_compensation')
    };

    // Latest balance sheet items
    const latest = quarterly[0] || {};
    const BALANCE = {
      cash: latest.cash,
      total_debt: latest.total_debt,
      stockholder_equity: latest.stockholder_equity,
      total_assets: latest.total_assets,
      shares_out: latest.shares_outstanding
    };

    log(`[FundMetrics] TTM Revenue: ${TTM.revenue}, CFO: ${TTM.cfo}, Net Income:
${TTM.net_income}`);
    log(`[FundMetrics] Balance - Cash: ${BALANCE.cash}, Debt: $
{BALANCE.total_debt}, Equity: ${BALANCE.stockholder_equity}`);

    // Calculate EBITDA if missing
    if (!isNum(TTM.ebit) && isNum(TTM.operating_income)) {
      TTM.ebit = TTM.operating_income;
    }
    const ebitda = isNum(TTM.ebit) && isNum(TTM.depreciation) ? TTM.ebit +
TTM.depreciation : TTM.ebit;

    // Calculate FCF
    const fcf = isNum(TTM.cfo) && isNum(TTM.capex) ? TTM.cfo -
Math.abs(TTM.capex) : null;
```

```
    // Build annual series for CAGR
    const revenueSeries = annual
      .filter(a => isNum(a.revenue))
      .map(a => ({ date: a.period_end, value: a.revenue }))
      .sort((a, b) => a.date.localeCompare(b.date));

    const epsSeries = annual
      .filter(a => isNum(a.eps_diluted) || (isNum(a.net_income) &&
isNum(a.shares_diluted)))
      .map(a => ({
        date: a.period_end,
        value: isNum(a.eps_diluted) ? a.eps_diluted : a.net_income /
a.shares_diluted
      }))
      .sort((a, b) => a.date.localeCompare(b.date));

    // Build operating margin series
    const operatingMargins = annual
      .filter(a => isNum(a.operating_income) && isNum(a.revenue) && a.revenue >
0)
      .map(a => (a.operating_income / a.revenue) * 100);

    log(`[FundMetrics] Revenue series: ${revenueSeries.length} points, EPS
series: ${epsSeries.length} points`);

    // --- Compute all fundamental metrics ---
    const patch = {
      ticker_symbol: symbol,
      ticker: symbol, // Both keys for compatibility
      as_of_date: new Date().toISOString().slice(0, 10),
      data_source: 'computed:fundamentals',

      // Quality metrics
      fcf_margin_pct: isNum(fcf) ? 100 * safeDiv(fcf, TTM.revenue) : null,
      cfo_to_ni: safeDiv(TTM.cfo, TTM.net_income),
      fcf_to_ebit: isNum(fcf) ? safeDiv(fcf, TTM.ebit) : null,
      accruals_ratio: safeDiv(TTM.net_income - TTM.cfo, BALANCE.total_assets),
      margin_stdev_5y_pct: operatingMargins.length > 1 ? stdev(operatingMargins)
: null,

      // ROIC
      roic_pct: (() => {
        const nopat = isNum(TTM.net_income) && isNum(TTM.interest_expense)
          ? TTM.net_income + TTM.interest_expense
          : null;
        const investedCapital = isNum(BALANCE.total_debt) &&
isNum(BALANCE.stockholder_equity)
          ? BALANCE.total_debt + BALANCE.stockholder_equity
          : null;
        return 100 * safeDiv(nopat, investedCapital);
      })(),

      // Capital Allocation metrics
      debt_to_equity: safeDiv(BALANCE.total_debt, BALANCE.stockholder_equity),
      netdebt_to_ebitda: safeDiv(BALANCE.total_debt - BALANCE.cash, ebitda),
      net_debt_to_ebitda: safeDiv(BALANCE.total_debt - BALANCE.cash, ebitda), //
Alias
      interest_coverage_x: (() => {
        const coverage = safeDiv(TTM.ebit, TTM.interest_expense);
        if (isNum(coverage)) return Math.min(coverage, 99);
        if (isNum(TTM.ebit) && TTM.ebit > 0 && Math.abs(TTM.interest_expense ||
0) < 1e-6) return 99;
        return null;
      })(),
```

```
      interest_coverage: (() => { // Alias
        const coverage = safeDiv(TTM.ebit, TTM.interest_expense);
        if (isNum(coverage)) return Math.min(coverage, 99);
        if (isNum(TTM.ebit) && TTM.ebit > 0 && Math.abs(TTM.interest_expense ||
0) < 1e-6) return 99;
        return null;
      })(),

      // Buyback yield
      buyback_yield_pct: (() => {
        const sharesSeries = annual
          .filter(a => isNum(a.shares_diluted))
          .map(a => a.shares_diluted);
        if (sharesSeries.length >= 2) {
          const t = sharesSeries[0];
          const t_minus_1 = sharesSeries[1];
          return 100 * safeDiv(t_minus_1 - t, t_minus_1);
        }
        return null;
      })(),

      // SBC ratio
      sbc_to_sales_pct: 100 * safeDiv(TTM.sbc, TTM.revenue),

      // Growth metrics
      eps_cagr_5y_pct: cagr(epsSeries, 5),
      eps_cagr_3y_pct: cagr(epsSeries, 3),
      revenue_cagr_5y_pct: cagr(revenueSeries, 5),
      revenue_cagr_3y_pct: cagr(revenueSeries, 3),
      rev_cagr_3y_pct: cagr(revenueSeries, 3), // Alias

      // TTM fundamentals for reference
      revenue_ttm: TTM.revenue,
      net_income_ttm: TTM.net_income,
      cfo_ttm: TTM.cfo,
      capex_ttm: TTM.capex,
      ebit_ttm: TTM.ebit,
      ebitda_ttm: ebitda,
      fcf_ttm: fcf,
      sbc_ttm: TTM.sbc,

      // Balance sheet
      cash: BALANCE.cash,
      total_debt: BALANCE.total_debt,
      equity: BALANCE.stockholder_equity,
      total_assets: BALANCE.total_assets,
      shares_out: BALANCE.shares_out
    };

    // Filter to only numeric values
    const safePatch = Object.fromEntries(
      Object.entries(patch).filter(([k, v]) =>
        k === 'ticker_symbol' || k === 'ticker' || k === 'as_of_date' || k ===
'data_source' ||
        (typeof v === 'number' && Number.isFinite(v))
      )
    );

    log(`[FundMetrics] Computed ${Object.keys(safePatch).length} valid fields`);

    // Upsert to Metrics
    if (metricsRow.id) {
      const updatePayload = { ...safePatch };
      delete updatePayload.ticker_symbol; // Don't update the key field
```

```
      await base44.asServiceRole.entities.Metrics.update(metricsRow.id,
updatePayload);
      log(`[FundMetrics] Updated existing metrics row`);
    } else {
      await base44.asServiceRole.entities.Metrics.create(safePatch);
      log(`[FundMetrics] Created new metrics row`);
    }

    return new Response(JSON.stringify({
      ok: true,
      success: true,
      message: `Fundamental metrics computed for ${symbol}`,
      summary: {
        ticker: symbol,
        metrics_computed: Object.keys(safePatch).filter(k =>
          !['ticker_symbol', 'ticker', 'as_of_date', 'data_source'].includes(k)
        ).length
      },
      logs
    }), {
      status: 200,
      headers: { 'Content-Type': 'application/json' }
    });

  } catch (error) {
    log(`[FundMetrics][FATAL] ${error.message}`);
    return new Response(JSON.stringify({
      ok: false,
      success: false,
      stage: 'fundamental_metrics',
      error: error.message,
      logs
    }), {
      status: 500,
      headers: { 'Content-Type': 'application/json' }
    });
  }
});



import { createClientFromRequest } from 'npm:@base44/sdk@0.5.0';

const isNum = (v) => v !== null && v !== undefined && Number.isFinite(v);
const safeDiv = (a, b) => (isNum(a) && isNum(b) && Math.abs(b) > 1e-9) ? a / b :
null;

const stdev = (arr) => {
    const filtered = arr.filter(isNum);
    if (filtered.length < 2) return null;
    const n = filtered.length;
    const mean = filtered.reduce((a, b) => a + b) / n;
    const variance = filtered.map(x => Math.pow(x - mean, 2)).reduce((a, b) => a
+ b) / (n - 1);
    return Math.sqrt(variance);
};

const cagr = (series, years) => {
    if (!series || series.length < 2) return null;
    const end = series[series.length - 1];
    const endDate = new Date(end.date);
    const startCutoff = new Date(endDate);
    startCutoff.setFullYear(startCutoff.getFullYear() - years);
```

```
    const start = series.reduce((acc, s) => {
        const sDate = new Date(s.date);
        return (sDate <= startCutoff && (!acc || sDate > new Date(acc.date))) ?
s : acc;
    }, null);

    if (!start || !isNum(start.value) || start.value <= 0) return null;

    const actualYears = (endDate.getTime() - new Date(start.date).getTime()) /
(365.25 * 24 * 3600 * 1000);
    if (actualYears < years - 0.5) return null;

    return (Math.pow(end.value / start.value, 1 / actualYears) - 1) * 100;
};

Deno.serve(async (req) => {
  const base44 = createClientFromRequest(req);
  const logs = [];
  const log = (msg) => logs.push(`[${new Date().toISOString().split('T')
[1].slice(0, -1)}] ${msg}`);

  try {
    if (!(await base44.auth.isAuthenticated())) {
      return new Response(JSON.stringify({ ok: false, error: 'Unauthorized' }),
{ status: 401 });
    }

    const { ticker } = await req.json();
    const symbol = String(ticker || '').trim().toUpperCase();
    if (!symbol) {
      return new Response(JSON.stringify({ ok: false, error: 'ticker
required' }), { status: 400 });
    }

    log(`[FundMetrics] Starting computation for ${symbol}`);

    // Fetch all necessary data
    const [metricsRow, financials] = await Promise.all([
      (async () => (await
base44.asServiceRole.entities.Metrics.filter({ ticker_symbol: symbol }))[0] || {
ticker_symbol: symbol, ticker: symbol })(),
      base44.asServiceRole.entities.FinancialsHistory.filter({ ticker: symbol },
'-period_end', 40)
    ]);

    if (!financials || financials.length === 0) {
      log(`[FundMetrics] No financial history found for ${symbol}`);
      return new Response(JSON.stringify({
        ok: false,
        error: 'No financial history available',
        logs
      }), { status: 400 });
    }

    const quarterly = financials.filter(f => f.freq === 'quarterly').slice(0,
8);
    const annual = financials.filter(f => f.freq === 'annual').slice(0, 6);

    log(`[FundMetrics] Found ${quarterly.length} quarterly and ${annual.length}
annual records`);

    // Build TTM aggregates
    const ttmSum = (field) => {
```

```
      const values = quarterly.slice(0, 4).map(q => q[field]).filter(isNum);
      return values.length === 4 ? values.reduce((a, b) => a + b, 0) : null;
    };

    const TTM = {
      revenue: ttmSum('revenue'),
      net_income: ttmSum('net_income'),
      cfo: ttmSum('cfo'),
      capex: ttmSum('capex'),
      operating_income: ttmSum('operating_income'),
      ebit: ttmSum('ebit'),
      interest_expense: ttmSum('interest_expense'),
      depreciation: ttmSum('depreciation'),
      sbc: ttmSum('stock_based_compensation')
    };

    // Latest balance sheet items
    const latest = quarterly[0] || {};
    const BALANCE = {
      cash: latest.cash,
      total_debt: latest.total_debt,
      stockholder_equity: latest.stockholder_equity,
      total_assets: latest.total_assets,
      shares_out: latest.shares_outstanding
    };

    log(`[FundMetrics] TTM Revenue: ${TTM.revenue}, CFO: ${TTM.cfo}, Net Income:
${TTM.net_income}`);
    log(`[FundMetrics] Balance - Cash: ${BALANCE.cash}, Debt: $
{BALANCE.total_debt}, Equity: ${BALANCE.stockholder_equity}`);

    // Calculate EBITDA if missing
    if (!isNum(TTM.ebit) && isNum(TTM.operating_income)) {
      TTM.ebit = TTM.operating_income;
    }
    const ebitda = isNum(TTM.ebit) && isNum(TTM.depreciation) ? TTM.ebit +
TTM.depreciation : TTM.ebit;

    // Calculate FCF
    const fcf = isNum(TTM.cfo) && isNum(TTM.capex) ? TTM.cfo -
Math.abs(TTM.capex) : null;

    // Build annual series for CAGR
    const revenueSeries = annual
      .filter(a => isNum(a.revenue))
      .map(a => ({ date: a.period_end, value: a.revenue }))
      .sort((a, b) => a.date.localeCompare(b.date));

    const epsSeries = annual
      .filter(a => isNum(a.eps_diluted) || (isNum(a.net_income) &&
isNum(a.shares_diluted)))
      .map(a => ({
        date: a.period_end,
        value: isNum(a.eps_diluted) ? a.eps_diluted : a.net_income /
a.shares_diluted
      }))
      .sort((a, b) => a.date.localeCompare(b.date));

    // Build operating margin series
    const operatingMargins = annual
      .filter(a => isNum(a.operating_income) && isNum(a.revenue) && a.revenue >
0)
      .map(a => (a.operating_income / a.revenue) * 100);
```

```
    log(`[FundMetrics] Revenue series: ${revenueSeries.length} points, EPS
series: ${epsSeries.length} points`);

    // --- Compute all fundamental metrics ---
    const patch = {
      ticker_symbol: symbol,
      ticker: symbol, // Both keys for compatibility
      as_of_date: new Date().toISOString().slice(0, 10),
      data_source: 'computed:fundamentals',

      // Quality metrics
      fcf_margin_pct: isNum(fcf) ? 100 * safeDiv(fcf, TTM.revenue) : null,
      cfo_to_ni: safeDiv(TTM.cfo, TTM.net_income),
      fcf_to_ebit: isNum(fcf) ? safeDiv(fcf, TTM.ebit) : null,
      accruals_ratio: safeDiv(TTM.net_income - TTM.cfo, BALANCE.total_assets),
      margin_stdev_5y_pct: operatingMargins.length > 1 ? stdev(operatingMargins)
: null,

      // ROIC
      roic_pct: (() => {
        const nopat = isNum(TTM.net_income) && isNum(TTM.interest_expense)
          ? TTM.net_income + TTM.interest_expense
          : null;
        const investedCapital = isNum(BALANCE.total_debt) &&
isNum(BALANCE.stockholder_equity)
          ? BALANCE.total_debt + BALANCE.stockholder_equity
          : null;
        return 100 * safeDiv(nopat, investedCapital);
      })(),

      // Capital Allocation metrics
      debt_to_equity: safeDiv(BALANCE.total_debt, BALANCE.stockholder_equity),
      netdebt_to_ebitda: safeDiv(BALANCE.total_debt - BALANCE.cash, ebitda),
      net_debt_to_ebitda: safeDiv(BALANCE.total_debt - BALANCE.cash, ebitda), //
Alias
      interest_coverage_x: (() => {
        const coverage = safeDiv(TTM.ebit, TTM.interest_expense);
        if (isNum(coverage)) return Math.min(coverage, 99);
        if (isNum(TTM.ebit) && TTM.ebit > 0 && Math.abs(TTM.interest_expense ||
0) < 1e-6) return 99;
        return null;
      })(),
      interest_coverage: (() => { // Alias
        const coverage = safeDiv(TTM.ebit, TTM.interest_expense);
        if (isNum(coverage)) return Math.min(coverage, 99);
        if (isNum(TTM.ebit) && TTM.ebit > 0 && Math.abs(TTM.interest_expense ||
0) < 1e-6) return 99;
        return null;
      })(),

      // Buyback yield
      buyback_yield_pct: (() => {
        const sharesSeries = annual
          .filter(a => isNum(a.shares_diluted))
          .map(a => a.shares_diluted);
        if (sharesSeries.length >= 2) {
          const t = sharesSeries[0];
          const t_minus_1 = sharesSeries[1];
          return 100 * safeDiv(t_minus_1 - t, t_minus_1);
        }
        return null;
      })(),

      // SBC ratio
```

```
      sbc_to_sales_pct: 100 * safeDiv(TTM.sbc, TTM.revenue),

      // Growth metrics
      eps_cagr_5y_pct: cagr(epsSeries, 5),
      eps_cagr_3y_pct: cagr(epsSeries, 3),
      revenue_cagr_5y_pct: cagr(revenueSeries, 5),
      revenue_cagr_3y_pct: cagr(revenueSeries, 3),
      rev_cagr_3y_pct: cagr(revenueSeries, 3), // Alias

      // TTM fundamentals for reference
      revenue_ttm: TTM.revenue,
      net_income_ttm: TTM.net_income,
      cfo_ttm: TTM.cfo,
      capex_ttm: TTM.capex,
      ebit_ttm: TTM.ebit,
      ebitda_ttm: ebitda,
      fcf_ttm: fcf,
      sbc_ttm: TTM.sbc,

      // Balance sheet
      cash: BALANCE.cash,
      total_debt: BALANCE.total_debt,
      equity: BALANCE.stockholder_equity,
      total_assets: BALANCE.total_assets,
      shares_out: BALANCE.shares_out
    };

    // Filter to only numeric values
    const safePatch = Object.fromEntries(
      Object.entries(patch).filter(([k, v]) =>
        k === 'ticker_symbol' || k === 'ticker' || k === 'as_of_date' || k ===
'data_source' ||
        (typeof v === 'number' && Number.isFinite(v))
      )
    );

    log(`[FundMetrics] Computed ${Object.keys(safePatch).length} valid fields`);

    // Upsert to Metrics
    if (metricsRow.id) {
      const updatePayload = { ...safePatch };
      delete updatePayload.ticker_symbol; // Don't update the key field
      await base44.asServiceRole.entities.Metrics.update(metricsRow.id,
updatePayload);
      log(`[FundMetrics] Updated existing metrics row`);
    } else {
      await base44.asServiceRole.entities.Metrics.create(safePatch);
      log(`[FundMetrics] Created new metrics row`);
    }

    return new Response(JSON.stringify({
      ok: true,
      success: true,
      message: `Fundamental metrics computed for ${symbol}`,
      summary: {
        ticker: symbol,
        metrics_computed: Object.keys(safePatch).filter(k =>
          ['ticker_symbol', 'ticker', 'as_of_date', 'data_source'].includes(k)
        ).length
      },
      logs
    }), {
      status: 200,
      headers: { 'Content-Type': 'application/json' }
```

```
    });

  } catch (error) {
    log(`[FundMetrics][FATAL] ${error.message}`);
    return new Response(JSON.stringify({
      ok: false,
      success: false,
      stage: 'fundamental_metrics',
      error: error.message,
      logs
    }), {
      status: 500,
      headers: { 'Content-Type': 'application/json' }
    });
  }
});
```


```
{
  "computeFundamentalMetrics.ts": "import { createClientFromRequest } from
'npm:@base44/sdk@0.5.0';\n\nconst isNum = (v) => v !== null && v !== undefined
&& Number.isFinite(v);\nconst safeDiv = (a, b) => (isNum(a) && isNum(b) &&
Math.abs(b) > 1e-9) ? a / b : null;\n\nconst stdev = (arr) => {\n    const
filtered = arr.filter(isNum);\n    if (filtered.length < 2) return null;\n
const n = filtered.length;\n    const mean = filtered.reduce((a, b) => a + b) /
n;\n    const variance = filtered.map(x => Math.pow(x - mean, 2)).reduce((a, b)
=> a + b) / (n - 1);\n    return Math.sqrt(variance);\n};\n\nconst cagr =
(series, years) => {\n    if (!series || series.length < 2) return null;\n
const end = series[series.length - 1];\n    const endDate = new
Date(end.date);\n    const startCutoff = new Date(endDate);\n
startCutoff.setFullYear(startCutoff.getFullYear() - years);\n    \n    const
start = series.reduce((acc, s) => {\n        const sDate = new Date(s.date);\n
return (sDate <= startCutoff && (!acc || sDate > new Date(acc.date))) ? s :
acc;\n    }, null);\n    \n    if (!start || !isNum(start.value) || start.value
<= 0) return null;\n    \n    const actualYears = (endDate.getTime() - new
Date(start.date).getTime()) / (365.25 * 24 * 3600 * 1000);\n    if (actualYears
< years - 0.5) return null;\n    \n    return (Math.pow(end.value / start.value,
1 / actualYears) - 1) * 100;\n};\n\nDeno.serve(async (req) => {\n  const base44
= createClientFromRequest(req);\n  const logs = [];\n  const log = (msg) =>
logs.push(`[${new Date().toISOString().split('T')[1].slice(0, -1)}] $
{msg}`);\n\n  try {\n    if (!(await base44.auth.isAuthenticated())) {\n
return new Response(JSON.stringify({ ok: false, error: 'Unauthorized' }),
{ status: 401 });\n    }\n\n    const { ticker } = await req.json();\n    const
symbol = String(ticker || '').trim().toUpperCase();\n    if (!symbol) {\n
return new Response(JSON.stringify({ ok: false, error: 'ticker required' }),
{ status: 400 });\n    }\n\n    log(`[FundMetrics] Starting computation for $
{symbol}`);\n\n    // Fetch all necessary data\n    const [metricsRow,
financials] = await Promise.all([\n      (async () => (await
base44.asServiceRole.entities.Metrics.filter({ ticker_symbol: symbol }))[0] || {
ticker_symbol: symbol, ticker: symbol })(),\n
base44.asServiceRole.entities.FinancialsHistory.filter({ ticker: symbol }, '-
period_end', 40)\n    ]);\n\n    if (!financials || financials.length === 0) {\n
log(`[FundMetrics] No financial history found for ${symbol}`);\n      return new
Response(JSON.stringify({ \n        ok: false, \n        error: 'No financial
history available',\n        logs \n      }), { status: 400 });\n    }\n\n
const quarterly = financials.filter(f => f.freq === 'quarterly').slice(0, 8);\n
const annual = financials.filter(f => f.freq === 'annual').slice(0, 6);\n\n
log(`[FundMetrics] Found ${quarterly.length} quarterly and ${annual.length}
annual records`);\n\n    // Build TTM aggregates\n    const ttmSum = (field) =>
{\n      const values = quarterly.slice(0, 4).map(q =>
q[field]).filter(isNum);\n      return values.length === 4 ? values.reduce((a,
```

```
b) => a + b, 0) : null;\n    };\n\n    const TTM = {\n      revenue:
ttmSum('revenue'),\n      net_income: ttmSum('net_income'),\n      cfo:
ttmSum('cfo'),\n      capex: ttmSum('capex'),\n      operating_income:
ttmSum('operating_income'),\n      ebit: ttmSum('ebit'),\n
interest_expense: ttmSum('interest_expense'),\n      depreciation:
ttmSum('depreciation'),\n      sbc:
ttmSum('stock_based_compensation')\n    };\n\n    // Latest balance sheet
items\n    const latest = quarterly[0] || {};\n    const BALANCE = {\n
cash: latest.cash,\n      total_debt: latest.total_debt,\n
stockholder_equity: latest.stockholder_equity,\n      total_assets:
latest.total_assets,\n      shares_out: latest.shares_outstanding\n    };\n\n
log(`[FundMetrics] TTM Revenue: ${TTM.revenue}, CFO: ${TTM.cfo}, Net Income: $
{TTM.net_income}`);\n    log(`[FundMetrics] Balance - Cash: ${BALANCE.cash},
Debt: ${BALANCE.total_debt}, Equity: ${BALANCE.stockholder_equity}`);\n\n    //
Calculate EBITDA if missing\n    if (!isNum(TTM.ebit) &&
isNum(TTM.operating_income)) {\n      TTM.ebit = TTM.operating_income;\n    }\n
const ebitda = isNum(TTM.ebit) && isNum(TTM.depreciation) ? TTM.ebit +
TTM.depreciation : TTM.ebit;\n\n    // Calculate FCF\n    const fcf =
isNum(TTM.cfo) && isNum(TTM.capex) ? TTM.cfo - Math.abs(TTM.capex) : null;\n\n
// Build annual series for CAGR\n    const revenueSeries = annual\n
.filter(a => isNum(a.revenue))\n      .map(a => ({ date: a.period_end, value:
a.revenue }))\n      .sort((a, b) => a.date.localeCompare(b.date));\n\n    const
epsSeries = annual\n      .filter(a => isNum(a.eps_diluted) ||
(isNum(a.net_income) && isNum(a.shares_diluted)))\n      .map(a => ({\n
date: a.period_end,\n        value: isNum(a.eps_diluted) ? a.eps_diluted :
a.net_income / a.shares_diluted\n      }))\n      .sort((a, b) =>
a.date.localeCompare(b.date));\n\n    // Build operating margin series\n
const operatingMargins = annual\n      .filter(a => isNum(a.operating_income) &&
isNum(a.revenue) && a.revenue > 0)\n      .map(a => (a.operating_income /
a.revenue) * 100);\n\n    log(`[FundMetrics] Revenue series: $
{revenueSeries.length} points, EPS series: ${epsSeries.length}
points`);\n\n    // --- Compute all fundamental metrics ---\n    const patch =
{\n      ticker_symbol: symbol,\n      ticker: symbol, // Both keys for
compatibility\n      as_of_date: new Date().toISOString().slice(0, 10),\n
data_source: 'computed:fundamentals',\n\n      // Quality metrics\n
fcf_margin_pct: isNum(fcf) ? 100 * safeDiv(fcf, TTM.revenue) : null,\n
cfo_to_ni: safeDiv(TTM.cfo, TTM.net_income),\n      fcf_to_ebit: isNum(fcf) ?
safeDiv(fcf, TTM.ebit) : null,\n      accruals_ratio: safeDiv(TTM.net_income -
TTM.cfo, BALANCE.total_assets),\n      margin_stdev_5y_pct:
operatingMargins.length > 1 ? stdev(operatingMargins) : null,\n\n      // ROIC\n
roic_pct: (() => {\n        const nopat = isNum(TTM.net_income) &&
isNum(TTM.interest_expense) \n          ? TTM.net_income +
TTM.interest_expense \n          : null;\n        const investedCapital =
isNum(BALANCE.total_debt) && isNum(BALANCE.stockholder_equity)\n          ?
BALANCE.total_debt + BALANCE.stockholder_equity\n          : null;\n
return 100 * safeDiv(nopat, investedCapital);\n      })(),\n\n      // Capital
Allocation metrics\n      debt_to_equity: safeDiv(BALANCE.total_debt,
BALANCE.stockholder_equity),\n      netdebt_to_ebitda:
safeDiv(BALANCE.total_debt - BALANCE.cash, ebitda),\n      net_debt_to_ebitda:
safeDiv(BALANCE.total_debt - BALANCE.cash, ebitda), // Alias\n
interest_coverage_x: (() => {\n        const coverage = safeDiv(TTM.ebit,
TTM.interest_expense);\n        if (isNum(coverage)) return Math.min(coverage,
99);\n        if (isNum(TTM.ebit) && TTM.ebit > 0 &&
Math.abs(TTM.interest_expense || 0) < 1e-6) return 99;\n        return null;\n
})(),\n      interest_coverage: (() => { // Alias\n        const coverage =
safeDiv(TTM.ebit, TTM.interest_expense);\n        if (isNum(coverage)) return
Math.min(coverage, 99);\n        if (isNum(TTM.ebit) && TTM.ebit > 0 &&
Math.abs(TTM.interest_expense || 0) < 1e-6) return 99;\n        return null;\n
})(),\n\n      // Buyback yield\n      buyback_yield_pct: (() => {\n
const sharesSeries = annual\n          .filter(a => isNum(a.shares_diluted))\n
.map(a => a.shares_diluted);\n        if (sharesSeries.length >= 2) {\n
const t = sharesSeries[0];\n          const t_minus_1 = sharesSeries[1];\n
return 100 * safeDiv(t_minus_1 - t, t_minus_1);\n        }\n        return
null;\n      })(),\n\n      // SBC ratio\n      sbc_to_sales_pct: 100 *
```

safeDiv(TTM.sbc, TTM.revenue),\n\n   // Growth metrics\n
eps_cagr_5y_pct: cagr(epsSeries, 5),\n   eps_cagr_3y_pct: cagr(epsSeries,
3),\n   revenue_cagr_5y_pct: cagr(revenueSeries, 5),\n
revenue_cagr_3y_pct: cagr(revenueSeries, 3),\n   rev_cagr_3y_pct:
cagr(revenueSeries, 3), // Alias\n\n   // TTM fundamentals for reference\n
revenue_ttm: TTM.revenue,\n   net_income_ttm: TTM.net_income,\n   cfo_ttm:
TTM.cfo,\n   capex_ttm: TTM.capex,\n   ebit_ttm: TTM.ebit,\n
ebitda_ttm: ebitda,\n   fcf_ttm: fcf,\n   sbc_ttm: TTM.sbc,\n\n   //
Balance sheet\n   cash: BALANCE.cash,\n   total_debt:
BALANCE.total_debt,\n   equity: BALANCE.stockholder_equity,\n
total_assets: BALANCE.total_assets,\n   shares_out:
BALANCE.shares_out\n   };\n\n   // Filter to only numeric values\n   const
safePatch = Object.fromEntries(\n   Object.entries(patch).filter(([k, v])
=> \n   k === 'ticker_symbol' || k === 'ticker' || k === 'as_of_date' || k
=== 'data_source' ||\n   (typeof v === 'number' && Number.isFinite(v))\n
)\n   );\n\n   log(`[FundMetrics] Computed ${Object.keys(safePatch).length}
valid fields`);\n\n   // Upsert to Metrics\n   if (metricsRow.id) {\n
const updatePayload = { ...safePatch };\n   delete
updatePayload.ticker_symbol; // Don't update the key field\n   await
base44.asServiceRole.entities.Metrics.update(metricsRow.id, updatePayload);\n
log(`[FundMetrics] Updated existing metrics row`);\n   } else {\n   await
base44.asServiceRole.entities.Metrics.create(safePatch);\n
log(`[FundMetrics] Created new metrics row`);\n   }\n\n   return new
Response(JSON.stringify({\n   ok: true,\n   success: true,\n   message:
`Fundamental metrics computed for ${symbol}`,\n   summary: {\n   ticker:
symbol,\n   metrics_computed: Object.keys(safePatch).filter(k => \n
![‘ticker_symbol', 'ticker', 'as_of_date',
'data_source'].includes(k)\n   ).length\n   },\n   logs\n   }), {\n
status: 200,\n   headers: { 'Content-Type': 'application/json' }\n   });\n\n
} catch (error) {\n   log(`[FundMetrics][FATAL] ${error.message}`);\n   return
new Response(JSON.stringify({\n   ok: false,\n   success: false,\n
stage: 'fundamental_metrics',\n   error: error.message,\n   logs\n   }),
{\n   status: 500,\n   headers: { 'Content-Type': 'application/json' }\n
});\n }\n});",
  "computeGrowthMetrics.ts": "import { createClientFromRequest } from
'npm:@base44/sdk@0.5.0';\n// --- INLINED HELPERS ---\nconst isNum = (v) => v !==
null && v !== undefined && typeof v === 'number' && Number.isFinite(v);\nconst
pushLog = (logs, level, step, msg) => logs.push(`[${level}] [${new
Date().toISOString()}] [${step}] ${msg}`);\nconst persistMetrics = async
(base44, ticker, patch, logs) => { const writes = []; const skipped = []; const
finalPatch = {}; Object.entries(patch).forEach(([key, value]) => { if
(isNum(value)) { finalPatch[key] = value; writes.push(`metrics.${key}`);
pushLog(logs, 'UPSERT', 'persist', `metrics.${key}=${value.toFixed(3)}`); } else
{ skipped.push(`metrics.${key}`); pushLog(logs, 'SKIP', 'persist', `metrics.$
{key} reason=value is not a finite number`); } }); if
(Object.keys(finalPatch).length === 0) { pushLog(logs, 'INFO', 'persist', 'No
new valid metrics to save.'); return { writes, skipped }; } const [existing] =
await base44.asServiceRole.entities.Metrics.filter({ ticker_symbol: ticker });
if (existing?.id) { await
base44.asServiceRole.entities.Metrics.update(existing.id, finalPatch); } else
{ await base44.asServiceRole.entities.Metrics.create({ ...finalPatch,
ticker_symbol: ticker, ticker: ticker }); } return { writes,
skipped }; };\nconst cagr = (series, years) => { if (!series || series.length <
2) return null; const end = series[series.length - 1]; const endDate = new
Date(end.date); const startCutoff = new Date(endDate);
startCutoff.setFullYear(startCutoff.getFullYear() - years); const start =
series.reduce((acc, s) => { const sDate = new Date(s.date); if (sDate <=
startCutoff && (!acc || sDate > new Date(acc.date))) return s; return acc; },
null); if (!start || !isNum(start.value) || start.value <= 0) return null; const
actualYears = (endDate.getTime() - new Date(start.date).getTime()) / (365.25 *
24 * 3600 * 1000); if (actualYears < years - 0.5) return null; return
(Math.pow(end.value / start.value, 1 / actualYears) - 1) * 100; };\n// --- END
HELPERS ---\n\nDeno.serve(async (req) => {\n   const logs = [];\n   const
base44 = createClientFromRequest(req);\n   try {\n   if (!(await

```
base44.auth.isAuthenticated())) {\n            pushLog(logs, 'FATAL', 'auth',
'Unauthorized');\n            return new Response(JSON.stringify({ ok: false,
error: 'Unauthorized', logs }), { status: 401, headers: { 'Content-Type':
'application/json' } });\n        }\n        const { ticker } = await
req.json().catch(() => ({}));\n        if (!ticker) throw new Error(\"Ticker is
required.\");\n        pushLog(logs, 'INIT', 'growth', `Starting for $
{ticker}`);\n\n        const financials = await
base44.asServiceRole.entities.FinancialsHistory.filter({ ticker, freq:
'annual' }, '-period_end');\n        pushLog(logs, 'FETCH', 'growth',
`FinancialsHistory count=${financials.length}`);\n        \n        const
epsSeries = financials.map(r => ({ date: r.period_end, value:
r.eps_diluted })).filter(p => isNum(p.value));\n        const revenueSeries =
financials.map(r => ({ date: r.period_end, value: r.revenue })).filter(p =>
isNum(p.value));\n\n        const patch = {\n            eps_cagr_5y_pct:
cagr(epsSeries, 5),\n            eps_cagr_3y_pct: cagr(epsSeries, 3),\n
revenue_cagr_5y_pct: cagr(revenueSeries, 5),\n            revenue_cagr_3y_pct:
cagr(revenueSeries, 3),\n        };\n        \n        const { writes, skipped }
= await persistMetrics(base44, ticker, patch, logs);\n        return new
Response(JSON.stringify({ ok: true, ticker, writes, skipped, logs }), { status:
200, headers: { 'Content-Type': 'application/json' } });\n    } catch (err) {\n
pushLog(logs, 'FATAL', 'growth', err.message);\n        return new
Response(JSON.stringify({ ok: false, error: err.message, logs }), { status: 200,
headers: { 'Content-Type': 'application/json' } });\n    }\n});"
}
```

```
// FILE: functions/backfillMetricsTickerFk.ts
// ==============================================================

import { createClientFromRequest } from 'npm:@base44/sdk@0.5.0';

Deno.serve(async (req) => {
  const base44 = createClientFromRequest(req);

  try {
    if (!(await base44.auth.isAuthenticated())) {
      return new Response(JSON.stringify({ ok: false, error: 'Unauthorized' }),
{ status: 401 });
    }

    const metrics = await base44.asServiceRole.entities.Metrics.list();
    let updated = 0;

    for (const m of metrics) {
      if (!m.ticker && m.ticker_symbol) {
        const [t] = await base44.asServiceRole.entities.Ticker.filter({ symbol:
m.ticker_symbol }, undefined, 1);
        if (t?.id) {
          await base44.asServiceRole.entities.Metrics.update(m.id, { ticker:
t.id });
          updated++;
        }
      }
    }

    return new Response(JSON.stringify({
      ok: true,
      updated,
      message: `Updated ${updated} Metrics rows with missing ticker FK`
    }), { status: 200 });
```

```
  } catch (error) {
    return new Response(JSON.stringify({
      ok: false,
      error: error.message,
      stage: 'backfillFK'
    }), { status: 500 });
  }
});

// FILE: functions/computeAndSavePriceMetrics.ts
// ============================================================

import { createClientFromRequest } from 'npm:@base44/sdk@0.5.0';

const isNum = (v) => v != null && typeof v === 'number' && Number.isFinite(v);

function monthEndSeries(days) {
  // group by YYYY-MM, keep last
  const map = new Map();
  for (const d of days) {
    if (!d?.date || !Number.isFinite(d.close_adj)) continue;
    const key = d.date.slice(0,7);
    const prev = map.get(key);
    if (!prev || d.date > prev.date) map.set(key, d);
  }
  return Array.from(map.values()).sort((a,b)=>a.date.localeCompare(b.date));
}

function median(nums){
  const a=[...nums].sort((x,y)=>x-y);
  const m=Math.floor(a.length/2);
  return a.length%2?a[m]:(a[m-1]+a[m])/2;
}

function updateField(patch, key, value) {
  if (isNum(value)) patch[key] = value;
}

Deno.serve(async (req) => {
  const base44 = createClientFromRequest(req);

  try {
    if (!(await base44.auth.isAuthenticated())) {
      return new Response(JSON.stringify({ ok: false, error: 'Unauthorized' }),
{ status: 401 });
    }

    const { ticker } = await req.json();
    const symbol = String(ticker || '').trim().toUpperCase();
    if (!symbol) {
      return new Response(JSON.stringify({ ok: false, error: 'ticker
required' }), { status: 400 });
    }

    const [metricsRow, daily] = await Promise.all([
      (await base44.asServiceRole.entities.Metrics.filter({ ticker_symbol:
symbol }))[0] || { ticker_symbol: symbol },
      base44.asServiceRole.entities.PricesHistory.filter({ ticker: symbol }, '-
date', 1500)
    ]);

    if (!daily?.length) {
```

```
      return new Response(JSON.stringify({ ok: false, error: 'No price history
found' }), { status: 400 });
    }

    const latest = daily[0];
    const eps_ttm = metricsRow.eps_ttm;
    const current_pe = (isNum(latest.close_adj) && isNum(eps_ttm) && eps_ttm >
0) ? latest.close_adj / eps_ttm : null;

    // Calculate P/E history for 5Y stats
    const monthly = monthEndSeries(daily).slice(-60); // last 60 months
    const peSeries = monthly
      .map(d => (isNum(eps_ttm) && eps_ttm>0 ? d.close_adj / eps_ttm : null))
      .filter(v => Number.isFinite(v));

    const patch = {
      ticker_symbol: symbol,
      ticker: symbol,
      price_current: latest.close_adj,
      pe_ttm: current_pe
    };

    updateField(patch, 'pe_5y_low', peSeries.length ? Math.min(...peSeries) :
null);
    updateField(patch, 'pe_5y_high', peSeries.length ? Math.max(...peSeries) :
null);
    updateField(patch, 'pe_5y_median', peSeries.length ? median(peSeries) :
null);

    // PEG 5Y if growth data exists
    if (isNum(current_pe) && isNum(metricsRow.eps_cagr_5y_pct) &&
metricsRow.eps_cagr_5y_pct > 0) {
      updateField(patch, 'peg_5y', current_pe / (metricsRow.eps_cagr_5y_pct /
100));
    }

    // Multi-horizon P/E history
    const now = new Date();
    const dates = [
      new Date(now.getFullYear() - 1, now.getMonth(), now.getDate()),
      new Date(now.getFullYear() - 2, now.getMonth(), now.getDate()),
      new Date(now.getFullYear() - 3, now.getMonth(), now.getDate())
    ];

    dates.forEach((targetDate, i) => {
      const field = ['pe_12m', 'pe_24m', 'pe_36m'][i];
      const closest = daily.reduce((best, d) => {
        const daysDiff = Math.abs(new Date(d.date) - targetDate) / (1000 * 60 *
60 * 24);
        const bestDiff = Math.abs(new Date(best.date) - targetDate) / (1000 * 60
* 60 * 24);
        return daysDiff < bestDiff ? d : best;
      });
      if (closest && isNum(closest.close_adj) && isNum(eps_ttm) && eps_ttm > 0)
{
        updateField(patch, field, closest.close_adj / eps_ttm);
      }
    });

    const safePatch = Object.fromEntries(
      Object.entries(patch).filter(([, v]) => typeof v === 'number' &&
Number.isFinite(v) || typeof v === 'string')
    );
```

```
    if (metricsRow.id) {
      await base44.asServiceRole.entities.Metrics.update(metricsRow.id,
safePatch);
    } else {
      await base44.asServiceRole.entities.Metrics.create({ ...metricsRow,
...safePatch });
    }

    return new Response(JSON.stringify({ ok: true, updated:
Object.keys(safePatch) }), { status: 200 });

  } catch (error) {
    return new Response(JSON.stringify({
      ok: false,
      stage: 'price-metrics',
      error: error.message
    }), { status: 500 });
  }
});

// FILE: functions/computeFundamentalMetrics.ts
// ============================================================

import { createClientFromRequest } from 'npm:@base44/sdk@0.5.0';

const isNum = (v) => v !== null && v !== undefined && Number.isFinite(v);
const safeDiv = (a, b) => (isNum(a) && isNum(b) && Math.abs(b) > 1e-9) ? a / b :
null;

const stdev = (arr) => {
    const filtered = arr.filter(isNum);
    if (filtered.length < 2) return null;
    const n = filtered.length;
    const mean = filtered.reduce((a, b) => a + b) / n;
    const variance = filtered.map(x => Math.pow(x - mean, 2)).reduce((a, b) => a
+ b) / (n - 1);
    return Math.sqrt(variance);
};

const cagr = (series, years) => {
    if (!series || series.length < 2) return null;
    const end = series[series.length - 1];
    const endDate = new Date(end.date);
    const startCutoff = new Date(endDate);
    startCutoff.setFullYear(startCutoff.getFullYear() - years);

    const start = series.reduce((acc, s) => {
        const sDate = new Date(s.date);
        return (sDate <= startCutoff && (!acc || sDate > new Date(acc.date))) ?
s : acc;
    }, null);

    if (!start || !isNum(start.value) || start.value <= 0) return null;

    const actualYears = (endDate.getTime() - new Date(start.date).getTime()) /
(365.25 * 24 * 3600 * 1000);
    if (actualYears < years - 0.5) return null;

    return (Math.pow(end.value / start.value, 1 / actualYears) - 1) * 100;
};

Deno.serve(async (req) => {
  const base44 = createClientFromRequest(req);
  const logs = [];
```

```
  const log = (msg) => logs.push(`[${new Date().toISOString().split('T')
[1].slice(0, -1)}] ${msg}`);

  try {
    if (!(await base44.auth.isAuthenticated())) {
      return new Response(JSON.stringify({ ok: false, error: 'Unauthorized' }),
{ status: 401 });
    }

    const { ticker } = await req.json();
    const symbol = String(ticker || '').trim().toUpperCase();
    if (!symbol) {
      return new Response(JSON.stringify({ ok: false, error: 'ticker
required' }), { status: 400 });
    }

    log(`[FundMetrics] Starting computation for ${symbol}`);

    // Fetch all necessary data
    const [metricsRow, financials] = await Promise.all([
      (async () => (await
base44.asServiceRole.entities.Metrics.filter({ ticker_symbol: symbol }))[0] || {
ticker_symbol: symbol, ticker: symbol })(),
      base44.asServiceRole.entities.FinancialsHistory.filter({ ticker: symbol },
'-period_end', 40)
    ]);

    if (!financials || financials.length === 0) {
      log(`[FundMetrics] No financial history found for ${symbol}`);
      return new Response(JSON.stringify({
        ok: false,
        error: 'No financial history available',
        logs
      }), { status: 400 });
    }

    const quarterly = financials.filter(f => f.freq === 'quarterly').slice(0,
8);
    const annual = financials.filter(f => f.freq === 'annual').slice(0, 6);

    log(`[FundMetrics] Found ${quarterly.length} quarterly and ${annual.length}
annual records`);

    // Build TTM aggregates
    const ttmSum = (field) => {
      const values = quarterly.slice(0, 4).map(q => q[field]).filter(isNum);
      return values.length === 4 ? values.reduce((a, b) => a + b, 0) : null;
    };

    const TTM = {
      revenue: ttmSum('revenue'),
      net_income: ttmSum('net_income'),
      cfo: ttmSum('cfo'),
      capex: ttmSum('capex'),
      operating_income: ttmSum('operating_income'),
      ebit: ttmSum('ebit'),
      interest_expense: ttmSum('interest_expense'),
      depreciation: ttmSum('depreciation'),
      sbc: ttmSum('stock_based_compensation')
    };

    // Latest balance sheet items
    const latest = quarterly[0] || {};
    const BALANCE = {
```

```
      cash: latest.cash,
      total_debt: latest.total_debt,
      stockholder_equity: latest.stockholder_equity,
      total_assets: latest.total_assets,
      shares_out: latest.shares_outstanding
    };

    log(`[FundMetrics] TTM Revenue: ${TTM.revenue}, CFO: ${TTM.cfo}, Net Income:
${TTM.net_income}`);
    log(`[FundMetrics] Balance - Cash: ${BALANCE.cash}, Debt: $
{BALANCE.total_debt}, Equity: ${BALANCE.stockholder_equity}`);

    // Calculate EBITDA if missing
    if (!isNum(TTM.ebit) && isNum(TTM.operating_income)) {
      TTM.ebit = TTM.operating_income;
    }
    const ebitda = isNum(TTM.ebit) && isNum(TTM.depreciation) ? TTM.ebit +
TTM.depreciation : TTM.ebit;

    // Calculate FCF
    const fcf = isNum(TTM.cfo) && isNum(TTM.capex) ? TTM.cfo -
Math.abs(TTM.capex) : null;

    // Build annual series for CAGR
    const revenueSeries = annual
      .filter(a => isNum(a.revenue))
      .map(a => ({ date: a.period_end, value: a.revenue }))
      .sort((a, b) => a.date.localeCompare(b.date));

    const epsSeries = annual
      .filter(a => isNum(a.eps_diluted) || (isNum(a.net_income) &&
isNum(a.shares_diluted)))
      .map(a => ({
        date: a.period_end,
        value: isNum(a.eps_diluted) ? a.eps_diluted : a.net_income /
a.shares_diluted
      }))
      .sort((a, b) => a.date.localeCompare(b.date));

    // Build operating margin series
    const operatingMargins = annual
      .filter(a => isNum(a.operating_income) && isNum(a.revenue) && a.revenue >
0)
      .map(a => (a.operating_income / a.revenue) * 100);

    log(`[FundMetrics] Revenue series: ${revenueSeries.length} points, EPS
series: ${epsSeries.length} points`);

    // --- Compute all fundamental metrics ---
    const patch = {
      ticker_symbol: symbol,
      ticker: symbol, // Both keys for compatibility
      as_of_date: new Date().toISOString().slice(0, 10),
      data_source: 'computed:fundamentals',

      // Quality metrics
      fcf_margin_pct: isNum(fcf) ? 100 * safeDiv(fcf, TTM.revenue) : null,
      cfo_to_ni: safeDiv(TTM.cfo, TTM.net_income),
      fcf_to_ebit: isNum(fcf) ? safeDiv(fcf, TTM.ebit) : null,
      accruals_ratio: safeDiv(TTM.net_income - TTM.cfo, BALANCE.total_assets),
      margin_stdev_5y_pct: operatingMargins.length > 1 ? stdev(operatingMargins)
: null,

      // ROIC
```

```
      roic_pct: (() => {
        const nopat = isNum(TTM.net_income) && isNum(TTM.interest_expense)
          ? TTM.net_income + TTM.interest_expense
          : null;
        const investedCapital = isNum(BALANCE.total_debt) &&
isNum(BALANCE.stockholder_equity)
          ? BALANCE.total_debt + BALANCE.stockholder_equity
          : null;
        return 100 * safeDiv(nopat, investedCapital);
      })(),

      // Capital Allocation metrics
      debt_to_equity: safeDiv(BALANCE.total_debt, BALANCE.stockholder_equity),
      netdebt_to_ebitda: safeDiv(BALANCE.total_debt - BALANCE.cash, ebitda),
      net_debt_to_ebitda: safeDiv(BALANCE.total_debt - BALANCE.cash, ebitda), //
Alias
      interest_coverage_x: (() => {
        const coverage = safeDiv(TTM.ebit, TTM.interest_expense);
        if (isNum(coverage)) return Math.min(coverage, 99);
        if (isNum(TTM.ebit) && TTM.ebit > 0 && Math.abs(TTM.interest_expense ||
0) < 1e-6) return 99;
        return null;
      })(),
      interest_coverage: (() => { // Alias
        const coverage = safeDiv(TTM.ebit, TTM.interest_expense);
        if (isNum(coverage)) return Math.min(coverage, 99);
        if (isNum(TTM.ebit) && TTM.ebit > 0 && Math.abs(TTM.interest_expense ||
0) < 1e-6) return 99;
        return null;
      })(),

      // Buyback yield
      buyback_yield_pct: (() => {
        const sharesSeries = annual
          .filter(a => isNum(a.shares_diluted))
          .map(a => a.shares_diluted);
        if (sharesSeries.length >= 2) {
          const t = sharesSeries[0];
          const t_minus_1 = sharesSeries[1];
          return 100 * safeDiv(t_minus_1 - t, t_minus_1);
        }
        return null;
      })(),

      // SBC ratio
      sbc_to_sales_pct: 100 * safeDiv(TTM.sbc, TTM.revenue),

      // Growth metrics
      eps_cagr_5y_pct: cagr(epsSeries, 5),
      eps_cagr_3y_pct: cagr(epsSeries, 3),
      revenue_cagr_5y_pct: cagr(revenueSeries, 5),
      revenue_cagr_3y_pct: cagr(revenueSeries, 3),
      rev_cagr_3y_pct: cagr(revenueSeries, 3), // Alias

      // TTM fundamentals for reference
      revenue_ttm: TTM.revenue,
      net_income_ttm: TTM.net_income,
      cfo_ttm: TTM.cfo,
      capex_ttm: TTM.capex,
      ebit_ttm: TTM.ebit,
      ebitda_ttm: ebitda,
      fcf_ttm: fcf,
      sbc_ttm: TTM.sbc,
```

```
      // Balance sheet
      cash: BALANCE.cash,
      total_debt: BALANCE.total_debt,
      equity: BALANCE.stockholder_equity,
      total_assets: BALANCE.total_assets,
      shares_out: BALANCE.shares_out
    };

    // Filter to only numeric values
    const safePatch = Object.fromEntries(
      Object.entries(patch).filter(([k, v]) =>
        k === 'ticker_symbol' || k === 'ticker' || k === 'as_of_date' || k ===
'data_source' ||
        (typeof v === 'number' && Number.isFinite(v))
      )
    );

    log(`[FundMetrics] Computed ${Object.keys(safePatch).length} valid fields`);

    // Upsert to Metrics
    if (metricsRow.id) {
      const updatePayload = { ...safePatch };
      delete updatePayload.ticker_symbol; // Don't update the key field
      await base44.asServiceRole.entities.Metrics.update(metricsRow.id,
updatePayload);
      log(`[FundMetrics] Updated existing metrics row`);
    } else {
      await base44.asServiceRole.entities.Metrics.create(safePatch);
      log(`[FundMetrics] Created new metrics row`);
    }

    return new Response(JSON.stringify({
      ok: true,
      success: true,
      message: `Fundamental metrics computed for ${symbol}`,
      summary: {
        ticker: symbol,
        metrics_computed: Object.keys(safePatch).filter(k =>
          ![ 'ticker_symbol', 'ticker', 'as_of_date', 'data_source'].includes(k)
        ).length
      },
      logs
    }), {
      status: 200,
      headers: { 'Content-Type': 'application/json' }
    });

  } catch (error) {
    log(`[FundMetrics][FATAL] ${error.message}`);
    return new Response(JSON.stringify({
      ok: false,
      success: false,
      stage: 'fundamental_metrics',
      error: error.message,
      logs
    }), {
      status: 500,
      headers: { 'Content-Type': 'application/json' }
    });
  }
});

// FILE: functions/computeGrowthMetrics.ts
// ==============================================================
```

```
import { createClientFromRequest } from 'npm:@base44/sdk@0.5.0';
// --- INLINED HELPERS ---
const isNum = (v) => v !== null && v !== undefined && typeof v === 'number' &&
Number.isFinite(v);
const pushLog = (logs, level, step, msg) => logs.push(`[${level}] [${new
Date().toISOString()}] [${step}] ${msg}`);
const persistMetrics = async (base44, ticker, patch, logs) => { const writes =
[]; const skipped = []; const finalPatch = {};
Object.entries(patch).forEach(([key, value]) => { if (isNum(value))
{ finalPatch[key] = value; writes.push(`metrics.${key}`); pushLog(logs,
'UPSERT', 'persist', `metrics.${key}=${value.toFixed(3)}`); } else
{ skipped.push(`metrics.${key}`); pushLog(logs, 'SKIP', 'persist', `metrics.$
{key} reason=value is not a finite number`); } }); if
(Object.keys(finalPatch).length === 0) { pushLog(logs, 'INFO', 'persist', 'No
new valid metrics to save.'); return { writes, skipped }; } const [existing] =
await base44.asServiceRole.entities.Metrics.filter({ ticker_symbol: ticker });
if (existing?.id) { await
base44.asServiceRole.entities.Metrics.update(existing.id, finalPatch); } else
{ await base44.asServiceRole.entities.Metrics.create({ ...finalPatch,
ticker_symbol: ticker, ticker: ticker }); } return { writes, skipped }; };
const cagr = (series, years) => { if (!series || series.length < 2) return null;
const end = series[series.length - 1]; const endDate = new Date(end.date); const
startCutoff = new Date(endDate);
startCutoff.setFullYear(startCutoff.getFullYear() - years); const start =
series.reduce((acc, s) => { const sDate = new Date(s.date); if (sDate <=
startCutoff && (!acc || sDate > new Date(acc.date))) return s; return acc; },
null); if (!start || !isNum(start.value) || start.value <= 0) return null; const
actualYears = (endDate.getTime() - new Date(start.date).getTime()) / (365.25 *
24 * 3600 * 1000); if (actualYears < years - 0.5) return null; return
(Math.pow(end.value / start.value, 1 / actualYears) - 1) * 100; };
// --- END HELPERS ---

Deno.serve(async (req) => {
    const logs = [];
    const base44 = createClientFromRequest(req);
    try {
        if (!(await base44.auth.isAuthenticated())) {
            pushLog(logs, 'FATAL', 'auth', 'Unauthorized');
            return new Response(JSON.stringify({ ok: false, error:
'Unauthorized', logs }), { status: 401, headers: { 'Content-Type':
'application/json' } });
        }
        const { ticker } = await req.json().catch(() => ({}));
        if (!ticker) throw new Error("Ticker is required.");
        pushLog(logs, 'INIT', 'growth', `Starting for ${ticker}`);

        const financials = await
base44.asServiceRole.entities.FinancialsHistory.filter({ ticker, freq:
'annual' }, '-period_end');
        pushLog(logs, 'FETCH', 'growth', `FinancialsHistory count=$
{financials.length}`);

        const epsSeries = financials.map(r => ({ date: r.period_end, value:
r.eps_diluted })).filter(p => isNum(p.value));
        const revenueSeries = financials.map(r => ({ date: r.period_end, value:
r.revenue })).filter(p => isNum(p.value));

        const patch = {
            eps_cagr_5y_pct: cagr(epsSeries, 5),
            eps_cagr_3y_pct: cagr(epsSeries, 3),
            revenue_cagr_5y_pct: cagr(revenueSeries, 5),
            revenue_cagr_3y_pct: cagr(revenueSeries, 3),
        };
```

```
        const { writes, skipped } = await persistMetrics(base44, ticker, patch,
logs);
        return new Response(JSON.stringify({ ok: true, ticker, writes, skipped,
logs }), { status: 200, headers: { 'Content-Type': 'application/json' } });
    } catch (err) {
        pushLog(logs, 'FATAL', 'growth', err.message);
        return new Response(JSON.stringify({ ok: false, error: err.message, logs
}), { status: 200, headers: { 'Content-Type': 'application/json' } });
    }
});

// FILE: functions/computePriceMetrics.ts
// ==============================================================

import { createClientFromRequest } from 'npm:@base44/sdk@0.5.0';
// --- INLINED HELPERS ---
const isNum = (v) => v !== null && v !== undefined && typeof v === 'number' &&
Number.isFinite(v);
const safeDiv = (a, b) => (isNum(a) && isNum(b) && Math.abs(b) > 1e-9) ? a / b :
null;
const pushLog = (logs, level, step, msg) => logs.push(`[${level}] [${new
Date().toISOString()}] [${step}] ${msg}`);
const persistMetrics = async (base44, ticker, patch, logs) => { const writes =
[]; const skipped = []; const finalPatch = {};
Object.entries(patch).forEach(([key, value]) => { if (isNum(value))
{ finalPatch[key] = value; writes.push(`metrics.${key}`); pushLog(logs,
'UPSERT', 'persist', `metrics.${key}=${value.toFixed(3)}`); } else
{ skipped.push(`metrics.${key}`); pushLog(logs, 'SKIP', 'persist', `metrics.$
{key} reason=value is not a finite number`); } }); if
(Object.keys(finalPatch).length === 0) { pushLog(logs, 'INFO', 'persist', 'No
new valid metrics to save.'); return { writes, skipped }; } const [existing] =
await base44.asServiceRole.entities.Metrics.filter({ ticker_symbol: ticker });
if (existing?.id) { await
base44.asServiceRole.entities.Metrics.update(existing.id, finalPatch); } else
{ await base44.asServiceRole.entities.Metrics.create({ ...finalPatch,
ticker_symbol: ticker, ticker: ticker }); } return { writes, skipped }; };
// --- END HELPERS ---

Deno.serve(async (req) => {
    const logs = [];
    const base44 = createClientFromRequest(req);

    try {
        if (!(await base44.auth.isAuthenticated())) {
            pushLog(logs, 'FATAL', 'auth', 'Unauthorized');
            return new Response(JSON.stringify({ ok: false, error:
'Unauthorized', logs }), { status: 401, headers: { 'Content-Type':
'application/json' } });
        }
        const { ticker } = await req.json().catch(() => ({}));
        if (!ticker) throw new Error("Ticker is required.");
        pushLog(logs, 'INIT', 'price', `Starting for ${ticker}`);

        const fromDate = new Date(); fromDate.setFullYear(fromDate.getFullYear()
- 5);
        const [prices, financials, metrics] = await Promise.all([
            base44.asServiceRole.entities.PricesHistory.filter({ ticker, date: {
$gte: fromDate.toISOString().split('T')[0] } }, '-date'),
            base44.asServiceRole.entities.FinancialsHistory.filter({ ticker },
'-period_end'),
            base44.asServiceRole.entities.Metrics.filter({ ticker_symbol: ticker
})
        ]);
```

```
        pushLog(logs, 'FETCH', 'price', `PricesHistory count=${prices.length}`);
        pushLog(logs, 'FETCH', 'price', `FinancialsHistory count=$
{financials.length}`);
        pushLog(logs, 'FETCH', 'price', `Metrics count=${metrics.length}`);

        const latestPrice = prices[0]?.close_adj;
        const shares_latest =
financials.find(f=>isNum(f.shares_outstanding))?.shares_outstanding;
        const market_cap_latest = isNum(latestPrice) && isNum(shares_latest) ?
latestPrice * shares_latest : null;
        pushLog(logs, 'DERIVE', 'price', `market_cap_latest=$
{market_cap_latest?.toFixed(0)}`);

        const quarterlies = financials.filter(f=>f.freq === 'quarterly');
        const ebitda_ttm = quarterlies.slice(0,4).reduce((sum,q)=>sum+
(q.ebitda||0),0);
        const cash_latest = financials[0]?.cash;
        const total_debt_latest = financials[0]?.total_debt;
        const enterprise_value = market_cap_latest + (total_debt_latest -
cash_latest);
        pushLog(logs, 'DERIVE', 'price', `enterprise_value=$
{enterprise_value?.toFixed(0)}`);

        const fcf_ttm = quarterlies.slice(0,4).reduce((sum,q)=>sum+(q.cfo||0),0)
- quarterlies.slice(0,4).reduce((sum,q)=>sum+(q.capex||0),0);

        const patch = {
            ev_ebitda: safeDiv(enterprise_value, ebitda_ttm),
            fcf_yield_pct: 100 * safeDiv(fcf_ttm, market_cap_latest),
            peg_5y: safeDiv(metrics[0]?.pe_ttm, metrics[0]?.eps_cagr_5y_pct),
        };

        const { writes, skipped } = await persistMetrics(base44, ticker, patch,
logs);
        return new Response(JSON.stringify({ ok: true, ticker, writes, skipped,
logs }), { status: 200, headers: { 'Content-Type': 'application/json' } });

    } catch (err) {
        pushLog(logs, 'FATAL', 'price', err.message);
        return new Response(JSON.stringify({ ok: false, error: err.message, logs
}), { status: 200, headers: { 'Content-Type': 'application/json' } });
    }
});

// FILE: functions/computeQualityMetrics.ts
// ==============================================================

import { createClientFromRequest } from 'npm:@base44/sdk@0.5.0';
// --- INLINED HELPERS ---
const isNum = (v) => v !== null && v !== undefined && typeof v === 'number' &&
Number.isFinite(v);
const safeDiv = (a, b) => (isNum(a) && isNum(b) && Math.abs(b) > 1e-9) ? a / b :
null;
const pushLog = (logs, level, step, msg) => logs.push(`[${level}] [${new
Date().toISOString()}] [${step}] ${msg}`);
const persistMetrics = async (base44, ticker, patch, logs) => { const writes =
[]; const skipped = []; const finalPatch = {};
Object.entries(patch).forEach(([key, value]) => { if (isNum(value))
{ finalPatch[key] = value; writes.push(`metrics.${key}`); pushLog(logs,
'UPSERT', 'persist', `metrics.${key}=${value.toFixed(3)}`); } else
{ skipped.push(`metrics.${key}`); pushLog(logs, 'SKIP', 'persist', `metrics.$
{key} reason=value is not a finite number`); } }); if
(Object.keys(finalPatch).length === 0) { pushLog(logs, 'INFO', 'persist', 'No
new valid metrics to save.'); return { writes, skipped }; } const [existing] =
```

```
await base44.asServiceRole.entities.Metrics.filter({ ticker_symbol: ticker });
if (existing?.id) { await
base44.asServiceRole.entities.Metrics.update(existing.id, finalPatch); } else
{ await base44.asServiceRole.entities.Metrics.create({ ...finalPatch,
ticker_symbol: ticker, ticker: ticker }); } return { writes, skipped }; };
const stdev = (arr) => { if (!arr || arr.length < 2) return null; const n =
arr.length; const mean = arr.reduce((a, b) => a + b) / n; const variance =
arr.map(x => Math.pow(x - mean, 2)).reduce((a, b) => a + b) / (n - 1); return
Math.sqrt(variance); };
// --- END HELPERS ---

Deno.serve(async (req) => {
    const logs = [];
    const base44 = createClientFromRequest(req);
    try {
        if (!(await base44.auth.isAuthenticated())) {
            pushLog(logs, 'FATAL', 'auth', 'Unauthorized');
            return new Response(JSON.stringify({ ok: false, error:
'Unauthorized', logs }), { status: 401, headers: { 'Content-Type':
'application/json' } });
        }
        const { ticker } = await req.json().catch(() => ({}));
        if (!ticker) throw new Error("Ticker is required.");
        pushLog(logs, 'INIT', 'quality', `Starting for ${ticker}`);

        const financials = await
base44.asServiceRole.entities.FinancialsHistory.filter({ ticker }, '-
period_end');
        pushLog(logs, 'FETCH', 'quality', `FinancialsHistory count=$
{financials.length}`);

        const quarterlies = financials.filter(f => f.freq ===
'quarterly').slice(0, 20);
        const cfo_ttm = quarterlies.slice(0,4).reduce((s,q)=>s+(q.cfo||0),0);
        const capex_ttm = quarterlies.slice(0,4).reduce((s,q)=>s+(q.capex||
0),0);
        const revenue_ttm = quarterlies.slice(0,4).reduce((s,q)=>s+(q.revenue||
0),0);
        const net_income_ttm = quarterlies.slice(0,4).reduce((s,q)=>s+
(q.net_income||0),0);
        const ebit_ttm = quarterlies.slice(0,4).reduce((s,q)=>s+
(q.operating_income||0),0);
        const total_assets_latest =
financials.find(f=>isNum(f.total_assets))?.total_assets;
        const fcf_ttm = cfo_ttm - capex_ttm;

        const operatingMargins = quarterlies.map(q =>
safeDiv(q.operating_income, q.revenue)).filter(isNum);

        const patch = {
            fcf_margin_pct: 100 * safeDiv(fcf_ttm, revenue_ttm),
            cfo_to_ni: safeDiv(cfo_ttm, net_income_ttm),
            fcf_to_ebit: safeDiv(fcf_ttm, ebit_ttm),
            accruals_ratio: safeDiv(net_income_ttm - cfo_ttm,
total_assets_latest),
            margin_stdev_5y_pct: 100 * stdev(operatingMargins),
        };

        const { writes, skipped } = await persistMetrics(base44, ticker, patch,
logs);
        return new Response(JSON.stringify({ ok: true, ticker, writes, skipped,
logs }), { status: 200, headers: { 'Content-Type': 'application/json' } });
    } catch (err) {
        pushLog(logs, 'FATAL', 'quality', err.message);
```

```
        return new Response(JSON.stringify({ ok: false, error: err.message, logs
}), { status: 200, headers: { 'Content-Type': 'application/json' } });
    }
});

// FILE: functions/computeRiskMetrics.ts
// ============================================================

import { createClientFromRequest } from 'npm:@base44/sdk@0.5.0';
// --- INLINED HELPERS ---
const isNum = (v) => v !== null && v !== undefined && typeof v === 'number' &&
Number.isFinite(v);
const safeDiv = (a, b) => (isNum(a) && isNum(b) && Math.abs(b) > 1e-9) ? a / b :
null;
const pushLog = (logs, level, step, msg) => logs.push(`[${level}] [${new
Date().toISOString()}] [${step}] ${msg}`);
const persistMetrics = async (base44, ticker, patch, logs) => { const writes =
[]; const skipped = []; const finalPatch = {};
Object.entries(patch).forEach(([key, value]) => { if (isNum(value))
{ finalPatch[key] = value; writes.push(`metrics.${key}`); pushLog(logs,
'UPSERT', 'persist', `metrics.${key}=${value.toFixed(3)}`); } else
{ skipped.push(`metrics.${key}`); pushLog(logs, 'SKIP', 'persist', `metrics.$
{key} reason=value is not a finite number`); } }); if
(Object.keys(finalPatch).length === 0) { pushLog(logs, 'INFO', 'persist', 'No
new valid metrics to save.'); return { writes, skipped }; } const [existing] =
await base44.asServiceRole.entities.Metrics.filter({ ticker_symbol: ticker });
if (existing?.id) { await
base44.asServiceRole.entities.Metrics.update(existing.id, finalPatch); } else
{ await base44.asServiceRole.entities.Metrics.create({ ...finalPatch,
ticker_symbol: ticker, ticker: ticker }); } return { writes, skipped }; };
const stdev = (arr) => { if (!arr || arr.length < 2) return null; const n =
arr.length; const mean = arr.reduce((a, b) => a + b) / n; const variance =
arr.map(x => Math.pow(x - mean, 2)).reduce((a, b) => a + b) / (n - 1); return
Math.sqrt(variance); };
// --- END HELPERS ---

Deno.serve(async (req) => {
    const logs = [];
    const base44 = createClientFromRequest(req);
    try {
        if (!(await base44.auth.isAuthenticated())) {
            pushLog(logs, 'FATAL', 'auth', 'Unauthorized');
            return new Response(JSON.stringify({ ok: false, error:
'Unauthorized', logs }), { status: 401, headers: { 'Content-Type':
'application/json' } });
        }
        const { ticker } = await req.json().catch(() => ({}));
        if (!ticker) throw new Error("Ticker is required.");
        pushLog(logs, 'INIT', 'risk', `Starting for ${ticker}`);

        const fromDate = new Date(); fromDate.setFullYear(fromDate.getFullYear()
- 5);
        const [prices, financials] = await Promise.all([
            base44.asServiceRole.entities.PricesHistory.filter({ ticker, date: {
$gte: fromDate.toISOString().split('T')[0] } }, '-date'),
            base44.asServiceRole.entities.FinancialsHistory.filter({ ticker },
'-period_end')
        ]);
        pushLog(logs, 'FETCH', 'risk', `PricesHistory count=${prices.length},
FinancialsHistory count=${financials.length}`);

        const latestPrice = prices[0]?.close_adj;
        const shares_latest =
financials.find(f=>isNum(f.shares_outstanding))?.shares_outstanding;
```

```
        const market_cap_latest = isNum(latestPrice) && isNum(shares_latest) ?
latestPrice * shares_latest : null;

        const cash_latest = financials[0]?.cash;
        const total_debt_latest = financials[0]?.total_debt;

        const quarterlies = financials.filter(f=>f.freq ===
'quarterly').slice(0,20);
        const yoyRevenueGrowth = [];
        for(let i=0; i < quarterlies.length - 4; i++) {
            const current = quarterlies[i].revenue;
            const prior = quarterlies[i+4].revenue;
            if (isNum(current) && isNum(prior)) {
                yoyRevenueGrowth.push(safeDiv(current, prior) - 1);
            }
        }

        const patch = {
            net_cash_to_mktcap_pct: 100 * safeDiv(cash_latest -
total_debt_latest, market_cap_latest),
            cyclicality_pct: 100 * stdev(yoyRevenueGrowth.filter(isNum)),
        };

        const { writes, skipped } = await persistMetrics(base44, ticker, patch,
logs);
        return new Response(JSON.stringify({ ok: true, ticker, writes, skipped,
logs }), { status: 200, headers: { 'Content-Type': 'application/json' } });
    } catch (err) {
        pushLog(logs, 'FATAL', 'risk', err.message);
        return new Response(JSON.stringify({ ok: false, error: err.message, logs
}), { status: 200, headers: { 'Content-Type': 'application/json' } });
    }
});

// FILE: functions/dbHealthCheck.ts
// ============================================================

import { createClientFromRequest } from 'npm:@base44/sdk@0.5.0';

Deno.serve(async (req) => {
    const base44 = createClientFromRequest(req);

    if (!(await base44.auth.isAuthenticated())) {
        return new Response(JSON.stringify({ error: 'Unauthorized' }), { status:
401 });
    }

    try {
        const logs = [];
        const log = (msg) => logs.push(`[${new Date().toISOString().split('T')
[1].slice(0, -1)}] ${msg}`);

        log("[HealthCheck] Starting comprehensive database health check...");

        const results = {
            user_access: null,
            service_role_access: null,
            prices_table_access: null,
            environment_check: null,
            errors: []
        };

        // 1. Check user-level access
        try {
```

```
            log("[HealthCheck] Testing user-level database access...");
            const userTest = await base44.entities.PricesHistory.filter({},
null, 1);
            results.user_access = {
                status: 'OK',
                message: `Retrieved ${userTest?.length || 0} records`
            };
            log(`[HealthCheck] User access: OK`);
        } catch (userError) {
            results.user_access = {
                status: 'FAILED',
                message: userError.message
            };
            results.errors.push(`User access: ${userError.message}`);
            log(`[HealthCheck] User access: FAILED - ${userError.message}`);
        }

        // 2. Check service role access
        try {
            log("[HealthCheck] Testing service role database access...");
            const serviceTest = await
base44.asServiceRole.entities.PricesHistory.filter({}, null, 1);
            results.service_role_access = {
                status: 'OK',
                message: `Service role retrieved ${serviceTest?.length || 0}
records`
            };
            log(`[HealthCheck] Service role access: OK`);
        } catch (serviceError) {
            results.service_role_access = {
                status: 'FAILED',
                message: serviceError.message
            };
            results.errors.push(`Service role access: ${serviceError.message}`);
            log(`[HealthCheck] Service role access: FAILED - $
{serviceError.message}`);
        }

        // 3. Test write operations on PricesHistory
        try {
            log("[HealthCheck] Testing PricesHistory write permissions...");

            const testRecord = {
                ticker: 'HEALTH_TEST',
                date: new Date().toISOString().slice(0, 10),
                as_of_date: new Date().toISOString().slice(0, 10),
                source: 'health_check',
                close_adj: 100.00,
                open: 99.50,
                high: 101.00,
                low: 99.00,
                close: 100.50,
                volume: 1000
            };

            // Try to create with service role
            const created = await
base44.asServiceRole.entities.PricesHistory.create(testRecord);
            log(`[HealthCheck] Test record created with ID: ${created.id}`);

            // Try to update it
            const updated = await
base44.asServiceRole.entities.PricesHistory.update(created.id, {
                ...testRecord,
```

```
                close_adj: 101.00
            });
            log(`[HealthCheck] Test record updated successfully`);

            // Clean up - delete the test record
            await
base44.asServiceRole.entities.PricesHistory.delete(created.id);
            log(`[HealthCheck] Test record cleaned up`);

            results.prices_table_access = {
                status: 'OK',
                message: 'CREATE, UPDATE, DELETE operations successful'
            };

        } catch (writeError) {
            results.prices_table_access = {
                status: 'FAILED',
                message: writeError.message
            };
            results.errors.push(`PricesHistory write test: $
{writeError.message}`);
            log(`[HealthCheck] PricesHistory write test: FAILED - $
{writeError.message}`);
        }

        // 4. Environment check
        try {
            log("[HealthCheck] Checking environment variables...");

            const envCheck = {
                base44_app_id: !!Deno.env.get("BASE44_APP_ID"),
                has_required_env: true
            };

            results.environment_check = {
                status: 'OK',
                message: 'Environment variables present',
                details: envCheck
            };
            log(`[HealthCheck] Environment check: OK`);

        } catch (envError) {
            results.environment_check = {
                status: 'FAILED',
                message: envError.message
            };
            results.errors.push(`Environment check: ${envError.message}`);
            log(`[HealthCheck] Environment check: FAILED - $
{envError.message}`);
        }

        // Summary
        const overallStatus = results.errors.length === 0 ? 'HEALTHY' :
'ISSUES_DETECTED';
        log(`[HealthCheck] Overall status: ${overallStatus}`);

        return new Response(JSON.stringify({
            overall_status: overallStatus,
            timestamp: new Date().toISOString(),
            results,
            logs,
            recommendations: results.errors.length > 0 ? [
                "If service role access is failing, check RLS policies on
PricesHistory table",
```

```
                "Ensure service role has INSERT, UPDATE, SELECT permissions",
                "Verify BASE44_APP_ID environment variable is set",
                "Check that the service role is properly configured in the
Base44 platform"
            ] : ["All systems operational"]
        }), {
            status: results.errors.length === 0 ? 200 : 500,
            headers: { 'Content-Type': 'application/json' }
        });

    } catch (error) {
        return new Response(JSON.stringify({
            overall_status: 'CRITICAL_ERROR',
            error: error.message,
            timestamp: new Date().toISOString()
        }), {
            status: 500,
            headers: { 'Content-Type': 'application/json' }
        });
    }
});

// FILE: functions/deriveMetricsFromHistory.ts
// ============================================================

import { createClientFromRequest } from 'npm:@base44/sdk@0.5.0';

Deno.serve(async (req) => {
    const base44 = createClientFromRequest(req);

    if (!(await base44.auth.isAuthenticated())) {
        return new Response(JSON.stringify({ error: 'Unauthorized' }), {
            status: 401,
            headers: { 'Content-Type': 'application/json' }
        });
    }

    try {
        const { ticker } = await req.json();

        if (!ticker) {
            return new Response(JSON.stringify({ error: 'Ticker required' }), {
                status: 400,
                headers: { 'Content-Type': 'application/json' }
            });
        }

        // Fetch historical data
        const financials = await
base44.entities.FinancialsHistory.filter({ ticker });
        const prices = await base44.entities.PricesHistory.filter({ ticker });

        const derivedMetrics = {};

        // Calculate CAGR metrics
        const annualFinancials = financials
            .filter(f => f.freq === 'annual')
            .sort((a, b) => b.period.localeCompare(a.period));

        // Revenue CAGR 5Y
        const revenueCAGR5y = calculateCAGR(annualFinancials, 'revenue', 5);
        if (revenueCAGR5y !== null) {
            derivedMetrics.revenue_cagr_5y_pct = revenueCAGR5y * 100;
        }
```

```
        // Revenue CAGR 3Y
        const revenueCAGR3y = calculateCAGR(annualFinancials, 'revenue', 3);
        if (revenueCAGR3y !== null) {
            derivedMetrics.rev_cagr_3y_pct = revenueCAGR3y * 100;
        }

        // EPS CAGR 5Y
        const epsCAGR5y = calculateCAGR(annualFinancials, 'eps_basic', 5);
        if (epsCAGR5y !== null) {
            derivedMetrics.eps_cagr_5y_pct = epsCAGR5y * 100;
        }

        // EPS CAGR 3Y
        const epsCAGR3y = calculateCAGR(annualFinancials, 'eps_basic', 3);
        if (epsCAGR3y !== null) {
            derivedMetrics.eps_cagr_3y_pct = epsCAGR3y * 100;
        }

        // Calculate P/E bands from price history
        if (prices.length > 0 && annualFinancials.length > 0) {
            const sortedPrices = prices.sort((a, b) =>
a.date.localeCompare(b.date));
            const latestEPS = annualFinancials[0]?.eps_basic;

            if (latestEPS && latestEPS > 0) {
                const peValues = sortedPrices
                    .filter(p => p.close_adj > 0)
                    .map(p => p.close_adj / latestEPS)
                    .filter(pe => pe > 0 && pe < 200); // Filter outliers

                if (peValues.length > 0) {
                    derivedMetrics.pe_5y_low = Math.min(...peValues);
                    derivedMetrics.pe_5y_high = Math.max(...peValues);
                }
            }
        }

        // Calculate max drawdown
        if (prices.length > 0) {
            const sortedPrices = prices.sort((a, b) =>
a.date.localeCompare(b.date));
            const closes = sortedPrices.map(p => p.close_adj).filter(c => c >
0);

            if (closes.length > 0) {
                let maxDrawdown = 0;
                let peak = closes[0];

                for (const price of closes) {
                    if (price > peak) {
                        peak = price;
                    } else {
                        const drawdown = (peak - price) / peak;
                        maxDrawdown = Math.max(maxDrawdown, drawdown);
                    }
                }

                derivedMetrics.maxdrawdown_5y_pct = maxDrawdown * 100;
            }
        }

        // Calculate ROIC if data available
        const latestFinancial = annualFinancials[0];
```

```
        if (latestFinancial?.ebit && latestFinancial?.invested_capital &&
latestFinancial.invested_capital > 0) {
            derivedMetrics.roic_pct = (latestFinancial.ebit /
latestFinancial.invested_capital) * 100;
        }

        // Update metrics table if we have derived values
        if (Object.keys(derivedMetrics).length > 0) {
            const existingMetrics = await
base44.entities.Metrics.filter({ ticker_symbol: ticker });
            if (existingMetrics.length > 0) {
                await base44.entities.Metrics.update(existingMetrics[0].id,
derivedMetrics);
            }
        }

        return new Response(JSON.stringify({
            ticker,
            derived_fields: Object.keys(derivedMetrics),
            derived_count: Object.keys(derivedMetrics).length
        }), {
            status: 200,
            headers: { 'Content-Type': 'application/json' }
        });

    } catch (error) {
        console.error('Error deriving metrics:', error);
        return new Response(JSON.stringify({ error: error.message }), {
            status: 500,
            headers: { 'Content-Type': 'application/json' }
        });
    }
});

function calculateCAGR(financials, field, years) {
    if (!financials || financials.length < 2) return null;

    const latest = financials[0];
    const targetYear = new Date(latest.period).getFullYear() - years;

    const historical = financials.find(f => {
        const year = new Date(f.period).getFullYear();
        return year <= targetYear && f[field] != null;
    });

    if (!historical || !latest[field] || !historical[field] || historical[field]
<= 0) {
        return null;
    }

    const actualYears = (new Date(latest.period).getFullYear() - new
Date(historical.period).getFullYear());
    if (actualYears <= 0) return null;

    return Math.pow(latest[field] / historical[field], 1 / actualYears) - 1;
}

// FILE: functions/diagnoseMetricDependencies.ts
// ==============================================================

import { createClientFromRequest } from 'npm:@base44/sdk@0.5.0';

// --- Pure computation helpers (no side effects) ---
const isNum = (v) => v !== null && v !== undefined && Number.isFinite(v);
```

```
const safeDiv = (a, b) => (isNum(a) && isNum(b) && Math.abs(b) > 1e-9) ? a / b :
null;

const stdev = (arr) => {
    const filtered = arr.filter(isNum);
    if (filtered.length < 2) return null;
    const n = filtered.length;
    const mean = filtered.reduce((a, b) => a + b) / n;
    const variance = filtered.map(x => Math.pow(x - mean, 2)).reduce((a, b) => a
+ b) / (n - 1);
    return Math.sqrt(variance);
};

const cagr = (series, years) => {
    if (!series || series.length < 2) return null;
    const end = series[series.length - 1];
    const endDate = new Date(end.date);
    const startCutoff = new Date(endDate);
    startCutoff.setFullYear(startCutoff.getFullYear() - years);

    const start = series.reduce((acc, s) => {
        const sDate = new Date(s.date);
        return (sDate <= startCutoff && (!acc || sDate > new Date(acc.date))) ?
s : acc;
    }, null);

    if (!start || !isNum(start.value) || start.value <= 0) return null;

    const actualYears = (endDate.getTime() - new Date(start.date).getTime()) /
(365.25 * 24 * 3600 * 1000);
    if (actualYears < years - 0.5) return null;

    return (Math.pow(end.value / start.value, 1 / actualYears) - 1) * 100;
};

// --- Metric dependency definitions ---
const METRIC_DEFINITIONS = {
    // Quality metrics
    fcf_margin_pct: {
        category: 'Quality',
        label: 'FCF Margin %',
        rawFields: ['cfo', 'capex', 'revenue'],
        aggregates: ['CFO_TTM', 'Capex_TTM', 'Revenue_TTM'],
        compute: (agg) => {
            const fcf = isNum(agg.CFO_TTM) && isNum(agg.Capex_TTM) ? agg.CFO_TTM
- agg.Capex_TTM : null;
            return isNum(fcf) ? 100 * safeDiv(fcf, agg.Revenue_TTM) : null;
        }
    },
    cfo_to_ni: {
        category: 'Quality',
        label: 'CFO/NI',
        rawFields: ['cfo', 'net_income'],
        aggregates: ['CFO_TTM', 'NetIncome_TTM'],
        compute: (agg) => safeDiv(agg.CFO_TTM, agg.NetIncome_TTM)
    },
    fcf_to_ebit: {
        category: 'Quality',
        label: 'FCF/EBIT',
        rawFields: ['cfo', 'capex', 'operating_income'],
        aggregates: ['CFO_TTM', 'Capex_TTM', 'EBIT_TTM'],
        compute: (agg) => {
            const fcf = isNum(agg.CFO_TTM) && isNum(agg.Capex_TTM) ? agg.CFO_TTM
- agg.Capex_TTM : null;
```

```
            return isNum(fcf) ? safeDiv(fcf, agg.EBIT_TTM) : null;
        }
    },
    accruals_ratio: {
        category: 'Quality',
        label: 'Accruals Ratio',
        rawFields: ['net_income', 'cfo', 'total_assets'],
        aggregates: ['NetIncome_TTM', 'CFO_TTM', 'TotalAssets_Latest'],
        compute: (agg) => safeDiv(agg.NetIncome_TTM - agg.CFO_TTM,
agg.TotalAssets_Latest)
    },
    margin_stdev_5y_pct: {
        category: 'Quality',
        label: 'Margin Stdev 5Y %',
        rawFields: ['operating_income', 'revenue'],
        aggregates: ['OperatingMargin_Series'],
        compute: (agg) => stdev(agg.OperatingMargin_Series)
    },
    roic_pct: {
        category: 'Quality',
        label: 'ROIC %',
        rawFields: ['net_income', 'interest_expense', 'total_debt',
'stockholder_equity'],
        aggregates: ['NetIncome_TTM', 'InterestExpense_TTM', 'TotalDebt_Latest',
'Equity_Latest'],
        compute: (agg) => {
            const nopat = isNum(agg.NetIncome_TTM) &&
isNum(agg.InterestExpense_TTM)
                ? agg.NetIncome_TTM + agg.InterestExpense_TTM
                : null;
            const investedCapital = isNum(agg.TotalDebt_Latest) &&
isNum(agg.Equity_Latest)
                ? agg.TotalDebt_Latest + agg.Equity_Latest
                : null;
            return 100 * safeDiv(nopat, investedCapital);
        }
    },

    // Capital Allocation metrics
    debt_to_equity: {
        category: 'Capital Allocation',
        label: 'Debt/Equity',
        rawFields: ['total_debt', 'stockholder_equity'],
        aggregates: ['TotalDebt_Latest', 'Equity_Latest'],
        compute: (agg) => safeDiv(agg.TotalDebt_Latest, agg.Equity_Latest)
    },
    netdebt_to_ebitda: {
        category: 'Capital Allocation',
        label: 'Net Debt/EBITDA',
        rawFields: ['total_debt', 'cash', 'operating_income', 'depreciation'],
        aggregates: ['TotalDebt_Latest', 'Cash_Latest', 'EBITDA_TTM'],
        compute: (agg) => safeDiv(agg.TotalDebt_Latest - agg.Cash_Latest,
agg.EBITDA_TTM)
    },
    interest_coverage_x: {
        category: 'Capital Allocation',
        label: 'Interest Coverage',
        rawFields: ['operating_income', 'interest_expense'],
        aggregates: ['EBIT_TTM', 'InterestExpense_TTM'],
        compute: (agg) => {
            const coverage = safeDiv(agg.EBIT_TTM, agg.InterestExpense_TTM);
            if (isNum(coverage)) return Math.min(coverage, 99);
            if (isNum(agg.EBIT_TTM) && agg.EBIT_TTM > 0 &&
Math.abs(agg.InterestExpense_TTM || 0) < 1e-6) return 99;
```

```
            return null;
        }
    },
    buyback_yield_pct: {
        category: 'Capital Allocation',
        label: 'Buyback Yield %',
        rawFields: ['shares_diluted', 'purchase_of_stock'],
        aggregates: ['DilutedShares_Series', 'PurchaseOfStock_TTM',
'MarketCap'],
        compute: (agg) => {
            if (agg.DilutedShares_Series?.length >= 2) {
                const t =
agg.DilutedShares_Series[agg.DilutedShares_Series.length - 1];
                const t_minus_1 =
agg.DilutedShares_Series[agg.DilutedShares_Series.length - 2];
                if (isNum(t) && isNum(t_minus_1)) {
                    return 100 * safeDiv(t_minus_1 - t, t_minus_1);
                }
            }
            if (isNum(agg.PurchaseOfStock_TTM) && isNum(agg.MarketCap)) {
                return 100 * safeDiv(-agg.PurchaseOfStock_TTM, agg.MarketCap);
            }
            return null;
        }
    },

    // Growth metrics
    eps_cagr_5y_pct: {
        category: 'Growth',
        label: 'EPS CAGR 5Y %',
        rawFields: ['eps_diluted', 'net_income', 'shares_diluted'],
        aggregates: ['EPS_Series'],
        compute: (agg) => cagr(agg.EPS_Series, 5)
    },
    eps_cagr_3y_pct: {
        category: 'Growth',
        label: 'EPS CAGR 3Y %',
        rawFields: ['eps_diluted', 'net_income', 'shares_diluted'],
        aggregates: ['EPS_Series'],
        compute: (agg) => cagr(agg.EPS_Series, 3)
    },
    revenue_cagr_5y_pct: {
        category: 'Growth',
        label: 'Revenue CAGR 5Y %',
        rawFields: ['revenue'],
        aggregates: ['Revenue_Series'],
        compute: (agg) => cagr(agg.Revenue_Series, 5)
    },
    revenue_cagr_3y_pct: {
        category: 'Growth',
        label: 'Revenue CAGR 3Y %',
        rawFields: ['revenue'],
        aggregates: ['Revenue_Series'],
        compute: (agg) => cagr(agg.Revenue_Series, 3)
    },

    // Risk metrics
    net_cash_to_mktcap_pct: {
        category: 'Risk',
        label: 'Net Cash/MktCap %',
        rawFields: ['cash', 'total_debt'],
        aggregates: ['Cash_Latest', 'TotalDebt_Latest', 'MarketCap'],
        compute: (agg) => 100 * safeDiv(agg.Cash_Latest - agg.TotalDebt_Latest,
agg.MarketCap)
```

```
    }
};

// --- Main diagnostic function ---
Deno.serve(async (req) => {
    const base44 = createClientFromRequest(req);

    try {
        if (!(await base44.auth.isAuthenticated())) {
            return new Response(JSON.stringify({ error: 'Unauthorized' }),
{ status: 401 });
        }

        const { ticker } = await req.json();
        const symbol = String(ticker || '').trim().toUpperCase();
        if (!symbol) {
            return new Response(JSON.stringify({ error: 'ticker required' }),
{ status: 400 });
        }

        const report = {
            ticker: symbol,
            timestamp: new Date().toISOString(),
            summary: {
                computed_ok: 0,
                computed_null: 0,
                stored_ok: 0,
                stored_null: 0,
                mismatches: 0
            },
            financials_check: [],
            aggregates_check: [],
            metrics_check: [],
            recommendations: []
        };

        // 1. Fetch all data
        const [financials, prices, spyPrices, metricsRow] = await Promise.all([
            base44.asServiceRole.entities.FinancialsHistory.filter({ ticker:
symbol }, '-period_end', 100),
            base44.asServiceRole.entities.PricesHistory.filter({ ticker:
symbol }, '-date', 1500),
            base44.asServiceRole.entities.PricesHistory.filter({ ticker:
'SPY' }, '-date', 1500),
            base44.asServiceRole.entities.Metrics.filter({ ticker_symbol: symbol
}).then(arr => arr[0] || null)
        ]);

        if (financials.length === 0) {
            report.recommendations.push('âŒ CRITICAL: No FinancialsHistory
records found for this ticker');
            return new Response(JSON.stringify(report), { status: 200, headers:
{ 'Content-Type': 'application/json' } });
        }

        const quarterly = financials.filter(f => f.freq ===
'quarterly').sort((a, b) => b.period_end.localeCompare(a.period_end));
        const annual = financials.filter(f => f.freq === 'annual').sort((a, b)
=> a.period_end.localeCompare(b.period_end));

        // 2. Check FinancialsHistory completeness
        const expectedFields = [
            'revenue', 'net_income', 'cfo', 'capex', 'operating_income', 'ebit',
'ebitda',
```

```
            'total_debt', 'cash', 'stockholder_equity', 'total_assets',
            'eps_diluted', 'shares_diluted', 'interest_expense', 'depreciation',
'purchase_of_stock'
        ];

        const allFields = new Set();
        financials.forEach(f => Object.keys(f).forEach(k => allFields.add(k)));

        expectedFields.forEach(field => {
            const records = financials.filter(f => isNum(f[field]));
            const check = {
                field,
                status: records.length > 0 ? 'Present' : 'Missing',
                count: records.length,
                mostRecent: records[0]?.period_end || null
            };

            if (records.length === 0) {
                // Look for similar field names
                const aliases = Array.from(allFields).filter(f =>
                    f.toLowerCase().includes(field.replace(/_/g, '')) ||
                    field.replace(/_/g, '').includes(f.toLowerCase())
                );
                if (aliases.length > 0) {
                    check.possibleAliases = aliases;
                    report.recommendations.push(`âš ï¸ Field '${field}' missing
â€” possible aliases: ${aliases.join(', ')}`);
                } else {
                    report.recommendations.push(`âŒ Field '${field}' completely
missing from FinancialsHistory`);
                }
            } else if (records.length < 4) {
                report.recommendations.push(`âš ï¸ Field '${field}' has only $
{records.length} records (need 4+ for TTM)`);
            }

            report.financials_check.push(check);
        });

        // 3. Simulate aggregates
        const aggregates = {};

        // TTM aggregates
        const ttmFields = {
            CFO_TTM: 'cfo',
            Capex_TTM: 'capex',
            Revenue_TTM: 'revenue',
            NetIncome_TTM: 'net_income',
            EBIT_TTM: 'operating_income',
            InterestExpense_TTM: 'interest_expense',
            PurchaseOfStock_TTM: 'purchase_of_stock'
        };

        Object.entries(ttmFields).forEach(([aggName, field]) => {
            const values = quarterly.slice(0, 4).map(q =>
q[field]).filter(isNum);
            if (values.length === 4) {
                aggregates[aggName] = values.reduce((a, b) => a + b, 0);
                report.aggregates_check.push({
                    aggregate: aggName,
                    status: 'Success',
                    value: aggregates[aggName],
                    source: `Sum of last 4 quarters of '${field}'`
                });
```

```
            } else {
                aggregates[aggName] = null;
                report.aggregates_check.push({
                    aggregate: aggName,
                    status: 'Failed',
                    value: null,
                    reason: `Insufficient quarterly '${field}' data (need 4,
have ${values.length})`
                });
                report.recommendations.push(`âš ï¸ Cannot compute ${aggName}:
need 4 quarters of '${field}', have ${values.length}`);
            }
        });

        // EBITDA_TTM (special case)
        const depValues = quarterly.slice(0, 4).map(q =>
q.depreciation).filter(isNum);
        if (depValues.length === 4) {
            const dep_ttm = depValues.reduce((a, b) => a + b, 0);
            aggregates.EBITDA_TTM = isNum(aggregates.EBIT_TTM) ?
aggregates.EBIT_TTM + dep_ttm : null;
        } else {
            aggregates.EBITDA_TTM = null;
        }
        report.aggregates_check.push({
            aggregate: 'EBITDA_TTM',
            status: aggregates.EBITDA_TTM ? 'Success' : 'Failed',
            value: aggregates.EBITDA_TTM,
            reason: aggregates.EBITDA_TTM ? 'EBIT_TTM + Depreciation_TTM' :
'Missing EBIT or Depreciation'
        });

        // Latest balance sheet items
        const latestQ = quarterly[0] || {};
        aggregates.TotalDebt_Latest = latestQ.total_debt;
        aggregates.Cash_Latest = latestQ.cash;
        aggregates.Equity_Latest = latestQ.stockholder_equity;
        aggregates.TotalAssets_Latest = latestQ.total_assets;

        ['TotalDebt_Latest', 'Cash_Latest', 'Equity_Latest',
'TotalAssets_Latest'].forEach(aggName => {
            report.aggregates_check.push({
                aggregate: aggName,
                status: isNum(aggregates[aggName]) ? 'Success' : 'Failed',
                value: aggregates[aggName],
                source: `Latest quarterly balance sheet`
            });
        });

        // Annual series
        aggregates.EPS_Series = annual.map(a => ({
            date: a.period_end,
            value: a.eps_diluted || (isNum(a.net_income) &&
isNum(a.shares_diluted) ? a.net_income / a.shares_diluted : null)
        })).filter(p => isNum(p.value));

        aggregates.Revenue_Series = annual.map(a => ({
            date: a.period_end,
            value: a.revenue
        })).filter(p => isNum(p.value));

        aggregates.DilutedShares_Series = annual.map(a =>
a.shares_diluted).filter(isNum);
```

```
        aggregates.OperatingMargin_Series = annual
            .map(a => 100 * safeDiv(a.operating_income, a.revenue))
            .filter(isNum);

        report.aggregates_check.push({
            aggregate: 'EPS_Series',
            status: aggregates.EPS_Series.length >= 5 ? 'Success' :
'Insufficient',
            count: aggregates.EPS_Series.length,
            reason: aggregates.EPS_Series.length < 5 ? 'Need 5+ years for 5Y
CAGR' : null
        });

        report.aggregates_check.push({
            aggregate: 'Revenue_Series',
            status: aggregates.Revenue_Series.length >= 5 ? 'Success' :
'Insufficient',
            count: aggregates.Revenue_Series.length,
            reason: aggregates.Revenue_Series.length < 5 ? 'Need 5+ years for 5Y
CAGR' : null
        });

        // Market cap
        if (prices.length > 0 && isNum(latestQ.shares_outstanding)) {
            aggregates.MarketCap = prices[0].close_adj *
latestQ.shares_outstanding;
            report.aggregates_check.push({
                aggregate: 'MarketCap',
                status: 'Success',
                value: aggregates.MarketCap,
                source: 'Latest price Ã— shares_outstanding'
            });
        } else {
            aggregates.MarketCap = null;
            report.aggregates_check.push({
                aggregate: 'MarketCap',
                status: 'Failed',
                reason: prices.length === 0 ? 'No price data' : 'No
shares_outstanding'
            });
        }

        // 4. Simulate metric computations
        Object.entries(METRIC_DEFINITIONS).forEach(([metricKey, definition]) =>
{
            const simulatedValue = definition.compute(aggregates);
            const storedValue = metricsRow?.[metricKey];

            const check = {
                metric: metricKey,
                label: definition.label,
                category: definition.category,
                simulatedValue,
                storedValue,
                status: null,
                missingDependencies: [],
                guardTriggered: null
            };

            // Check dependencies
            definition.aggregates.forEach(agg => {
                if (!isNum(aggregates[agg])) {
                    check.missingDependencies.push(agg);
                }
```

```
            });

            // Determine status
            if (isNum(simulatedValue)) {
                report.summary.computed_ok++;
                if (isNum(storedValue)) {
                    report.summary.stored_ok++;
                    const diff = Math.abs(simulatedValue - storedValue);
                    check.status = diff < 0.01 ? 'Match' : 'Mismatch';
                    if (diff >= 0.01) {
                        report.summary.mismatches++;
                        report.recommendations.push(`âš ï¸ ${definition.label}:
Simulated=${simulatedValue.toFixed(2)}, Stored=${storedValue.toFixed(2)} â€”
values differ`);
                    }
                } else {
                    report.summary.stored_null++;
                    check.status = 'Simulated OK, Stored Null';
                    report.recommendations.push(`âŒ ${definition.label}:
Computed successfully (${simulatedValue.toFixed(2)}) but not stored in Metrics
â€” check persistence logic`);
                }
            } else {
                report.summary.computed_null++;
                check.status = 'Simulated Null';
                if (check.missingDependencies.length > 0) {
                    check.guardTriggered = `Missing dependencies: $
{check.missingDependencies.join(', ')}`;
                    report.recommendations.push(`âš ï¸ ${definition.label}:
Cannot compute due to missing ${check.missingDependencies.join(', ')}`);
                } else {
                    check.guardTriggered = 'Division by zero or invalid input
values';
                }
                if (!isNum(storedValue)) {
                    report.summary.stored_null++;
                } else {
                    report.summary.stored_ok++;
                }
            }

            report.metrics_check.push(check);
        });

        // 5. Final recommendations
        if (report.summary.computed_null > report.summary.computed_ok) {
            report.recommendations.unshift('âŒ CRITICAL: More metrics failed
than succeeded â€” check FinancialsHistory data quality');
        }

        if (report.summary.stored_null > 0 && report.summary.computed_ok > 0) {
            report.recommendations.unshift(`âš ï¸ ${report.summary.computed_ok}
metrics computed successfully but ${report.summary.stored_null} not persisted
â€” check compute function execution`);
        }

        return new Response(JSON.stringify(report, null, 2), {
            status: 200,
            headers: { 'Content-Type': 'application/json' }
        });

    } catch (error) {
        return new Response(JSON.stringify({
            error: error.message,
```

```
            stack: error.stack
        }), {
            status: 500,
            headers: { 'Content-Type': 'application/json' }
        });
    }
});

// FILE: functions/diagnosePeHistory.ts
// ============================================================

import { createClientFromRequest } from 'npm:@base44/sdk@0.5.0';

Deno.serve(async (req) => {
    const base44 = createClientFromRequest(req);

    if (!(await base44.auth.isAuthenticated())) {
        return new Response(JSON.stringify({ error: 'Unauthorized' }), { status:
401 });
    }

    try {
        const { ticker } = await req.json();
        if (!ticker) {
            return new Response(JSON.stringify({ error: 'ticker required' }),
{ status: 400 });
        }

        const tickerUpper = ticker.trim().toUpperCase();

        const [metrics, financials, prices] = await Promise.all([
            base44.entities.Metrics.filter({ ticker_symbol: tickerUpper }),
            base44.entities.FinancialsHistory.filter({ ticker: tickerUpper }, '-
period_end', 50),
            base44.entities.PricesHistory.filter({ ticker: tickerUpper }, '-
date', 5*252)
        ]);

        const issues = [];
        const diagnostics = {
            ticker: tickerUpper,
            timestamp: new Date().toISOString(),
            issues: issues,
            summary: {},
            data_coverage: {},
            pe_analysis: {}
        };

        // Check quarterly financials coverage
        const quarterlyFinancials = financials.filter(f => f.freq ===
'quarterly');
        diagnostics.data_coverage.financials_quarterly_count =
quarterlyFinancials.length;

        if (quarterlyFinancials.length === 0) {
            issues.push("âŒ No quarterly financials found");
        } else if (quarterlyFinancials.length < 4) {
            issues.push(`âš ï¸ Only ${quarterlyFinancials.length} quarterly
reports (need 4+ for TTM)`);
        }

        // Check TTM EPS points
        const ttmEpsPoints = [];
        if (quarterlyFinancials.length >= 4) {
```

```
            for (let i = 0; i <= quarterlyFinancials.length - 4; i++) {
                const window = quarterlyFinancials.slice(i, i + 4);
                const epsSum = window.reduce((s, w) => s + (w.eps_diluted || 0),
0);
                if (epsSum > 0) {
                    ttmEpsPoints.push({
                        date: window[0].period_end,
                        eps_ttm: epsSum
                    });
                }
            }
        }

        diagnostics.data_coverage.ttm_eps_points = {
            count: ttmEpsPoints.length,
            points: ttmEpsPoints
        };

        if (ttmEpsPoints.length === 0) {
            issues.push("âŒ No valid TTM EPS points computed");
        }

        // Check price coverage
        diagnostics.data_coverage.prices_count = prices.length;
        diagnostics.data_coverage.price_date_range = prices.length > 0 ? {
            latest: prices[0]?.date,
            earliest: prices[prices.length - 1]?.date
        } : null;

        if (prices.length === 0) {
            issues.push("âŒ No price history found");
        } else if (prices.length < 252) {
            issues.push(`âš ï¸ Limited price history: ${prices.length} days
(recommend 5Y = ~1260 days)`);
        }

        // Check P/E anchor calculations
        const metricsRow = metrics[0] || {};
        const peAnchors = {
            pe_12m: { value: metricsRow.pe_12m, has_price_date: !!prices[0],
has_eps_ttm_date: ttmEpsPoints.length > 0 },
            pe_24m: { value: metricsRow.pe_24m, has_price_date: !!prices[0],
has_eps_ttm_date: ttmEpsPoints.length > 0 },
            pe_36m: { value: metricsRow.pe_36m, has_price_date: !!prices[0],
has_eps_ttm_date: ttmEpsPoints.length > 0 }
        };

        diagnostics.pe_analysis.anchors = peAnchors;

        Object.entries(peAnchors).forEach(([key, anchor]) => {
            const finite = typeof anchor.value === 'number' &&
Number.isFinite(anchor.value);
            if (!finite && anchor.has_price_date && anchor.has_eps_ttm_date) {
                issues.push(`âš ï¸ ${key} not computed despite having price &
EPS data`);
            }
        });

        // Check 5Y distribution
        const fiveYearDistribution = {
            pe_5y_low: metricsRow.pe_5y_low,
            pe_5y_median: metricsRow.pe_5y_median,
            pe_5y_high: metricsRow.pe_5y_high
        };
```

```
        const validDistValues = Object.values(fiveYearDistribution).filter(v =>
typeof v === 'number' && Number.isFinite(v));
        diagnostics.pe_analysis.five_year_distribution = {
            ...fiveYearDistribution,
            pe_count: validDistValues.length,
            expected_trading_days: Math.min(prices.length, 5 * 252)
        };

        if (validDistValues.length === 0 && prices.length > 252 &&
ttmEpsPoints.length > 0) {
            issues.push("âŒ 5Y P/E distribution not computed despite sufficient
data");
        }

        // Summary
        diagnostics.summary = {
            has_sufficient_data: quarterlyFinancials.length >= 4 &&
prices.length > 252 && ttmEpsPoints.length > 0,
            critical_issues_count: issues.filter(i =>
i.startsWith('âŒ')).length,
            warning_issues_count: issues.filter(i =>
i.startsWith('âš ï¸')).length,
            overall_status: issues.length === 0 ? "âœ… All checks passed" : `$
{issues.length} issues found`
        };

        return new Response(JSON.stringify(diagnostics, null, 2), {
            status: 200,
            headers: { 'Content-Type': 'application/json' }
        });

    } catch (error) {
        return new Response(JSON.stringify({
            error: error.message,
            ticker: ticker || 'unknown'
        }), {
            status: 500,
            headers: { 'Content-Type': 'application/json' }
        });
    }
});

// FILE: functions/diagnoseTickerPipeline.ts
// ============================================================

import { createClientFromRequest } from 'npm:@base44/sdk@0.5.0';

const isNum = (v) => v!=null && typeof v==='number' && Number.isFinite(v);
const REQ = [

'pe_ttm','pe_fwd','ev_ebitda','pe_12m','pe_24m','pe_36m','pe_5y_low','pe_5y_high
','pe_5y_median',
  'cfo_to_ni','fcf_margin_pct','fcf_to_ebit','accruals_ratio',
  'debt_to_equity','netdebt_to_ebitda','interest_coverage_x',

'eps_cagr_5y_pct','eps_cagr_3y_pct','revenue_cagr_5y_pct','revenue_cagr_3y_pct'
];
const pick = (o,k)=>Object.fromEntries(k.map(x=>[x,o?.[x]]));

Deno.serve(async (req)=>{
  const b = createClientFromRequest(req); const logs=[];
  const log=(m)=>{logs.push(m); console.log('[diagnose]',m);};
  try{
```

```
    if(!(await b.auth.isAuthenticated())) return new
Response(JSON.stringify({ok:false,error:'Unauthorized'}),{status:401});
    const body = await req.json().catch(()=>({}));
    const ticker = String(body?.ticker ?? '').trim().toUpperCase();
    const force = !!body?.force;
    if(!ticker) return new Response(JSON.stringify({ok:false,error:'ticker
required'}),{status:400});

    const beforeRow = (await
b.asServiceRole.entities.Metrics.filter({ ticker_symbol: ticker }))[0] ?? {};
    const before = pick(beforeRow, REQ);

    await b.functions.invoke('fixMetricFieldNames', {});
    await b.functions.invoke('computeFundamentalMetrics', { ticker });
    await b.functions.invoke('computeAndSavePriceMetrics', { ticker });
    if (force) await b.functions.invoke('onboardOrRefreshTicker', { ticker,
force: true });

    const afterRow = (await
b.asServiceRole.entities.Metrics.filter({ ticker_symbol: ticker }))[0] ?? {};
    const after = pick(afterRow, REQ);
    const missing = REQ.filter(k => !isNum(after[k]));
    const filled_now = REQ.filter(k => !isNum(before[k]) && isNum(after[k]));

    return new Response(JSON.stringify({ ok:true, ticker, filled_now,
still_missing: missing, snapshot:{before,after}, logs }), { status:200 });
  }catch(e){
    return new Response(JSON.stringify({ ok:false, stage:'diagnose',
error:e.message, logs }), { status:500 });
  }
});

// FILE: functions/diagnosticMetricsCompleteness.ts
// ==============================================================

import { createClientFromRequest } from 'npm:@base44/sdk@0.5.0';

const isNum = (v) => v !== null && v !== undefined && Number.isFinite(v);

Deno.serve(async (req) => {
  const base44 = createClientFromRequest(req);

  try {
    if (!(await base44.auth.isAuthenticated())) {
      return new Response(JSON.stringify({ error: 'Unauthorized' }), { status:
401 });
    }

    const { ticker } = await req.json();
    const symbol = String(ticker || '').trim().toUpperCase();
    if (!symbol) {
      return new Response(JSON.stringify({ error: 'ticker required' }),
{ status: 400 });
    }

    const report = {
      ticker: symbol,
      timestamp: new Date().toISOString(),
      steps: [],
      summary: {
        functionsExecuted: 0,
        functionsSucceeded: 0,
        functionsFailed: 0,
        metricsComputed: 0,
```

```
        metricsNull: 0,
        criticalMissing: []
      }
    };

    // Step 1: Run fixMetricFieldNames
    report.steps.push({ step: 1, name: 'fixMetricFieldNames', status: 'starting'
});
    try {
      const { data: fixResult } = await
base44.functions.invoke('fixMetricFieldNames', {});
      report.steps[0].status = 'success';
      report.steps[0].result = fixResult;
      report.summary.functionsExecuted++;
      report.summary.functionsSucceeded++;
    } catch (error) {
      report.steps[0].status = 'failed';
      report.steps[0].error = error.message;
      report.summary.functionsExecuted++;
      report.summary.functionsFailed++;
    }

    // Step 2: Check current Metrics state
    report.steps.push({ step: 2, name: 'checkCurrentMetrics', status: 'starting'
});
    const [metricsRow] = await
base44.asServiceRole.entities.Metrics.filter({ ticker_symbol: symbol });
    if (!metricsRow) {
      report.steps[1].status = 'warning';
      report.steps[1].message = 'No Metrics row exists for this ticker';
      report.steps[1].availableFields = [];
    } else {
      report.steps[1].status = 'success';
      const fields = Object.keys(metricsRow).filter(k => !k.startsWith('_') && k
!== 'id' && k !== 'created_date' && k !== 'updated_date');
      const populatedFields = fields.filter(k => metricsRow[k] !== null &&
metricsRow[k] !== undefined);
      report.steps[1].totalFields = fields.length;
      report.steps[1].populatedFields = populatedFields.length;
      report.steps[1].emptyFields = fields.length - populatedFields.length;
      report.steps[1].samplePopulated = populatedFields.slice(0, 10);
      report.steps[1].sampleEmpty = fields.filter(k => !
populatedFields.includes(k)).slice(0, 10);
    }

    // Step 3: Check FinancialsHistory availability
    report.steps.push({ step: 3, name: 'checkFinancialsHistory', status:
'starting' });
    const financials = await
base44.asServiceRole.entities.FinancialsHistory.filter({ ticker: symbol }, '-
period_end', 40);
    const quarterly = financials.filter(f => f.freq === 'quarterly');
    const annual = financials.filter(f => f.freq === 'annual');

    if (financials.length === 0) {
      report.steps[2].status = 'critical';
      report.steps[2].message = 'No financial history found';
      report.summary.criticalMissing.push('FinancialsHistory is empty');
    } else {
      report.steps[2].status = 'success';
      report.steps[2].totalRecords = financials.length;
      report.steps[2].quarterlyRecords = quarterly.length;
      report.steps[2].annualRecords = annual.length;
```

```
      // Check which fields are available
      const sampleRecord = financials[0];
      const availableFields = Object.keys(sampleRecord).filter(k => !
k.startsWith('_') && sampleRecord[k] !== null);
      report.steps[2].availableFields = availableFields;

      // Check for critical fields
      const criticalFields = [
        'revenue', 'net_income', 'cfo', 'capex', 'operating_income',
        'ebit', 'total_debt', 'cash', 'stockholder_equity', 'eps_diluted'
      ];
      const missingCritical = criticalFields.filter(f => !
availableFields.includes(f));
      if (missingCritical.length > 0) {
        report.steps[2].missingCriticalFields = missingCritical;
        report.summary.criticalMissing.push(...missingCritical.map(f =>
`FinancialsHistory.${f}`));
      }
    }

    // Step 4: Check PricesHistory availability
    report.steps.push({ step: 4, name: 'checkPricesHistory', status:
'starting' });
    const prices = await
base44.asServiceRole.entities.PricesHistory.filter({ ticker: symbol }, '-date',
100);
    if (prices.length === 0) {
      report.steps[3].status = 'critical';
      report.steps[3].message = 'No price history found';
      report.summary.criticalMissing.push('PricesHistory is empty');
    } else {
      report.steps[3].status = 'success';
      report.steps[3].totalRecords = prices.length;
      report.steps[3].latestPrice = prices[0]?.close_adj;
      report.steps[3].latestDate = prices[0]?.date;
      report.steps[3].oldestDate = prices[prices.length - 1]?.date;
    }

    // Step 5: Run computeQualityMetrics
    report.steps.push({ step: 5, name: 'computeQualityMetrics', status:
'starting' });
    try {
      const { data: qualityResult } = await
base44.functions.invoke('computeQualityMetrics', { ticker: symbol });
      report.steps[4].status = qualityResult?.ok ? 'success' : 'partial';
      report.steps[4].result = qualityResult;

      if (qualityResult?.computed) {
        const computed = Object.entries(qualityResult.computed).filter(([,v]) =>
isNum(v));
        const nullValues = Object.entries(qualityResult.computed).filter(([,v])
=> !isNum(v));
        report.steps[4].metricsComputed = computed.length;
        report.steps[4].metricsNull = nullValues.length;
        report.steps[4].computedMetrics = computed.map(([k]) => k);
        report.steps[4].nullMetrics = nullValues.map(([k]) => k);
        report.summary.metricsComputed += computed.length;
        report.summary.metricsNull += nullValues.length;
      }

      report.summary.functionsExecuted++;
      report.summary.functionsSucceeded++;
    } catch (error) {
      report.steps[4].status = 'failed';
```

```
      report.steps[4].error = error.message;
      report.summary.functionsExecuted++;
      report.summary.functionsFailed++;
    }

    // Step 6: Run computeCapitalAllocationMetrics
    report.steps.push({ step: 6, name: 'computeCapitalAllocationMetrics',
status: 'starting' });
    try {
      const { data: capAllocResult } = await
base44.functions.invoke('computeCapitalAllocationMetrics', { ticker: symbol });
      report.steps[5].status = capAllocResult?.ok ? 'success' : 'partial';
      report.steps[5].result = capAllocResult;

      if (capAllocResult?.computed) {
        const computed = Object.entries(capAllocResult.computed).filter(([,v])
=> isNum(v));
        const nullValues = Object.entries(capAllocResult.computed).filter(([,v])
=> !isNum(v));
        report.steps[5].metricsComputed = computed.length;
        report.steps[5].metricsNull = nullValues.length;
        report.steps[5].computedMetrics = computed.map(([k]) => k);
        report.steps[5].nullMetrics = nullValues.map(([k]) => k);
        report.summary.metricsComputed += computed.length;
        report.summary.metricsNull += nullValues.length;
      }

      report.summary.functionsExecuted++;
      report.summary.functionsSucceeded++;
    } catch (error) {
      report.steps[5].status = 'failed';
      report.steps[5].error = error.message;
      report.summary.functionsExecuted++;
      report.summary.functionsFailed++;
    }

    // Step 7: Run computeGrowthMetrics
    report.steps.push({ step: 7, name: 'computeGrowthMetrics', status:
'starting' });
    try {
      const { data: growthResult } = await
base44.functions.invoke('computeGrowthMetrics', { ticker: symbol });
      report.steps[6].status = growthResult?.ok ? 'success' : 'partial';
      report.steps[6].result = growthResult;

      if (growthResult?.computed) {
        const computed = Object.entries(growthResult.computed).filter(([,v]) =>
isNum(v));
        const nullValues = Object.entries(growthResult.computed).filter(([,v])
=> !isNum(v));
        report.steps[6].metricsComputed = computed.length;
        report.steps[6].metricsNull = nullValues.length;
        report.steps[6].computedMetrics = computed.map(([k]) => k);
        report.steps[6].nullMetrics = nullValues.map(([k]) => k);
        report.summary.metricsComputed += computed.length;
        report.summary.metricsNull += nullValues.length;
      }

      report.summary.functionsExecuted++;
      report.summary.functionsSucceeded++;
    } catch (error) {
      report.steps[6].status = 'failed';
      report.steps[6].error = error.message;
      report.summary.functionsExecuted++;
```

```
      report.summary.functionsFailed++;
    }

    // Step 8: Run computeRiskMetrics
    report.steps.push({ step: 8, name: 'computeRiskMetrics', status:
'starting' });
    try {
      const { data: riskResult } = await
base44.functions.invoke('computeRiskMetrics', { ticker: symbol });
      report.steps[7].status = riskResult?.ok ? 'success' : 'partial';
      report.steps[7].result = riskResult;

      if (riskResult?.computed) {
        const computed = Object.entries(riskResult.computed).filter(([,v]) =>
isNum(v));
        const nullValues = Object.entries(riskResult.computed).filter(([,v])
=> !isNum(v));
        report.steps[7].metricsComputed = computed.length;
        report.steps[7].metricsNull = nullValues.length;
        report.steps[7].computedMetrics = computed.map(([k]) => k);
        report.steps[7].nullMetrics = nullValues.map(([k]) => k);
        report.summary.metricsComputed += computed.length;
        report.summary.metricsNull += nullValues.length;
      }

      report.summary.functionsExecuted++;
      report.summary.functionsSucceeded++;
    } catch (error) {
      report.steps[7].status = 'failed';
      report.steps[7].error = error.message;
      report.summary.functionsExecuted++;
      report.summary.functionsFailed++;
    }

    // Step 9: Run computeAndSavePriceMetrics
    report.steps.push({ step: 9, name: 'computeAndSavePriceMetrics', status:
'starting' });
    try {
      const { data: priceResult } = await
base44.functions.invoke('computeAndSavePriceMetrics', { ticker: symbol });
      report.steps[8].status = priceResult?.ok ? 'success' : 'partial';
      report.steps[8].result = priceResult;

      if (priceResult?.updated) {
        report.steps[8].metricsComputed = priceResult.updated.length;
        report.steps[8].computedMetrics = priceResult.updated;
        report.summary.metricsComputed += priceResult.updated.length;
      }

      report.summary.functionsExecuted++;
      report.summary.functionsSucceeded++;
    } catch (error) {
      report.steps[8].status = 'failed';
      report.steps[8].error = error.message;
      report.summary.functionsExecuted++;
      report.summary.functionsFailed++;
    }

    // Step 10: Final Metrics check
    report.steps.push({ step: 10, name: 'finalMetricsCheck', status:
'starting' });
    const [finalMetrics] = await
base44.asServiceRole.entities.Metrics.filter({ ticker_symbol: symbol });
    if (finalMetrics) {
```

```
      const qualityFields = ['cfo_to_ni', 'fcf_margin_pct', 'fcf_to_ebit',
'accruals_ratio', 'margin_stdev_5y_pct', 'roic_pct'];
      const capitalFields = ['debt_to_equity', 'netdebt_to_ebitda',
'interest_coverage_x', 'buyback_yield_pct'];
      const growthFields = ['eps_cagr_5y_pct', 'revenue_cagr_5y_pct',
'eps_cagr_3y_pct', 'revenue_cagr_3y_pct'];
      const riskFields = ['net_cash_to_mktcap_pct', 'beta_5y',
'max_drawdown_5y_pct'];
      const priceFields = ['price_current', 'pe_ttm', 'pe_5y_low', 'pe_5y_high',
'pe_5y_median'];

      const checkCategory = (name, fields) => {
        const populated = fields.filter(f => isNum(finalMetrics[f]));
        const missing = fields.filter(f => !isNum(finalMetrics[f]));
        return {
          category: name,
          total: fields.length,
          populated: populated.length,
          missing: missing.length,
          populatedFields: populated,
          missingFields: missing
        };
      };

      report.steps[9].status = 'success';
      report.steps[9].categories = {
        quality: checkCategory('Quality', qualityFields),
        capital: checkCategory('Capital Allocation', capitalFields),
        growth: checkCategory('Growth', growthFields),
        risk: checkCategory('Risk', riskFields),
        price: checkCategory('Price', priceFields)
      };
    } else {
      report.steps[9].status = 'failed';
      report.steps[9].message = 'Metrics row still does not exist after all
computations';
    }

    // Generate recommendations
    report.recommendations = [];
    if (report.summary.criticalMissing.length > 0) {
      report.recommendations.push({
        priority: 'HIGH',
        issue: 'Missing critical data',
        details: report.summary.criticalMissing,
        action: 'Run data fetch functions (runYahooFundamentalsEtl,
runYahooEtlPipeline) to populate FinancialsHistory and PricesHistory'
      });
    }

    if (report.summary.metricsNull > report.summary.metricsComputed) {
      report.recommendations.push({
        priority: 'MEDIUM',
        issue: 'Most metrics are null',
        details: `${report.summary.metricsNull} null vs $
{report.summary.metricsComputed} computed`,
        action: 'Check if guard conditions (division by zero, missing
denominators) are excluding valid data'
      });
    }

    if (report.summary.functionsFailed > 0) {
      report.recommendations.push({
        priority: 'HIGH',
```

```
        issue: 'Some computation functions failed',
        details: `${report.summary.functionsFailed} out of $
{report.summary.functionsExecuted} functions failed`,
        action: 'Check function logs for specific errors'
      });
    }

    return new Response(JSON.stringify(report, null, 2), {
      status: 200,
      headers: { 'Content-Type': 'application/json' }
    });

  } catch (error) {
    return new Response(JSON.stringify({
      error: true,
      message: error.message,
      stack: error.stack
    }), {
      status: 500,
      headers: { 'Content-Type': 'application/json' }
    });
  }
});

// FILE: functions/diagnosticsFundamentals.ts
// ============================================================

import { createClientFromRequest } from 'npm:@base44/sdk@0.5.0';

Deno.serve(async (req) => {
    const base44 = createClientFromRequest(req);
    if (!(await base44.auth.isAuthenticated())) {
        return new Response(JSON.stringify({ error: 'Unauthorized' }), { status:
401 });
    }

    try {
        const { ticker } = await req.json();
        const tickerUpper = ticker.trim().toUpperCase();

        const logs = [];
        const log = (msg) => logs.push(`[${new Date().toISOString().split('T')
[1].slice(0, -1)}] ${msg}`);

        log(`[Diagnostics] Starting fundamentals diagnostics for $
{tickerUpper}...`);

        // --- 1. QUARTERLY FUNDAMENTALS CHECK ---
        log(`[Diagnostics][Stage:quarterly] Querying FinancialsHistory for
quarterly data...`);

        const quarterlies = await base44.entities.FinancialsHistory.filter({
            ticker: tickerUpper,
            freq: 'quarterly'
        });

        if (!quarterlies || quarterlies.length === 0) {
            log(`[Diagnostics][Stage:quarterly] âŒ NO quarterly data found for $
{tickerUpper}`);
        } else {
            const sorted = [...quarterlies].sort((a, b) => new
Date(b.period_end) - new Date(a.period_end));
            const last8 = sorted.slice(0, 8);
```

```
            log(`[Diagnostics][Stage:quarterly] âœ… Found ${quarterlies.length}
quarterly records. Showing latest 8:`);
            log(`[Diagnostics][Stage:quarterly] ` + '-'.repeat(180));
            log(`[Diagnostics][Stage:quarterly] ${'Period'.padEnd(12)} $
{'Revenue'.padEnd(10)} ${'NetInc'.padEnd(10)} ${'OpInc'.padEnd(10)} $
{'EBIT'.padEnd(10)} ${'CFO'.padEnd(10)} ${'CapEx'.padEnd(10)} $
{'SBC'.padEnd(10)} ${'Cash'.padEnd(10)} ${'DebtS'.padEnd(8)} $
{'DebtL'.padEnd(10)} ${'Equity'.padEnd(12)} ${'Assets'.padEnd(12)} $
{'Interest'.padEnd(10)} ${'Shares'.padEnd(12)}`);
            log(`[Diagnostics][Stage:quarterly] ` + '-'.repeat(180));

            const fmt = (v) => {
                if (v == null) return 'null'.padEnd(10);
                if (typeof v === 'number') {
                    if (Math.abs(v) >= 1e9) return `$
{(v/1e9).toFixed(1)}B`.padEnd(10);
                    if (Math.abs(v) >= 1e6) return `$
{(v/1e6).toFixed(0)}M`.padEnd(10);
                    if (Math.abs(v) >= 1e3) return `$
{(v/1e3).toFixed(0)}K`.padEnd(10);
                    return v.toFixed(0).padEnd(10);
                }
                return String(v).slice(0, 8).padEnd(10);
            };

            last8.forEach(q => {
                const row = `${q.period_end.padEnd(12)} ${fmt(q.revenue)} $
{fmt(q.net_income)} ${fmt(q.operating_income)} ${fmt(q.ebit)} ${fmt(q.cfo)} $
{fmt(q.capex)} ${fmt(q.sbc)} ${fmt(q.cash)} ${fmt(q.debt_short)} $
{fmt(q.debt_long)} ${fmt(q.equity)} ${fmt(q.total_assets)} $
{fmt(q.interest_expense)} ${fmt(q.shares_out)}`;
                log(`[Diagnostics][Stage:quarterly] ${row}`);
            });
        }

        // --- 2. TTM SNAPSHOT CHECK ---
        log(`[Diagnostics][Stage:ttm] Querying FinancialsHistory for TTM
snapshot...`);

        const ttmRecords = await base44.entities.FinancialsHistory.filter({
            ticker: tickerUpper,
            freq: 'ttm'
        });

        if (!ttmRecords || ttmRecords.length === 0) {
            log(`[Diagnostics][Stage:ttm] âŒ NO TTM snapshot found for $
{tickerUpper}`);
        } else {
            const latest = ttmRecords.sort((a, b) => new Date(b.as_of_date) -
new Date(a.as_of_date))[0];
            log(`[Diagnostics][Stage:ttm] âœ… Found TTM snapshot from $
{latest.as_of_date}:`);
            log(`[Diagnostics][Stage:ttm] ` + '='.repeat(100));

            const fmtTTM = (v) => v == null ? 'null' : (typeof v === 'number' ?
(Math.abs(v) >= 1e9 ? `${(v/1e9).toFixed(2)}B` : Math.abs(v) >= 1e6 ? `$
{(v/1e6).toFixed(1)}M` : v.toFixed(0)) : String(v));

            const ttmFields = [
                ['revenue_ttm', latest.revenue_ttm],
                ['cfo_ttm', latest.cfo_ttm],
                ['capex_ttm', latest.capex_ttm],
                ['fcf_ttm', latest.fcf_ttm],
                ['sbc_ttm', latest.sbc_ttm],
```

```
                ['ebitda_ttm', latest.ebitda_ttm],
                ['ebit_ttm', latest.ebit_ttm],
                ['cash', latest.cash],
                ['total_debt', latest.total_debt],
                ['equity', latest.equity],
                ['total_assets', latest.total_assets],
                ['shares_out', latest.shares_out],
                ['market_cap', latest.market_cap],
                ['as_of_date', latest.as_of_date]
            ];

            ttmFields.forEach(([field, value]) => {
                log(`[Diagnostics][Stage:ttm] ${field.padEnd(15)}: $
{fmtTTM(value)}`);
            });
        }

        // --- 3. EXISTING METRICS CHECK ---
        log(`[Diagnostics][Stage:metrics] Checking existing Metrics record...`);

        const existingMetrics = await
base44.entities.Metrics.filter({ ticker_symbol: tickerUpper });

        if (!existingMetrics || existingMetrics.length === 0) {
            log(`[Diagnostics][Stage:metrics] âŒ NO Metrics record found for $
{tickerUpper}`);
        } else {
            const metrics = existingMetrics[0];
            log(`[Diagnostics][Stage:metrics] âœ… Found Metrics record. Key
ratios:`);

            const ratioFields = [
                ['fcf_margin_pct', metrics.fcf_margin_pct],
                ['fcf_yield_pct', metrics.fcf_yield_pct],
                ['sbc_to_sales_pct', metrics.sbc_to_sales_pct],
                ['netcash_to_mktcap_pct', metrics.netcash_to_mktcap_pct],
                ['debt_to_equity', metrics.debt_to_equity],
                ['netdebt_to_ebitda', metrics.netdebt_to_ebitda],
                ['interest_coverage_x', metrics.interest_coverage_x],
                ['cfo_to_ni', metrics.cfo_to_ni],
                ['fcf_to_ebit', metrics.fcf_to_ebit],
                ['market_cap', metrics.market_cap],
                ['price_current', metrics.price_current]
            ];

            ratioFields.forEach(([field, value]) => {
                const display = value == null ? 'null' :
(field.includes('pct') ? `${value.toFixed(2)}%` : value.toFixed(2));
                log(`[Diagnostics][Stage:metrics] ${field.padEnd(20)}: $
{display}`);
            });
        }

        // --- 4. RECOMMENDATIONS ---
        log(`[Diagnostics][Stage:recommendations] Analysis & Recommendations:`);

        if (!quarterlies || quarterlies.length < 4) {
            log(`[Diagnostics][Stage:recommendations] ðŸ"¥ CRITICAL: Need to run
Yahoo Fundamentals ETL first - insufficient quarterly data`);
        } else if (!ttmRecords || ttmRecords.length === 0) {
            log(`[Diagnostics][Stage:recommendations] ðŸ"¥ CRITICAL: TTM
snapshot missing - fundamentals ETL didn't build TTM correctly`);
        } else if (!existingMetrics || existingMetrics.length === 0) {
```

```
            log(`[Diagnostics][Stage:recommendations] âš ï¸  WARNING: No Metrics
record - need to run Fundamental Metrics computation`);
        } else {
            const metrics = existingMetrics[0];
            const nullCount = [metrics.fcf_margin_pct, metrics.fcf_yield_pct,
metrics.sbc_to_sales_pct, metrics.netcash_to_mktcap_pct].filter(v => v ==
null).length;
            if (nullCount > 0) {
                log(`[Diagnostics][Stage:recommendations] âš ï¸  WARNING: $
{nullCount}/4 core ratios are null - need to re-run Fundamental Metrics`);
            } else {
                log(`[Diagnostics][Stage:recommendations] âœ… GOOD: Core ratios
computed successfully`);
            }
        }

        log(`[Diagnostics] Complete.`);

        return new Response(JSON.stringify({
            success: true,
            logs,
            summary: {
                quarterly_records: quarterlies?.length || 0,
                ttm_snapshots: ttmRecords?.length || 0,
                metrics_records: existingMetrics?.length || 0
            }
        }), {
            status: 200,
            headers: { 'Content-Type': 'application/json' }
        });

    } catch (error) {
        return new Response(JSON.stringify({
            success: false,
            message: error.message,
            logs: [`FATAL: ${error.message}`]
        }), {
            status: 500,
            headers: { 'Content-Type': 'application/json' }
        });
    }
});

// FILE: functions/diagnosticsPriceCoverage.ts
// ============================================================

import { createClientFromRequest } from 'npm:@base44/sdk@0.5.0';

Deno.serve(async (req) => {
    const base44 = createClientFromRequest(req);

    if (!(await base44.auth.isAuthenticated())) {
        return new Response(JSON.stringify({ error: 'Unauthorized' }), { status:
401 });
    }

    const logs = [];
    const log = (msg) => logs.push(`[${new Date().toISOString().split('T')
[1].slice(0, -1)}] ${msg}`);

    try {
        const { ticker } = await req.json();
        const tickerUpper = ticker.trim().toUpperCase();
```

```
        log(`[PriceCoverage] Starting diagnostics for ${tickerUpper}...`);

        // Fetch all price history for the ticker
        const priceHistory = await
base44.asServiceRole.entities.PricesHistory.filter({
            ticker: tickerUpper
        }, '-date');

        if (!priceHistory || priceHistory.length === 0) {
            return new Response(JSON.stringify({
                success: false,
                ticker: tickerUpper,
                message: 'No price history found',
                recommendation: `Run 5-year price backfill:
syncHistoryPricesYahoo(${tickerUpper})`,
                logs
            }), { status: 200 });
        }

        // Basic coverage stats
        const sortedPrices = priceHistory
            .filter(p => p.date)
            .sort((a, b) => new Date(a.date) - new Date(b.date));

        const oldestDate = sortedPrices[0]?.date;
        const newestDate = sortedPrices[sortedPrices.length - 1]?.date;
        const totalRows = sortedPrices.length;

        log(`[PriceCoverage] Found ${totalRows} price records from ${oldestDate}
to ${newestDate}`);

        // Check coverage for critical months
        const now = new Date();
        const criticalMonths = [12, 24, 36];
        const monthEndResults = {};
        const missingMonths = [];

        for (const monthsAgo of criticalMonths) {
            const targetDate = new Date(now);
            targetDate.setMonth(targetDate.getMonth() - monthsAgo);

            const monthKey = `${targetDate.getFullYear()}-$
{String(targetDate.getMonth() + 1).padStart(2, '0')}`;

            // Find month-end price
            const start = new Date(targetDate.getFullYear(),
targetDate.getMonth(), 1);
            const end = new Date(targetDate.getFullYear(), targetDate.getMonth()
+ 1, 0);

            const inMonth = sortedPrices.filter(row => {
                const rowDate = new Date(row.date);
                return rowDate >= start && rowDate <= end;
            });

            if (inMonth.length === 0) {
                missingMonths.push(monthsAgo);
                monthEndResults[`${monthsAgo}m_ago`] = {
                    found: false,
                    target_month: monthKey,
                    price: null,
                    date: null
                };
            } else {
```

```
                // Get the last trading day of the month
                const monthEnd = inMonth.reduce((latest, current) => {
                    const latestDate = new Date(latest.date);
                    const currentDate = new Date(current.date);
                    return currentDate > latestDate ? current : latest;
                });

                monthEndResults[`${monthsAgo}m_ago`] = {
                    found: true,
                    target_month: monthKey,
                    price: monthEnd.close,
                    price_adj: monthEnd.close_adj,
                    date: monthEnd.date,
                    days_in_month: inMonth.length
                };
            }
        }

        // Overall assessment
        const yearsBack = 4;
        const cutoffDate = new Date(now);
        cutoffDate.setFullYear(cutoffDate.getFullYear() - yearsBack);

        const hasMinimumCoverage = new Date(oldestDate) <= cutoffDate;

        const assessment = {
            ticker: tickerUpper,
            coverage_adequate: hasMinimumCoverage && missingMonths.length === 0,
            total_price_records: totalRows,
            date_range: {
                oldest: oldestDate,
                newest: newestDate,
                years_covered: ((new Date(newestDate) - new Date(oldestDate)) /
(365 * 24 * 60 * 60 * 1000)).toFixed(1)
            },
            critical_months: monthEndResults,
            missing_months: missingMonths,
            recommendation: missingMonths.length > 0
                ? `Run 5-year backfill to fill missing months: $
{missingMonths.join(', ')} months ago`
                : 'Coverage is sufficient for P/E calculations'
        };

        log(`[PriceCoverage] Assessment: ${assessment.coverage_adequate ?
'ADEQUATE' : 'NEEDS_BACKFILL'}`);
        log(`[PriceCoverage] Missing months: ${missingMonths.length ?
missingMonths.join(', ') : 'None'}`);

        return new Response(JSON.stringify({
            success: true,
            ...assessment,
            logs
        }), {
            status: 200,
            headers: { 'Content-Type': 'application/json' }
        });

    } catch (error) {
        log(`[PriceCoverage][FATAL] ${error.message}`);
        return new Response(JSON.stringify({
            success: false,
            error: error.message,
            logs
        }), {
```

```
            status: 500,
            headers: { 'Content-Type': 'application/json' }
        });
    }
});

// FILE: functions/finnhubCandles5y.ts
// ============================================================

import { createClientFromRequest } from 'npm:@base44/sdk@0.5.0';

function corsHeaders() {
  return {
    "Access-Control-Allow-Origin": "*",
    "Access-Control-Allow-Methods": "POST, GET, OPTIONS",
    "Access-Control-Allow-Headers": "Content-Type, Authorization",
    "Access-Control-Allow-Credentials": "true",
    "Content-Type": "application/json",
  };
}

async function getCandlesData(input = {}) {
  try {
    const ticker = String(input?.ticker ?? input?.symbol ??
'').trim().toUpperCase();
    if (!/^[A-Z.\-]{1,10}$/.test(ticker)) {
      return { status: 'invalid_symbol', message: 'Bad ticker format' };
    }

    const token = Deno.env.get("FINNHUB_API_KEY") ?? "";
    const USE_SANDBOX = (Deno.env.get("FINNHUB_USE_SANDBOX") === "true") ||
token.startsWith("sandbox_");
    const BASE = USE_SANDBOX ? "https://sandbox.finnhub.io" :
"https://finnhub.io";

    const now = Math.floor(Date.now() / 1000);
    const windows = [5 * 365 * 24 * 60 * 60, 365 * 24 * 60 * 60, 90 * 24 * 60 *
60];
    let lastPlanError = null;

    for (const windowSeconds of windows) {
      const from = now - windowSeconds;
      const url = new URL(`${BASE}/api/v1/stock/candle`);
      url.searchParams.set("symbol", ticker);
      url.searchParams.set("resolution", "D");
      url.searchParams.set("from", String(from));
      url.searchParams.set("to", String(now));
      url.searchParams.set("token", token);

      try {
        const resp = await fetch(url.toString(), { method: "GET" });
        const http_status = resp.status;
        const text = await resp.text();
        let data = {};
        try {
          data = text ? JSON.parse(text) : {};
        } catch (e) {
          // Hard fail on parse error, as it's unexpected
          return { status: "http_error", message: "Failed to parse Finnhub
response", debug: { windowSeconds, http_status, bodySnippet: text.slice(0,
120) }};
        }
```

```
        if (resp.ok && data.s === 'ok' && Array.isArray(data.t) &&
Array.isArray(data.c)) {
            const daily_prices = data.t.map((ts, i) => ({
              date: new Date(ts * 1000).toISOString().slice(0, 10),
              close_adj: Number(data.c[i]),
            })).filter(r => Number.isFinite(r.close_adj));

            if (daily_prices.length > 0) {
              return { status: "ok", http_status, finnhub_status: "ok", ticker,
count: daily_prices.length, daily_prices, windowSeconds };
            }
        }

        if (data.s === 'no_data') {
          continue; // Try next smaller window
        }

        if (http_status === 403 || http_status === 429) {
          lastPlanError = { windowSeconds, http_status, finnhub_status:
data.s ?? 'unknown' };
          continue; // Try next smaller window, might be plan-restricted
        }

        // For other non-2xx errors, fail fast
        if (!resp.ok) {
           return { status: "http_error", http_status, message: "HTTP error from
Finnhub", debug: { bodySnippet: text.slice(0,120) }};
        }
      } catch (fetchErr) {
        // Network-level error
        return { status: "exception", message: String(fetchErr.message ||
fetchErr), debug: { windowSeconds } };
      }
    }

    // If loop finishes, all windows failed
    if (lastPlanError) {
      return { status: "plan_restricted", http_status:
lastPlanError.http_status, message: "Candles blocked by plan/rate limit", debug:
{ triedWindows: windows.map(w => `${w/86400}d`) } };
    }

    return { status: "no_data", message: "Finnhub returned no data for all tried
windows", debug: { triedWindows: windows.map(w => `${w/86400}d`) } };

  } catch (err) {
    return { status: "exception", message: String(err.message || err) };
  }
}

Deno.serve(async (req) => {
  // Handle preflight to avoid 405
  if (req.method === 'OPTIONS') {
    return new Response(null, { status: 204, headers: corsHeaders() });
  }

  if (req.method !== 'POST' && req.method !== 'GET') {
    return new Response(JSON.stringify({ error: 'Method Not Allowed' }),
{ status: 405, headers: corsHeaders() });
  }

  // Auth (for GET we still require auth cookie/session)
  const base44 = createClientFromRequest(req);
  if (!(await base44.auth.isAuthenticated())) {
```

```
    return new Response(JSON.stringify({ error: 'Unauthorized' }), { status:
401, headers: corsHeaders() });
  }

  let input = {};
  if (req.method === 'POST') {
    try { input = await req.json(); } catch {
      return new Response(JSON.stringify({ error: 'Bad JSON body' }), { status:
400, headers: corsHeaders() });
    }
  } else if (req.method === 'GET') {
    const u = new URL(req.url);
    input = { ticker: (u.searchParams.get('ticker') || '').toUpperCase() };
  }

  const result = await getCandlesData(input);
  return new Response(JSON.stringify(result), { status: 200, headers:
corsHeaders() });
});

// FILE: functions/finnhubFinancialsReported.ts
// ============================================================

import { createClientFromRequest } from 'npm:@base44/sdk@0.5.0';

Deno.serve(async (req) => {
    const base44 = createClientFromRequest(req);

    if (!(await base44.auth.isAuthenticated())) {
        return new Response(JSON.stringify({ error: 'Unauthorized' }), {
            status: 401,
            headers: { 'Content-Type': 'application/json' }
        });
    }

    try {
        const { ticker } = await req.json();
        if (!ticker) {
            return new Response(JSON.stringify({ error: 'Ticker required' }), {
                status: 400,
                headers: { 'Content-Type': 'application/json' }
            });
        }

        // Use the secret FINNHUB_API_KEY
        const FINNHUB_KEY = Deno.env.get("FINNHUB_API_KEY");
        if (!FINNHUB_KEY) {
            throw new Error("FINNHUB_API_KEY environment variable is not set");
        }

        const url = `https://finnhub.io/api/v1/stock/financials-reported?
symbol=${ticker}&token=${FINNHUB_KEY}`;

        console.log(`Fetching financials for ${ticker}`);

        const response = await fetch(url);
        const responseText = await response.text();

        if (!response.ok) {
            console.error(`Finnhub API error: ${response.status} $
{response.statusText}`, responseText);
            throw new Error(`Finnhub API error: ${response.status} $
{response.statusText}`);
        }
```

```
        let data;
        try {
            data = JSON.parse(responseText);
        } catch (e) {
            console.error(`Finnhub returned non-JSON response for ${ticker}:`,
responseText);
            return new Response(JSON.stringify({
                error: 'Invalid JSON response from Finnhub',
                responseText: responseText
            }), {
                status: 502, // Bad Gateway
                headers: { 'Content-Type': 'application/json' }
            });
        }

        console.log(`Finnhub financials response for ${ticker}: $
{data.data?.length || 0} filings`);

        return new Response(JSON.stringify({
            ticker,
            finjson: data
        }), {
            status: 200,
            headers: { 'Content-Type': 'application/json' }
        });

    } catch (error) {
        console.error('Error fetching financials:', error);
        return new Response(JSON.stringify({ error: error.message }), {
            status: 500,
            headers: { 'Content-Type': 'application/json' }
        });
    }
});

// FILE: functions/finnhubHealth.ts
// ==============================================================

import { createClientFromRequest } from 'npm:@base44/sdk@0.5.0';

function corsHeaders() {
  return {
    "Access-Control-Allow-Origin": "*",
    "Access-Control-Allow-Methods": "POST, GET, OPTIONS",
    "Access-Control-Allow-Headers": "Content-Type, Authorization",
    "Access-Control-Allow-Credentials": "true",
    "Content-Type": "application/json",
  };
}

Deno.serve(async (req) => {
  // Handle preflight
  if (req.method === 'OPTIONS') {
    return new Response(null, { status: 204, headers: corsHeaders() });
  }

  // Auth
  const base44 = createClientFromRequest(req);
  if (!(await base44.auth.isAuthenticated())) {
    return new Response(JSON.stringify({ error: 'Unauthorized' }), { status:
401, headers: corsHeaders() });
  }
```

```
  try {
    const FINNHUB_KEY = Deno.env.get("FINNHUB_API_KEY") ||
"d2g4fd1r01qkv5nfnjvgd2g4fd1r01qkv5nfnk00";
    const url = `https://finnhub.io/api/v1/quote?symbol=AAPL&token=$
{FINNHUB_KEY}`;

    const resp = await fetch(url);
    const http_status = resp.status;
    const body = await resp.json();

    if (!resp.ok) {
      return new Response(JSON.stringify({ ok: false, http_status, body }),
{ status: 200, headers: corsHeaders() });
    }

    return new Response(JSON.stringify({ ok: true, http_status, body }),
{ status: 200, headers: corsHeaders() });

  } catch (err) {
    return new Response(JSON.stringify({ ok: false, error: String(err.message ||
err) }), { status: 500, headers: corsHeaders() });
  }
});

// FILE: functions/fixMetricFieldNames.ts
// ==============================================================

import { createClientFromRequest } from 'npm:@base44/sdk@0.5.0';

Deno.serve(async (req) => {
  const base44 = createClientFromRequest(req);

  try {
    if (!(await base44.auth.isAuthenticated())) {
      return new Response(JSON.stringify({ ok: false, error: 'Unauthorized' }),
{ status: 401 });
    }

    const metrics = await base44.asServiceRole.entities.Metrics.list();
    let updated_count = 0;
    const sample_updates = [];

    for (const m of metrics) {
      const upd = {};

      // Fill missing ticker field
      if (!m.ticker && m.ticker_symbol) {
        upd.ticker = m.ticker_symbol;
      }

      // Add back-compat aliases
      if (m.interest_coverage_x != null && m.interest_coverage == null) {
        upd.interest_coverage = m.interest_coverage_x;
      }

      if (m.netdebt_to_ebitda != null && m.net_debt_to_ebitda == null) {
        upd.net_debt_to_ebitda = m.netdebt_to_ebitda;
      }

      if (Object.keys(upd).length) {
        await base44.asServiceRole.entities.Metrics.update(m.id, upd);
        updated_count++;

        if (sample_updates.length < 5) {
```

```
          sample_updates.push({
            ticker: m.ticker_symbol,
            updated_fields: Object.keys(upd)
          });
        }
      }
    }

    return new Response(JSON.stringify({
      ok: true,
      updated_count,
      sample_updates
    }), { status: 200 });

  } catch (error) {
    return new Response(JSON.stringify({
      ok: false,
      error: error.message
    }), { status: 500 });
  }
});

// FILE: functions/onboardOrRefreshTicker.ts
// ============================================================

/**
 * Main entry point for onboarding or refreshing a stock's metrics.
 * This is the primary function to be called by the user or other services.
 *
 * Inputs: { ticker: string }
 * Outputs: A diagnostic object from the core pipeline execution.
 */
import { createClientFromRequest } from 'npm:@base44/sdk@0.5.0';

async function upsertTicker(base44, symbol) {
    const s = symbol.trim().toUpperCase();
    const [existing] = await
base44.asServiceRole.entities.Ticker.filter({ symbol: s });
    if (existing?.id) return existing;

    return await base44.asServiceRole.entities.Ticker.create({
        symbol: s,
        name: s, // Placeholder, can be enriched later
        sector: null,
        exchange: null
    });
}

Deno.serve(async (req) => {
    const base44 = createClientFromRequest(req);
    try {
        if (!(await base44.auth.isAuthenticated())) {
            return new Response(JSON.stringify({ error: 'Unauthorized' }),
{ status: 401 });
        }
        const { ticker } = await req.json();
        if (!ticker || typeof ticker !== 'string') {
            return new Response(JSON.stringify({ error: 'Ticker symbol (string)
is required.' }), { status: 400 });
        }
        const symbol = ticker.trim().toUpperCase();

        // Stage 0: Upsert Ticker
        await upsertTicker(base44, symbol);
```

```
        // Invoke the core pipeline
        const { data, error } = await
base44.functions.invoke('runDeterministicPipeline', { ticker: symbol });

        if (error) throw new Error(error.message || 'Pipeline invocation
failed');

        return new Response(JSON.stringify(data), { status: 200, headers:
{ 'Content-Type': 'application/json' } });

    } catch (e) {
        return new Response(JSON.stringify({ error: e.message, stack:
e.stack }), { status: 500, headers: { 'Content-Type': 'application/json' } });
    }
});

// FILE: functions/peekMetrics.ts
// ============================================================

import { createClientFromRequest } from 'npm:@base44/sdk@0.5.0';

const REQUIRED_KEYS = [
  // Valuation
  'pe_ttm', 'pe_5y_low', 'pe_5y_median', 'pe_5y_high',
  // Quality
  'roic_pct', 'fcf_margin_pct', 'cfo_to_ni', 'fcf_to_ebit', 'accruals_ratio',
'margin_stdev_5y_pct',
  // Capital Allocation
  'buyback_yield_pct', 'debt_to_equity', 'net_debt_to_ebitda',
'interest_coverage_x',
  // Growth
  'eps_cagr_5y_pct', 'rev_cagr_5y_pct', 'eps_cagr_3y_pct', 'rev_cagr_3y_pct',
];

function finiteOrNull(v) {
  if (v === null || v === undefined) return null;
  if (typeof v !== 'number') return null;
  if (!Number.isFinite(v)) return null;
  return v;
}

Deno.serve(async (request) => {
  const base44 = createClientFromRequest(request);

  if (!(await base44.auth.isAuthenticated())) {
    return new Response(JSON.stringify({ error: 'Unauthorized' }), { status:
401, headers: { 'Content-Type': 'application/json' } });
  }

  const { ticker } = await request.json();
  if (!ticker) {
    return new Response(JSON.stringify({ error: 'ticker required' }), { status:
400, headers: { 'Content-Type': 'application/json' } });
  }
  const tickerUpper = ticker.trim().toUpperCase();

  const rows = await
base44.asServiceRole.entities.Metrics.filter({ ticker_symbol: tickerUpper });
  const row = rows[0];

  if (!row) {
    return new Response(JSON.stringify({ ticker: tickerUpper, found: false,
missing_all: REQUIRED_KEYS }), {
```

```
        status: 200,
        headers: { 'content-type': 'application/json; charset=utf-8' },
    });
  }

  const cleaned = {};
  const missing = [];
  const non_finite = [];

  for (const k of REQUIRED_KEYS) {
    const raw = row[k];
    if (raw === undefined || raw === null) {
      missing.push(k);
    } else {
        const val = finiteOrNull(raw);
        if (val === null) {
            non_finite.push(k);
        }
        cleaned[k] = val;
    }
  }

  const payload = {
    ticker: tickerUpper,
    found: true,
    missing_fields: missing,
    non_finite_fields: non_finite,
    sample: cleaned,
  };

  return new Response(JSON.stringify(payload), {
    status: 200,
    headers: { 'content-type': 'application/json; charset=utf-8' },
  });
});

// FILE: functions/proxyAdminCall.ts
// ==============================================================

// Purpose: Accept Authorization: Bearer <token>, verify it, then invoke allowed
backend functions.
// NO cookies, NO `credentials: 'include'`, NO ALLOWED_ORIGINS list.
// CORS is minimal and permissive since we are not sending cookies.

import { createClientFromRequest } from 'npm:@base44/sdk@0.5.0';

const ALLOWED_FUNCTIONS = new Set([
  'onboardOrRefreshTicker',
  'runDeterministicOrchestrator',
  'computePriceMetrics',
  'computeQualityMetrics',
  'computeCapitalAllocationMetrics',
  'computeGrowthMetrics',
  'computeRiskMetrics',
  '__ping', // debug route
]);

function corsHeaders(req) {
  const origin = req.headers.get('Origin') || '*';
  return {
    'Content-Type': 'application/json',
    'Access-Control-Allow-Origin': origin,        // no cookies -> * would be
fine; keep echo for safety
    'Access-Control-Allow-Methods': 'POST, OPTIONS',
```

```
    'Access-Control-Allow-Headers': 'Content-Type, Authorization',
  };
}
const json = (req, body, status = 200) =>
  new Response(JSON.stringify(body), { status, headers: corsHeaders(req) });

// Safely extract bearer
function getBearer(req) {
  const hdr = req.headers.get('authorization') ||
req.headers.get('Authorization') || '';
  const m = hdr.match(/^Bearer\s+(.+)$/i);
  return m ? m[1] : null;
}

Deno.serve(async (req) => {
  if (req.method === 'OPTIONS') return json(req, { ok: true });

  let parsed = {};
  try { parsed = await req.json(); } catch (_) { /* ignore */ }
  const fn  = String(parsed?.fn ?? '');
  const payload = parsed?.payload ?? {};

  if (!ALLOWED_FUNCTIONS.has(fn)) {
    return json(req, { ok: false, error: `Unknown or disallowed function: ${fn}`
}, 404);
  }

  // 1) Require bearer token
  const bearer = getBearer(req);
  if (!bearer) {
    return json(req, { ok: false, error: 'Missing Authorization: Bearer
token' }, 401);
  }

  // 2) Create Base44 client. It will automatically use the Authorization header
for auth checks.
  const base44 = createClientFromRequest(req);

  // isAuthenticated() implicitly reads the header from the request object.
  const authed = await base44.auth.isAuthenticated().catch(() => false);
  if (!authed) {
    return json(req, { ok: false, error: 'Unauthorized (invalid/expired
token)' }, 401);
  }

  // 3) Debug route to prove the flow works
  if (fn === '__ping') {
    return json(req, { ok: true, whoami: 'authenticated', ts: new
Date().toISOString() });
  }

  // 4) Invoke underlying function, forwarding the user's identity via the
Authorization header.
  const { data, error } = await base44.asServiceRole.functions.invoke(fn,
payload);

  if (error) {
    return json(req, { ok: false, error: error.message ?? String(error), data:
data ?? null }, 200);
  }
  return json(req, { ok: true, ...(data || {}) });
});

// FILE: functions/recomputeAllTickers.ts
```

```
// ============================================================

import { createClientFromRequest } from 'npm:@base44/sdk@0.5.0';

Deno.serve(async (req) => {
  const base44 = createClientFromRequest(req);
  try {
    if (!(await base44.auth.isAuthenticated())) {
      return new Response(JSON.stringify({ ok:false, error:'Unauthorized' }),
{ status:401 });
    }
    const tickers = await base44.asServiceRole.entities.Ticker.list();
    const results = [];
    const pageSize = 10;

    for (let i=0; i<tickers.length; i+=pageSize) {
      const batch = tickers.slice(i, i+pageSize);
      for (const t of batch) {
        const symbol = t.symbol;
        if (!symbol) continue;
        try {
          const f = await base44.functions.invoke('computeFundamentalMetrics', {
ticker: symbol });
          const p = await base44.functions.invoke('computeAndSavePriceMetrics',
{ ticker: symbol });
          results.push({ symbol, fundamentals_ok: !!f?.data?.ok,
price_metrics_ok: !!p?.data?.ok });
        } catch (e) {
          results.push({ symbol, fundamentals_ok:false, price_metrics_ok:false,
error:e.message });
        }
      }
      await new Promise(r => setTimeout(r, 100));
    }
    const successful = results.filter(r => r.fundamentals_ok &&
r.price_metrics_ok).length;
    const failed = results.length - successful;
    return new Response(JSON.stringify({ ok:true, successful, failed }),
{ status:200 });
  } catch (e) {
    return new Response(JSON.stringify({ ok:false, stage:'recomputeAll',
error:e.message }), { status:500 });
  }
});

// FILE: functions/runDeterministicOrchestrator.ts
// ============================================================

import { createClientFromRequest } from 'npm:@base44/sdk@0.5.0';

const json = (body, status = 200) =>
  new Response(JSON.stringify(body), {
    status,
    headers: { 'Content-Type': 'application/json' },
  });

Deno.serve(async (req) => {
  const logs = [];
  const log = (m) => logs.push(m);

  try {
    const base44 = createClientFromRequest(req);
    if (!(await base44.auth.isAuthenticated())) {
      return json({ ok: false, error: 'Unauthorized' }, 401);
```

```
    }

    const body = await req.json().catch(() => ({}));
    const ticker = String(body?.ticker ?? '').trim().toUpperCase();
    if (!ticker) return json({ ok: false, error: 'ticker required' }, 400);

    log(`[INIT] runDeterministicOrchestrator for ${ticker}`);

    const call = async (fn) => {
      log(`[CALL] ${fn}`);
      const { data, error } = await base44.functions.invoke(fn, { ticker });
      if (error) {
        log(`[ERROR] ${fn}: ${error.message ?? error}`);
        return { name: fn, ok: false, writes: [], skipped: [], error:
String(error) };
      }
      if (!data?.ok) {
        log(`[WARN] ${fn} returned not ok`);
        return { name: fn, ok: false, writes: data?.writes ?? [], skipped:
data?.skipped ?? [], error: data?.error };
      }
      log(`[OK] ${fn} wrote ${data?.writes?.length ?? 0} fields`);
      return { name: fn, ok: true, writes: data?.writes ?? [], skipped:
data?.skipped ?? [] };
    };

    // Order: price â†’ quality â†’ capital â†’ growth â†’ risk
    const order = [
      'computePriceMetrics',
      'computeQualityMetrics',
      'computeCapitalAllocationMetrics',
      'computeGrowthMetrics',
      'computeRiskMetrics',
    ];

    const stages = [];
    for (const fn of order) stages.push(await call(fn));

    const ok = stages.every(s => s.ok);
    log(`[DONE] Orchestrator ${ok ? 'OK' : 'PARTIAL/ERROR'}`);

    return json({ ok, stages, logs });
  } catch (e) {
    logs.push(`[FATAL] ${e?.message ?? e}`);
    return json({ ok: false, error: String(e?.message ?? e), logs }, 500);
  }
});

// FILE: functions/runDeterministicPipeline.ts
// ============================================================

/**
 * Core pipeline for computing all deterministic financial metrics.
 * This function fetches raw data, builds aggregates, computes all metrics, and
persists them.
 *
 * It is a self-contained function with all necessary helpers and computation
modules.
 *
 * Inputs: { ticker: string }
 * Outputs: A diagnostic object with all computed values and status.
 * Side-effect: Persists all valid, computed metrics to the Metrics entity.
 */
import { createClientFromRequest } from 'npm:@base44/sdk@0.5.0';
```

```
// --- Start of inlined library: metricComputations.js ---

// --- Helpers ---
const isNum = (v) => v !== null && v !== undefined && Number.isFinite(v);
const safeDiv = (a, b) => (isNum(a) && isNum(b) && Math.abs(b) > 1e-9) ? a / b :
null;
const stdev = (arr) => {
    if (!arr || arr.length < 2) return null;
    const numericArr = arr.filter(isNum);
    const n = numericArr.length;
    if (n < 2) return null;
    const mean = numericArr.reduce((a, b) => a + b) / n;
    const variance = numericArr.map(x => Math.pow(x - mean, 2)).reduce((a, b) =>
a + b) / (n - 1);
    return Math.sqrt(variance);
};
const cagr = (series, years) => {
    if (!series || series.length < 2) return null;
    const end = series[series.length - 1];
    const endDate = new Date(end.date);
    const startCutoff = new Date(endDate);
    startCutoff.setFullYear(startCutoff.getFullYear() - years);

    const start = series.reduce((acc, s) => (new Date(s.date) <= startCutoff ? s
: acc), null);
    if (!start || !isNum(start.value) || start.value <= 0) return null;

    const actualYears = (endDate.getTime() - new Date(start.date).getTime()) /
(365.25 * 24 * 3600 * 1000);
    if (actualYears < years - 0.5) return null;

    return (Math.pow(end.value / start.value, 1 / actualYears) - 1) * 100;
};
const yoy = (arr) => {
    const changes = [];
    if (!arr) return changes;
    for (let i = 1; i < arr.length; i++) {
        const change = safeDiv(arr[i], arr[i - 1]) - 1;
        if (isNum(change)) changes.push(change);
    }
    return changes;
};

// --- Computation Modules ---
function computeQuality(data) {
    const FCF = isNum(data.cfoTTM) && isNum(data.capexTTM) ? data.cfoTTM -
data.capexTTM : null;
    return {
        fcf_margin_pct: isNum(FCF) ? 100 * safeDiv(FCF, data.revenueTTM) : null,
        cfo_to_ni: safeDiv(data.cfoTTM, data.netIncomeTTM),
        fcf_to_ebit: isNum(FCF) ? safeDiv(FCF, data.ebitTTM) : null,
        accruals_ratio: safeDiv(data.netIncomeTTM - data.cfoTTM,
data.totalAssets),
        margin_stdev_5y_pct: data.operatingMarginSeries?.length > 1 ?
stdev(data.operatingMarginSeries) : null,
    };
}

function computeCapitalAllocation(data) {
    // Primary: Share count reduction
    let buybackYield = null;
    if (data.dilutedSharesSeries?.length >= 2) {
        const t = data.dilutedSharesSeries[data.dilutedSharesSeries.length - 1];
```

```
        const t_minus_1 =
data.dilutedSharesSeries[data.dilutedSharesSeries.length - 2];
        if (isNum(t) && isNum(t_minus_1)) {
            buybackYield = 100 * safeDiv(t_minus_1 - t, t_minus_1);
        }
    }
    // Fallback: Cash used for buybacks
    if (!isNum(buybackYield) && isNum(data.purchaseOfStockTTM) &&
isNum(data.marketCap)) {
        buybackYield = 100 * safeDiv(-data.purchaseOfStockTTM, data.marketCap);
    }

    const interestCoverage = safeDiv(data.ebitTTM, data.interestExpenseTTM);

    return {
        debt_to_equity: safeDiv(data.totalDebt, data.equity),
        netdebt_to_ebitda: safeDiv((data.totalDebt - data.cash),
data.ebitdaTTM),
        interest_coverage_x: isNum(interestCoverage) ?
Math.min(interestCoverage, 99) : (isNum(data.ebitTTM) && data.ebitTTM > 0 &&
Math.abs(data.interestExpenseTTM || 0) < 1e-6 ? 99 : null),
        buyback_yield_pct: buybackYield,
    };
}

function computeGrowth(data) {
    const epsSeries = data.epsSeries?.map(p => ({ date: p.date, value:
p.value }));
    const revenueSeries = data.revenueSeries?.map(p => ({ date: p.date, value:
p.value }));
    return {
        eps_cagr_5y_pct: cagr(epsSeries, 5),
        eps_cagr_3y_pct: cagr(epsSeries, 3),
        revenue_cagr_5y_pct: cagr(revenueSeries, 5),
        revenue_cagr_3y_pct: cagr(revenueSeries, 3),
    };
}

function computeRisk(data) {
    // Beta Calculation
    const stockReturns = [];
    const spyReturns = [];
    if (data.weeklyPrices?.length > 104 && data.spyWeeklyPrices?.length > 104) {
        const returnsMap = new Map(data.spyWeeklyPrices.map(p =>
[p.date.slice(0, 10), Math.log(p.close_adj)]));
        for (let i = 1; i < data.weeklyPrices.length; i++) {
            const prevStock = data.weeklyPrices[i-1];
            const currStock = data.weeklyPrices[i];
            const prevSpyLog = returnsMap.get(prevStock.date.slice(0,10));
            const currSpyLog = returnsMap.get(currStock.date.slice(0,10));

            if(isNum(prevSpyLog) && isNum(currSpyLog)){
                const stockLogReturn = Math.log(currStock.close_adj) -
Math.log(prevStock.close_adj);
                const spyLogReturn = currSpyLog - prevSpyLog;
                if(isNum(stockLogReturn) && isNum(spyLogReturn)){
                    stockReturns.push(stockLogReturn);
                    spyReturns.push(spyLogReturn);
                }
            }
        }
    }

    let beta = null;
```

```
    if (stockReturns.length > 104) {
        const n = stockReturns.length;
        const meanS = stockReturns.reduce((s, v) => s + v, 0) / n;
        const meanM = spyReturns.reduce((s, v) => s + v, 0) / n;
        const cov = stockReturns.map((s, i) => (s - meanS) * (spyReturns[i] -
meanM)).reduce((s, v) => s + v, 0);
        const varM = spyReturns.map(m => (m - meanM) ** 2).reduce((s, v) => s +
v, 0);
        beta = safeDiv(cov, varM);
    }

    // Max Drawdown
    let maxDrawdown = 0;
    let peak = -Infinity;
    if(data.dailyPrices?.length > 0){
        for(const price of data.dailyPrices){
            if(price.close_adj > peak) peak = price.close_adj;
            const drawdown = safeDiv(peak - price.close_adj, peak);
            if(drawdown > maxDrawdown) maxDrawdown = drawdown;
        }
    }

    return {
        net_cash_to_mktcap_pct: 100 * safeDiv((data.cash - data.totalDebt),
data.marketCap),
        beta_5y: beta,
        max_drawdown_5y_pct: maxDrawdown > 0 ? maxDrawdown * 100 : null,
        cyclicality_pct: data.revenueYoYChanges?.length > 1 ?
stdev(data.revenueYoYChanges) * 100 : null,
    };
}

// --- End of inlined library ---

// --- Core Pipeline Logic ---
const ttmSum = (q, field) => {
    if (!q || q.length < 4) return null;
    return q.slice(0, 4).reduce((sum, item) => sum + (isNum(item[field]) ?
item[field] : 0), 0);
};
const getLatest = (records, field) => records?.[0]?.[field] ?? null;
const weekEndSeries = (days) => {
    const map = new Map();
    for (const d of days) {
        if (!d?.date || !isNum(d.close_adj)) continue;
        const dt = new Date(d.date);
        dt.setDate(dt.getDate() - (dt.getDay() + 2) % 7); // Floor to Friday of
the week
        const weekKey = dt.toISOString().slice(0, 10);
        if (!map.has(weekKey) || new Date(d.date) > new
Date(map.get(weekKey).date)) {
            map.set(weekKey, d);
        }
    }
    return Array.from(map.values()).sort((a,b)=>a.date.localeCompare(b.date));
};


Deno.serve(async (req) => {
    const base44 = createClientFromRequest(req);
    const { ticker } = await req.json();
    const diagnostics = { ticker, inputs: {}, aggregates: {}, outputs: {} };

    try {
```

```
        // --- Stage 1: Fetch Primitives ---
        const fromDate = new Date();
        fromDate.setFullYear(fromDate.getFullYear() - 6); // 6 years for 5y CAGR
buffer
        const [prices, spyPrices, financials] = await Promise.all([
            base44.asServiceRole.entities.PricesHistory.filter({ ticker: ticker,
date: { $gte: fromDate.toISOString().split('T')[0] } }, '-date'),
            base44.asServiceRole.entities.PricesHistory.filter({ ticker: 'SPY',
date: { $gte: fromDate.toISOString().split('T')[0] } }, '-date'),
            base44.asServiceRole.entities.FinancialsHistory.filter({ ticker:
ticker }, '-period_end', 40),
        ]);
        diagnostics.inputs = { prices: prices.length, spyPrices:
spyPrices.length, financials: financials.length };
        if (spyPrices.length < 100) throw new Error("SPY price history is
missing or insufficient, cannot compute Beta.");

        // --- Stage 2: Build Canonical Aggregates ---
        const annual = financials.filter(f => f.freq === 'annual').sort((a, b)
=> a.period_end.localeCompare(b.period_end));
        const quarterly = financials.filter(f => f.freq ===
'quarterly').sort((a, b) => b.period_end.localeCompare(a.period_end));

        const aggregates = {
            cfoTTM: ttmSum(quarterly, 'cfo'),
            capexTTM: ttmSum(quarterly, 'capex'),
            ebitTTM: ttmSum(quarterly, 'operating_income'),
            depreciationTTM: ttmSum(quarterly, 'depreciation'),
            ebitdaTTM: ttmSum(quarterly, 'ebitda'),
            netIncomeTTM: ttmSum(quarterly, 'net_income'),
            revenueTTM: ttmSum(quarterly, 'revenue'),
            interestExpenseTTM: ttmSum(quarterly, 'interest_expense'),
            purchaseOfStockTTM: ttmSum(quarterly, 'purchase_of_stock'),
            cash: getLatest(quarterly, 'cash'),
            totalDebt: getLatest(quarterly, 'total_debt'),
            totalAssets: getLatest(quarterly, 'total_assets'),
            equity: getLatest(quarterly, 'stockholder_equity'),
            sharesOutstanding: getLatest(quarterly, 'shares_outstanding'),
            revenueSeries: annual.map(r => ({ date: r.period_end, value:
r.revenue })),
            epsSeries: annual.map(r => ({ date: r.period_end, value:
r.eps_diluted ?? safeDiv(r.net_income, r.shares_diluted) })),
            dilutedSharesSeries: annual.map(r => r.shares_diluted),
            operatingMarginSeries: annual.map(r => 100 *
safeDiv(r.operating_income, r.revenue)).filter(isNum),
            revenueYoYChanges: yoy(annual.map(r => r.revenue)),
            priceCurrent: prices[0]?.close_adj,
            marketCap: isNum(prices[0]?.close_adj) && isNum(getLatest(quarterly,
'shares_outstanding')) ? prices[0].close_adj * getLatest(quarterly,
'shares_outstanding') : null,
            dailyPrices: prices,
            weeklyPrices: weekEndSeries(prices),
            spyWeeklyPrices: weekEndSeries(spyPrices),
        };
        if(!isNum(aggregates.ebitdaTTM)) aggregates.ebitdaTTM =
isNum(aggregates.ebitTTM) && isNum(aggregates.depreciationTTM) ?
aggregates.ebitTTM + aggregates.depreciationTTM : null;
        diagnostics.aggregates = { ...aggregates, dailyPrices: `$
{aggregates.dailyPrices.length} rows`, weeklyPrices: `$
{aggregates.weeklyPrices.length} rows`, spyWeeklyPrices: `$
{aggregates.spyWeeklyPrices.length} rows`};

        // --- Stage 3: Compute ALL Derived Metrics ---
        const quality = computeQuality(aggregates);
```

```
        const capital = computeCapitalAllocation(aggregates);
        const growth = computeGrowth(aggregates);
        const risk = computeRisk(aggregates);
        diagnostics.outputs = { quality, capital, growth, risk };

        // --- Stage 4: Persist to Metrics ---
        const patch = {
            ...quality, ...capital, ...growth, ...risk,
            asOf: new Date().toISOString(),
            as_of_date: new Date().toISOString().split('T')[0],
            ticker_symbol: ticker,
            ticker: ticker,
        };
        const safePatch = Object.fromEntries(Object.entries(patch).filter(([,
v]) => v !== undefined && v !== null && (typeof v !== 'number' ||
Number.isFinite(v))));

        const [existingMetrics] = await
base44.asServiceRole.entities.Metrics.filter({ ticker_symbol: ticker });
        if (existingMetrics?.id) {
            await
base44.asServiceRole.entities.Metrics.update(existingMetrics.id, safePatch);
        } else {
            await base44.asServiceRole.entities.Metrics.create(safePatch);
        }
        diagnostics.persisted_fields = Object.keys(safePatch);

        return new Response(JSON.stringify(diagnostics), { status: 200, headers:
{ 'Content-Type': 'application/json' } });

    } catch (e) {
        diagnostics.error = e.message;
        diagnostics.stack = e.stack;
        return new Response(JSON.stringify(diagnostics), { status: 500, headers:
{ 'Content-Type': 'application/json' } });
    }
});

// FILE: functions/runFinnhubFundamentalsEtl.ts
// ==============================================================


/**
 * Finnhub Fundamentals ETL â€” ONE DROP-IN FILE
 * Fixes:
 * 1) Typo: PATTERN_SRCS -> PATTERNS (and all references)
 * 2) EBIT pattern reference now PATTERNS.EBIT
 * 3) PE log header clarifies NI/AvgShares
 * 4) Quarter sort is now correct even if Finnhub returns quarter as 1/2/3/4
 * 5) EPS per quarter derived from YTD diffs with proper quarter order (no
spurious negatives)
 * 6) Normalize diluted shares to raw units (auto-detect "in millions")
 * 7) Always overwrite fast-moving metrics for the Projection Tool (eps_ttm,
pe_ttm, pe_12m/24m/36m, current_pe)
 * 8) Ensure Projection Tool field names are written (`pe_12m`, `pe_24m`,
`pe_36m`, `current_pe`)
 * 9) Use split-adjusted close (not dividend-adjusted) for P/E calculations
 */

import { createClientFromRequest } from 'npm:@base44/sdk@0.5.0';

const FALLBACK_KEY = "d2g4fd1r01qkv5nfnjvgd2g4fd1r01qkv5nfnk00";
const FINNHUB_KEY = Deno.env.get("FINNHUB_API_KEY") || FALLBACK_KEY;
const BASE_URL = "https://finnhub.io/api/v1";
```

```
// ---------- Rate limit + retry ----------
let lastRequestTime = 0;
const REQUEST_INTERVAL_MS = 1100;
async function gatedFetch(url, log, maxRetries = 3) {
  for (let attempt = 1; attempt <= maxRetries; attempt++) {
    const now = Date.now();
    const lag = now - lastRequestTime;
    if (lag < REQUEST_INTERVAL_MS) await new Promise(r => setTimeout(r,
REQUEST_INTERVAL_MS - lag));
    lastRequestTime = Date.now();

    try {
      const res = await fetch(url);
      const remaining = res.headers.get('x-ratelimit-remaining');
      const reset = res.headers.get('x-ratelimit-reset');
      if (remaining) log(`[Finnhub][RateLimit] Remaining: ${remaining}, Reset: $
{reset ? new Date(parseInt(reset) * 1000).toLocaleTimeString() : 'N/A'}`);

      if (res.status === 429) {
        const backoff = Math.pow(2, attempt) * 1000 + Math.random() * 1000;
        log(`[Finnhub][429] backing off ${backoff.toFixed(0)}ms (attempt $
{attempt}/${maxRetries})`);
        if (attempt < maxRetries) { await new Promise(r => setTimeout(r,
backoff)); continue; }
        throw new Error(`Rate limited after ${maxRetries} attempts`);
      }
      if (!res.ok) throw new Error(`HTTP ${res.status}: ${res.statusText}`);
      return await res.json();
    } catch (e) {
      log(`[Finnhub] Attempt ${attempt}/${maxRetries} failed: ${e.message}`);
      if (attempt === maxRetries) throw e;
    }
  }
}

// ---------- Helpers ----------
const num = v => (typeof v === 'number' && Number.isFinite(v)) ? v
  : (typeof v === 'string' && v.trim() !== '' && !isNaN(+v)) ? +v : null;

function val(arr, patterns) {
  if (!Array.isArray(arr)) return null;
  for (const pattern of patterns) {
    for (const item of arr) {
      if (!item || typeof item !== 'object') continue;
      const concept = String(item.concept || '').toLowerCase();
      const label = String(item.label || '').toLowerCase();
      if (pattern.test(concept) || pattern.test(label)) {
        const n = num(item.value);
        if (n !== null) return n;
      }
    }
  }
  return null;
}

function normalizeShares(v) {
  if (v == null) return null;
  if (v >= 1e8) return v;          // already raw shares
  if (v > 0 && v < 1e6) return v * 1e6; // likely "in millions"
  return v;
}

// Quarter label normalizer: accepts "Q1"/"1"/1 -> "Q1"
```

```
const toQ = q => {
  const s = String(q).toUpperCase().replace('QUARTER', '').trim();
  const n = /^[Q]?([1-4])$/.exec(s)?.[1];
  return n ? `Q${n}` : (['Q1','Q2','Q3','Q4'].includes(s) ? s : 'Q1');
};
const qNum = q => ({Q1:1,Q2:2,Q3:3,Q4:4}[toQ(q)] || 1);

// ---------- Pattern sets ----------
const PATTERNS = {
  CFO: [/us-gaap_netcashprovidedbyusedinoperatingactivities/i,
/operating.*cash.*flow/, /net.*cash.*provided.*operating/],
  CapEx: [
    /paymentstoacquirepropertyplantandequipment/i,
    /paymentstoacquireproductiveassets/i,
    /acquisitionofpropertyplantandequipment/i,
    /(acquire|purchase).*property.*plant.*equipment/i,
    /productiveassets/i,
    /capital.*expend/i
  ],
  SBC: [/us-gaap_sharebasedcompensation/i, /share.*based.*comp/,
/stock.*based.*comp/],
  Depreciation: [/us-gaap_depreciationdepletionandamortization/i,
/deprecia(tion)?(?!.*tax)/, /deprecia.*amor[tz]/],
  InterestExp: [/interest.*expense/, /interest.*paid/],
  Cash: [/cash(?!.*flow)/, /cash.*equivalents/, /cash.*short.*term.*invest/i],
  ShortDebt: [/commercial.*paper/i, /short.*term.*debt/,
/current.*portion.*long.*term.*debt/],
  LongDebt: [/longtermdebtnoncurrent/i, /long.*term.*debt(?!.*current)/],
  TotalDebt: [/total.*debt/, /total.*interest.*bearing.*debt/i],
  Equity: [/total.*share.*holder.*equity/, /totalstockholdersequity/i],
  TotalAssets: [/total.*assets/],
  EBIT: [/us-gaap_earningsbeforeinterestandtaxes/i, /us-
gaap_operatingincomeloss/i, /^ebit$/, /earnings.*before.*interest.*tax/],
  NetIncome: [/us-gaap_netincomeloss/i, /^net.*income/],
  SharesDiluted: [/us-gaap_weightedaveragenumberofdilutedshare/i,
/weighted.*average.*shares.*diluted/],
};

// ---------- Fetch ----------
const fetchQuarterlyFinancials = (ticker, log) =>
  gatedFetch(`${BASE_URL}/stock/financials-reported?symbol=$
{ticker}&freq=quarterly&token=${FINNHUB_KEY}`, log);

const fetchAnnualFinancials = async (ticker, log) => {
  try { return await gatedFetch(`${BASE_URL}/stock/financials-reported?symbol=$
{ticker}&freq=annual&token=${FINNHUB_KEY}`, log); }
  catch (e) { log(`[Finnhub] Annual data fetch failed: ${e.message}`); return
null; }
};

// ---------- Normalize + Quarterize ----------
function normalizeAndQuarterizeData(quarterlyData, annualData, log) {
  const reports = [];
  if (quarterlyData?.data) {
    for (const r of quarterlyData.data) {
      if (!r.report) continue;
      const fy = r.year || new Date(r.endDate || r.acceptedDate ||
'').getFullYear();
      const quarter = toQ(r.quarter || r.fiscalQuarter || r.report?.fp ||
r?.period || 'Q1');
      const endDate = r.endDate || r.acceptedDate || r.period;
      if (!endDate) continue;
      const ic = r.report.ic || [], bs = r.report.bs || [], cf = r.report.cf ||
[];
```

```
      reports.push({ type: 'quarterly', fiscalYear: fy, quarter, endDate,
periodEnd: new Date(endDate).toISOString().slice(0,10), ic, bs, cf, report:
r.report });
    }
  }

  const annual = {};
  if (annualData?.data) {
    for (const r of annualData.data) {
      if (!r.report) continue;
      const fy = r.year || new Date(r.endDate || r.acceptedDate ||
'').getFullYear();
      annual[fy] = { ic: r.report.ic || [], cf: r.report.cf || [] };
    }
  }

  // Sort by FY desc then quarter desc for preview logs
  reports.sort((a,b) => (b.fiscalYear - a.fiscalYear) || (qNum(b.quarter) -
qNum(a.quarter)));
  if (reports.length) {
    const f = reports[0];
    log(`[Finnhub][IC] concepts=${f.ic.map(x=>x.concept).join(', ')}`);
    log(`[Finnhub][CF] concepts=${f.cf.map(x=>x.concept).join(', ')}`);
    log(`[Finnhub][BS] concepts=${f.bs.map(x=>x.concept).join(', ')}`);
  }

  const quarterized = [];
  const years = [...new Set(reports.map(r => r.fiscalYear))];

  for (const fy of years) {
    // For differencing, process in true chronological order Q1â†'Q4
    const yr = reports.filter(r => r.fiscalYear === fy).sort((a,b) =>
qNum(a.quarter) - qNum(b.quarter));
    if (!yr.length) continue;

    log(`[Quarterize] Processing FY ${fy} with ${yr.length} quarters`);
    const ytdFlows = {};
    const annualFlows = annual[fy] || { ic:[], cf:[] };
    const out = [];

    for (const r of yr) {
      const sharesRaw = val(r.ic, PATTERNS.SharesDiluted);
      const shares_diluted = normalizeShares(sharesRaw);

      const qd = {
        periodEnd: r.periodEnd, fiscalYear: fy, quarter: toQ(r.quarter),
        cfo:null, capex:null, sbc:null, depreciation:null, interest_exp:null,
        ebit:null, net_income:null, diluted_eps:null, shares_diluted,
        cash: val(r.bs, PATTERNS.Cash),
        short_debt: val(r.bs, PATTERNS.ShortDebt),
        long_debt: val(r.bs, PATTERNS.LongDebt),
        total_debt: val(r.bs, PATTERNS.TotalDebt),
        equity: val(r.bs, PATTERNS.Equity),
        total_assets: val(r.bs, PATTERNS.TotalAssets),
      };

      const flowDefs = [
        { key:'cfo', patterns:PATTERNS.CFO, sections:[r.cf] },
        { key:'capex', patterns:PATTERNS.CapEx, sections:[r.cf] },
        { key:'sbc', patterns:PATTERNS.SBC, sections:[r.cf] },
        { key:'depreciation', patterns:PATTERNS.Depreciation, sections:[r.cf] },
        { key:'interest_exp', patterns:PATTERNS.InterestExp, sections:[r.ic] },
        { key:'ebit', patterns:PATTERNS.EBIT, sections:[r.ic] },
        { key:'net_income', patterns:PATTERNS.NetIncome, sections:[r.ic] },
```

```
      ];

      for (const d of flowDefs) {
        let ytd = null;
        for (const sec of d.sections) { ytd = val(sec, d.patterns); if (ytd !=
null) break; }
        if (ytd != null) {
          const prev = ytdFlows[d.key] ?? 0;
          log(`[Quarterize][Probe] FY ${fy} ${qNum(r.quarter)} ${d.key}_YTD=$
{ytd} prev_YTD=${prev}`);
          qd[d.key] = (qNum(r.quarter) === 1) ? ytd : (ytd - prev);
          ytdFlows[d.key] = ytd;
        }
      }

      if (qd.net_income != null && shares_diluted && shares_diluted > 0) {
        qd.diluted_eps = qd.net_income / shares_diluted;
        log(`[EPS-Qtr][Calc] FY ${fy} ${toQ(r.quarter)} eps=${qd.diluted_eps}`);
      }
      out.push(qd);
    }

    // Synthesize Q4 if we have only 3 quarters and an annual
    if (out.length === 3 && (annualFlows.ic?.length || annualFlows.cf?.length))
{
      const annualVal = (p) => val([...(annualFlows.ic||[]), ...
(annualFlows.cf||[])], p);
      const y3 = { ...out.find(x => x.quarter === 'Q3') };
      if (y3) {
        const sQ4 = { ...y3, quarter:'Q4' };
        const defs = [
          ['cfo', PATTERNS.CFO],
          ['capex', PATTERNS.CapEx],
          ['sbc', PATTERNS.SBC],
          ['depreciation', PATTERNS.Depreciation],
          ['interest_exp', PATTERNS.InterestExp],
          ['ebit', PATTERNS.EBIT],
          ['net_income', PATTERNS.NetIncome],
        ];
        for (const [k, pats] of defs) {
          const a = annualVal(pats);
          const y = ytdFlows[k];
          sQ4[k] = (Number.isFinite(a) && Number.isFinite(y)) ? (a - y) : null;
          log(`[Q4] fy=${fy} ${k}: annual=${a} ytd_q3=${y} q4=${sQ4[k]}`);
        }
        if (sQ4.net_income != null && sQ4.shares_diluted && sQ4.shares_diluted >
0) {
          sQ4.diluted_eps = sQ4.net_income / sQ4.shares_diluted;
          log(`[EPS-Qtr][Calc] FY ${fy} Q4 eps=${sQ4.diluted_eps}`);
        } else sQ4.diluted_eps = null;

        out.push(sQ4);
        out.sort((a,b)=> qNum(a.quarter) - qNum(b.quarter));
      }
    }

    quarterized.push(...out);
    log(`[Quarterize] Flows differenced for FY ${fy}`);
  }

  quarterized.sort((a,b) => new Date(b.periodEnd).getTime() - new
Date(a.periodEnd).getTime());
  log(`[Finnhub] Quarterized ${quarterized.length} reports`);
  return quarterized;
```

```
}

// ---------- TTM ----------
function buildTTMMetrics(quarters, log) {
  const qAsc = quarters.slice().sort((a,b)=> new Date(a.periodEnd).getTime()-new
Date(b.periodEnd).getTime());
  const last4 = qAsc.slice(-4);
  if (last4.length < 4) { log('[TTM] Not enough quarters for TTM calculation.');
return {}; }

  log(`[TTM][Quarters] ${last4.map(q=>q.periodEnd).join(', ')}`);

  const sum = (f) => last4.some(q=>q[f]==null) ? null :
last4.reduce((s,q)=>s+q[f],0);
  const avg = (f) => last4.some(q=>q[f]==null) ? null :
last4.reduce((s,q)=>s+q[f],0)/4;
  const latest = last4[3];

  const ebit_ttm = sum('ebit');
  const depreciation_ttm = sum('depreciation');
  const ebitda_ttm = (ebit_ttm!=null && depreciation_ttm!=null) ? ebit_ttm +
depreciation_ttm : null;

  const net_income_ttm = sum('net_income');
  const cfo_ttm = sum('cfo');
  const capex_ttm = sum('capex');
  const sbc_ttm = sum('sbc');
  const interest_expense_ttm = sum('interest_exp');

  let eps_ttm = null;
  const shAvg = avg('shares_diluted');
  if (net_income_ttm != null && shAvg && shAvg > 0) eps_ttm = net_income_ttm /
shAvg;

  const total_debt_calc = (latest.short_debt || 0) + (latest.long_debt || 0);

  const out = {
    cfo_ttm, capex_ttm, sbc_ttm, depreciation_ttm, ebit_ttm, ebitda_ttm,
    net_income_ttm, interest_expense_ttm, eps_ttm,
    cash: latest.cash,
    total_debt: latest.total_debt || (total_debt_calc > 0 ? total_debt_calc :
null),
    equity: latest.equity,
    total_assets: latest.total_assets,
  };
  log(`[TTM] ni=${net_income_ttm} cfo=${cfo_ttm} capex=${capex_ttm} eps_ttm=$
{eps_ttm?.toFixed(4)}`);
  return out;
}

// ---------- PE history (DB prices + NI/AvgShares) ----------
async function calculateHistoricalPE_DB(base44, ticker, quarters, log) {
  log(`[PE-Hist] Calculating with source: prices=DB (Close),
eps_ttm=NI/AvgShares`);

  // Fetch 5 years of price history to ensure robust month-end coverage
  const fiveYearsAgo = new Date();
  fiveYearsAgo.setFullYear(fiveYearsAgo.getFullYear() - 5);

  const priceHistory = await
base44.asServiceRole.entities.PricesHistory.filter({
    ticker,
    date: { $gte: fiveYearsAgo.toISOString().slice(0, 10) }
  }, '-date');
```

```
  if (!priceHistory?.length) {
    log(`[PE-Hist] No price history found for ${ticker}. Consider running 5-year
price backfill.`);
    return { pe_ttm_12m_ago: null, pe_ttm_24m_ago: null, pe_ttm_36m_ago: null };
  }

  log(`[PE-Hist] Found ${priceHistory.length} price records for ${ticker}`);

  // Helper function for robust month-end selection
  function pickMonthEnd(priceRows, targetMonth) {
    const start = new Date(targetMonth.getFullYear(), targetMonth.getMonth(),
1);
    const end = new Date(targetMonth.getFullYear(), targetMonth.getMonth() + 1,
0);

    const inMonth = priceRows.filter(row => {
      const rowDate = new Date(row.date);
      return rowDate >= start && rowDate <= end;
    });

    if (inMonth.length === 0) return null;

    // Return the row with the latest date (last trading day of the month)
    return inMonth.reduce((latest, current) => {
      const latestDate = new Date(latest.date);
      const currentDate = new Date(current.date);
      return currentDate > latestDate ? current : latest;
    });
  }

  // Validate price coverage
  function validatePriceCoverage(priceRows, minMonthsToCheck) {
    const coverage = {
      hasMinimumCoverage: true,
      oldestDate: null,
      newestDate: null,
      totalRows: priceRows.length,
      missingMonths: [],
      recommendation: ''
    };

    if (priceRows.length === 0) {
      coverage.hasMinimumCoverage = false;
      coverage.recommendation = "No price history found. Run 5-year price
backfill.";
      return coverage;
    }

    coverage.newestDate = priceRows[0]?.date;
    coverage.oldestDate = priceRows[priceRows.length - 1]?.date;

    // Check for critical 12, 24, 36 month points
    const targetMonthsAgo = [12, 24, 36];
    for (const monthsAgo of targetMonthsAgo) {
      const targetDate = new Date();
      targetDate.setMonth(targetDate.getMonth() - monthsAgo);
      const monthEndPrice = pickMonthEnd(priceRows, targetDate);
      if (!monthEndPrice) {
        coverage.missingMonths.push(monthsAgo);
        coverage.hasMinimumCoverage = false;
      }
    }
```

```
    if (!coverage.hasMinimumCoverage) {
      coverage.recommendation = "Missing price data for key historical periods.
Consider running 5-year price backfill to ensure full coverage.";
    } else {
      coverage.recommendation = "Price history appears sufficient for P/E
calculations.";
    }

    return coverage;
  }

  const coverage = validatePriceCoverage(priceHistory, 4);
  log(`[PE-Hist] Price coverage: ${coverage.hasMinimumCoverage ? 'SUFFICIENT' :
'INSUFFICIENT'}`);
  log(`[PE-Hist] Date range: ${coverage.oldestDate} to ${coverage.newestDate} ($
{coverage.totalRows} rows)`);

  if (coverage.missingMonths.length > 0) {
    log(`[PE-Hist] Missing critical months: ${coverage.missingMonths.join(', ')}
months ago`);
    log(`[PE-Hist] Recommendation: ${coverage.recommendation}`);
  }

  const prices = priceHistory.sort((a,b)=> new Date(b.date).getTime()-new
Date(a.date).getTime());
  const qAsc = quarters.slice().sort((a,b)=> new Date(a.periodEnd).getTime()-new
Date(b.periodEnd).getTime());

  // Build monthly EPS TTM series
  const monthlyEpsTtm = {};
  for (let i=0;i<qAsc.length;i++){
    const w = qAsc.slice(Math.max(0,i-3), i+1);
    if (w.length===4){
      const ni = w.map(q=>q.net_income).filter(v=>v!=null);
      const sh = w.map(q=>q.shares_diluted).filter(v=>v!=null);
      if (ni.length===4 && sh.length===4){
        const eps = ni.reduce((s,v)=>s+v,0) / (sh.reduce((s,v)=>s+v,0)/4);
        if (eps!=null && eps>0){
          const d=new Date(w[3].periodEnd); const
key=d.toISOString().slice(0,7);
          monthlyEpsTtm[key]=eps;
          log(`[PE-Hist] EPS TTM for ${key}: $${eps.toFixed(4)}`);
        }
      }
    }
  }

  const MIN_EPS = 0.01;

  const peFrom = (price, epsTTM) => {
    if (!price || !epsTTM || epsTTM < MIN_EPS) return null;
    return price / epsTTM;
  };

  // Robust month-end selection function
  function peAt(monthsAgo) {
    const targetDate = new Date();
    targetDate.setMonth(targetDate.getMonth() - monthsAgo);

    const monthKey = `${targetDate.getFullYear()}-${String(targetDate.getMonth()
+ 1).padStart(2, '0')}`;

    // Find the closest EPS TTM for this period
    const keys = Object.keys(monthlyEpsTtm).sort().reverse();
```

```
    const epsKey = keys.find(k => k <= monthKey);

    if (!epsKey) {
      log(`[PE-Hist] No EPS TTM found for ${monthsAgo}m ago (${monthKey})`);
      return null;
    }

    const eps = monthlyEpsTtm[epsKey];

    // Find month-end price using robust selection
    const monthEndPrice = pickMonthEnd(prices, targetDate);

    if (!monthEndPrice) {
      log(`[PE-Hist] No month-end price found for ${monthsAgo}m ago ($
{monthKey})`);
      return null;
    }

    // Use split-adjusted close (not dividend-adjusted adj_close) for P/E
calculations
    const pe = peFrom(monthEndPrice.close, eps);

    log(`[PE-Hist] ${monthsAgo}m ago: price=$${monthEndPrice.close?.toFixed(2)}
(${monthEndPrice.date}), eps=$${eps.toFixed(4)}, pe=${pe?.toFixed(2)}x`);

    return pe;
  }

  const pe_12m = peAt(12);
  const pe_24m = peAt(24);
  const pe_36m = peAt(36);

  log(`[PE-Hist] Historical P/E summary: PE_12m=${pe_12m?.toFixed(2)}, PE_24m=$
{pe_24m?.toFixed(2)}, PE_36m=${pe_36m?.toFixed(2)}`);

  // Validate results
  const validCount = [pe_12m, pe_24m, pe_36m].filter(pe => pe !== null).length;
  log(`[PE-Hist] ${validCount}/3 historical P/E values successfully
calculated`);

  return {
    pe_ttm_12m_ago: pe_12m,
    pe_ttm_24m_ago: pe_24m,
    pe_ttm_36m_ago: pe_36m,
  };
}

// ---------- DB upsert ----------
async function upsertFinnhubMetrics(base44, ticker, metrics, log) {
  const existing = await
base44.asServiceRole.entities.Metrics.filter({ ticker_symbol: ticker });

  // Always-overwrite list for fast-moving / UI-critical fields
  const ALWAYS_UPDATE = new Set([
    'eps_ttm','pe_ttm','pe_12m','pe_24m','pe_36m','current_pe',
    'cfo_ttm','capex_ttm','sbc_ttm','depreciation_ttm','ebit_ttm','ebitda_ttm',
    'net_income_ttm','cash','total_debt','equity','total_assets'
  ]);

  const clean = Object.fromEntries(Object.entries(metrics).filter(([,v]) => v !
== null && v !== undefined));

  if (existing.length > 0) {
    const row = existing[0];
```

```
    const updates = {};
    for (const [k,v] of Object.entries(clean)) {
      if (ALWAYS_UPDATE.has(k) || row[k] === null || row[k] === undefined)
updates[k] = v;
    }
    if (Object.keys(updates).length) {
      const src = row.data_source || '';
      updates.data_source = src.includes('finnhub') ? src : (src ? `${src}
+finnhub` : 'finnhub');
      await base44.asServiceRole.entities.Metrics.update(row.id, updates);
      log(`[DB] Updated ${ticker} with fields: ${Object.keys(updates).join(',
')}`);
    } else {
      log(`[DB] No fields changed for ${ticker}.`);
    }
  } else {
    await base44.asServiceRole.entities.Metrics.create({
      ticker_symbol: ticker,
      as_of_date: new Date().toISOString().slice(0,10),
      data_source: 'finnhub',
      ...clean
    });
    log(`[DB] Created new metrics record for ${ticker}.`);
  }
}

// ---------- Latest price helper (from your PricesHistory table) ----------
async function getLatestPrice(base44, ticker, log) {
  const latest = await
base44.asServiceRole.entities.PricesHistory.filter({ ticker, limit: 1, order:
'date.desc' });
  if (latest?.[0]) {
    log(`[Price] Latest price for ${ticker}: ${latest[0].close_adj} ($
{latest[0].date})`);
    return latest[0].close_adj;
  }
  return null;
}

// ---------- Server ----------
Deno.serve(async (req) => {
  const base44 = createClientFromRequest(req);
  const logs = [];
  const log = (msg) => logs.push(`[${new Date().toISOString().split('T')
[1].slice(0,-1)}] ${msg}`);

  if (!(await base44.auth.isAuthenticated())) {
    return new Response(JSON.stringify({ error:'Unauthorized' }),
{ status:401 });
  }

  try {
    const { ticker } = await req.json();
    if (!ticker) throw new Error('Ticker is required');

    const T = ticker.trim().toUpperCase();
    log(`[START] Running Finnhub ETL for ${T}`);

    const [qData, aData] = await Promise.all([
      fetchQuarterlyFinancials(T, log),
      fetchAnnualFinancials(T, log)
    ]);

    const quarters = normalizeAndQuarterizeData(qData, aData, log);
```

```
    if (quarters.length < 4) {
      log(`[Finnhub][Warn] Not enough Finnhub quarterly data for ${T} (found $
{quarters.length}). Will process what is available.`);
    }

    const ttm = buildTTMMetrics(quarters, log);
    const { net_income_ttm, cfo_ttm, eps_ttm } = ttm;

    if (!Number.isFinite(net_income_ttm) || !Number.isFinite(cfo_ttm)) {
      log('[Finnhub][Warn] Invalid basic TTM metrics (NI, CFO). Dependent
calculations will be skipped.');
    }
    if (!Number.isFinite(eps_ttm) || eps_ttm <= 0) {
      log('[Finnhub][Warn] Invalid EPS TTM. P/E calculations will be skipped.');
    }

    const historicalPE = await calculateHistoricalPE_DB(base44, T, quarters,
log);

    // Compute latest price & current PE
    const priceToday = await getLatestPrice(base44, T, log);
    const pe_ttm = (priceToday && eps_ttm > 0) ? (priceToday / eps_ttm) : null;

    // Map to Projection Tool field names
    const projectionFields = {
      eps_ttm,                        // core for UI
      pe_ttm,                         // current PE (TTM)
      pe_12m: historicalPE.pe_ttm_12m_ago,
      pe_24m: historicalPE.pe_ttm_24m_ago,
      pe_36m: historicalPE.pe_ttm_36m_ago,
      current_pe: pe_ttm,
    };

    const allMetrics = { ...ttm, ...projectionFields };

    const fcf_ttm = (ttm.cfo_ttm != null && ttm.capex_ttm != null) ?
(ttm.cfo_ttm + ttm.capex_ttm) : null;

    const cov = {
      CFO: ttm.cfo_ttm!=null ? 'Y':'N',
      CapEx: ttm.capex_ttm!=null ? 'Y':'N',
      SBC: ttm.sbc_ttm!=null ? 'Y':'N',
      Dep: ttm.depreciation_ttm!=null ? 'Y':'N',
      Cash: ttm.cash!=null ? 'Y':'N',
      ShortDebt: quarters[0]?.short_debt!=null ? 'Y':'N',
      LongDebt: quarters[0]?.long_debt!=null ? 'Y':'N',
      TotalDebt: ttm.total_debt!=null ? 'Y':'N',
      Equity: ttm.equity!=null ? 'Y':'N',
      TotalAssets: ttm.total_assets!=null ? 'Y':'N',
      EBIT: ttm.ebit_ttm!=null ? 'Y':'N',
      NI: ttm.net_income_ttm!=null ? 'Y':'N',
      EPS_TTM: ttm.eps_ttm!=null ? 'Y':'N',
      PE_TTM: pe_ttm!=null ? 'Y':'N',
      PE_12M: projectionFields.pe_12m!=null ? 'Y':'N',
      PE_24M: projectionFields.pe_24m!=null ? 'Y':'N',
      PE_36M: projectionFields.pe_36m!=null ? 'Y':'N',
    };
    log(`[Coverage-Finnhub] ${Object.entries(cov).map(([k,v])=>`${k}=$
{v}`).join(', ')}`);

    await upsertFinnhubMetrics(base44, T, allMetrics, log);

    log(`[Verify] cfo_ttm=${ttm.cfo_ttm}, capex_ttm=${ttm.capex_ttm}, fcf_ttm=$
{fcf_ttm}, cash=${ttm.cash}, total_debt=${ttm.total_debt}, equity=${ttm.equity},
```

```
ebit_ttm=${ttm.ebit_ttm}, depreciation_ttm=${ttm.depreciation_ttm}, ebitda_ttm=$
{ttm.ebitda_ttm}, net_income_ttm=${ttm.net_income_ttm}, eps_ttm=$
{eps_ttm?.toFixed(4)}, pe_ttm=${pe_ttm?.toFixed(2)}, pe_12m=$
{projectionFields.pe_12m?.toFixed(2)}, pe_24m=$
{projectionFields.pe_24m?.toFixed(2)}, pe_36m=$
{projectionFields.pe_36m?.toFixed(2)}`);

    const quartersProcessed = quarters.length;
    const metricsPopulated =
['CFO','CapEx','SBC','Dep','Cash','ShortDebt','LongDebt','TotalDebt','Equity','T
otalAssets','EBIT','NI','EPS_TTM','PE_TTM','PE_12M','PE_24M','PE_36M']
      .map(k => cov[k]).filter(v => v==='Y').length;

    // A) Standardized success response
    return new Response(JSON.stringify({
        ok: true,
        success: true,
        message: `Finnhub ETL processed ${quartersProcessed} quarters for $
{T}.`, // Use T for ticker as it's the uppercase one used in logs
        quarters_processed: quartersProcessed,
        metrics_populated: metricsPopulated,
        summary: {
            quarters_processed: quartersProcessed,
            metrics_populated: metricsPopulated,
            data_source: 'finnhub',
        },
        logs
    }), {
        status: 200,
        headers: { 'Content-Type': 'application/json' }
    });
  } catch (error) {
    log(`[FinnhubETL][FATAL] ${error.message}`);
    console.error(error);
    // A) Standardized error response
    return new Response(JSON.stringify({
        ok: false,
        success: false,
        error: `Finnhub ETL failed: ${error.message}`,
        message: `Could not process Finnhub fundamentals: ${error.message}`,
        logs
    }), {
        status: 500,
        headers: { 'Content-Type': 'application/json' }
    });
  }
});


// FILE: functions/runFullOnboardRefresh.ts
// ============================================================

import { createClientFromRequest } from 'npm:@base44/sdk@0.5.0';

Deno.serve(async (req) => {
    const base44 = createClientFromRequest(req);
    const logs = [];
    const log = (msg) => logs.push(`[${new Date().toISOString().split('T')
[1].slice(0, -1)}] ${msg}`);

    if (!(await base44.auth.isAuthenticated())) {
        return new Response(JSON.stringify({ ok: false, error:
'Unauthorized' }), { status: 401 });
    }
```

```
    try {
        const { ticker } = await req.json();
        if (!ticker) throw new Error('Ticker is required');

        const tickerUpper = ticker.trim().toUpperCase();
        log(`[Orchestrator] Starting full onboard/refresh for ${tickerUpper}`);

        const summary = {
            ticker: tickerUpper,
            prices_count: 0,
            financials_q_count: 0,
            financials_a_count: 0,
            ttm_points: 0,
            metrics_written: 0,
            steps_completed: []
        };

        // Step 1: Yahoo Prices ETL (5Y historical)
        log(`[Step 1] Yahoo Prices ETL...`);
        try {
            const { data: pricesResult, error: pricesError } = await
base44.functions.invoke('runYahooEtlPipeline', { ticker: tickerUpper });
            if (pricesError || !pricesResult?.ok) throw new
Error(pricesError?.message || 'Yahoo prices ETL failed');

            summary.prices_count = pricesResult.prices_saved || 0;
            summary.steps_completed.push('yahoo_prices');
            log(`[Step 1] âœ" Saved ${summary.prices_count} price points`);
        } catch (error) {
            log(`[Step 1] âŒ ${error.message}`);
            throw new Error(`Yahoo prices ETL failed: ${error.message}`);
        }

        // Step 2: Yahoo Fundamentals ETL (quarterly + annual)
        log(`[Step 2] Yahoo Fundamentals ETL...`);
        try {
            const { data: fundResult, error: fundError } = await
base44.functions.invoke('runYahooFundamentalsEtl', {
                ticker: tickerUpper,
                force_backfill: true
            });
            if (fundError || !fundResult?.ok) throw new Error(fundError?.message
|| 'Yahoo fundamentals ETL failed');

            summary.financials_q_count = fundResult.quarterly_count || 0;
            summary.financials_a_count = fundResult.annual_count || 0;
            summary.steps_completed.push('yahoo_fundamentals');
            log(`[Step 2] âœ" Saved ${summary.financials_q_count} quarterly + $
{summary.financials_a_count} annual reports`);
        } catch (error) {
            log(`[Step 2] âŒ ${error.message}`);
            throw new Error(`Yahoo fundamentals ETL failed: ${error.message}`);
        }

        // Step 3: Compute Fundamental Metrics (TTM, ratios, P/E history)
        log(`[Step 3] Computing fundamental metrics...`);
        try {
            const { data: metricsResult, error: metricsError } = await
base44.functions.invoke('computeFundamentalMetrics', { ticker: tickerUpper });
            if (metricsError || !metricsResult?.ok) throw new
Error(metricsError?.message || 'Fundamental metrics computation failed');

            summary.ttm_points = metricsResult.ttm_points || 0;
```

```
            summary.metrics_written = metricsResult.metrics_count || 0;
            summary.steps_completed.push('fundamental_metrics');
            log(`[Step 3] âœ" Computed metrics with ${summary.ttm_points} TTM
EPS points`);
        } catch (error) {
            log(`[Step 3] âŒ ${error.message}`);
            throw new Error(`Fundamental metrics computation failed: $
{error.message}`);
        }

        // Step 4: Compute Price Metrics (current price, P/E TTM)
        log(`[Step 4] Computing price metrics...`);
        try {
            const { data: priceMetricsResult, error: priceMetricsError } = await
base44.functions.invoke('computeAndSavePriceMetrics', { ticker: tickerUpper });
            if (priceMetricsError || !priceMetricsResult?.ok) throw new
Error(priceMetricsError?.message || 'Price metrics computation failed');

            summary.steps_completed.push('price_metrics');
            log(`[Step 4] âœ" Updated price-derived metrics`);
        } catch (error) {
            log(`[Step 4] âŒ ${error.message}`);
            throw new Error(`Price metrics computation failed: $
{error.message}`);
        }

        log(`[Orchestrator] âœ… Full onboard/refresh completed for $
{tickerUpper}`);

        return new Response(JSON.stringify({
            ok: true,
            message: `Full onboard/refresh completed successfully for $
{tickerUpper}`,
            summary,
            logs
        }), {
            status: 200,
            headers: { 'Content-Type': 'application/json' }
        });

    } catch (error) {
        log(`[Orchestrator] âŒ FATAL: ${error.message}`);
        return new Response(JSON.stringify({
            ok: false,
            error: error.message,
            message: `Full onboard/refresh failed: ${error.message}`,
            summary,
            logs
        }), {
            status: 500,
            headers: { 'Content-Type': 'application/json' }
        });
    }
});

// FILE: functions/runFullRefresh.ts
// ============================================================

import { createClientFromRequest } from 'npm:@base44/sdk@0.5.0';

Deno.serve(async (req) => {
    const base44 = createClientFromRequest(req);
    const logs = [];
```

```
    const log = (msg) => logs.push(`[${new Date().toISOString().split('T')
[1].slice(0, -1)}] ${msg}`);

    if (!(await base44.auth.isAuthenticated())) {
        return new Response(JSON.stringify({ ok: false, success: false, error:
'Unauthorized' }), { status: 401 });
    }

    let ticker = '';
    const results = {
        status: "ok", // 'ok' or 'partial'
        errors: [],
        steps: {}
    };

    try {
        const body = await req.json();
        ticker = (body.ticker || '').trim().toUpperCase();
        if (!ticker) throw new Error('Ticker is required for full refresh.');

        const step = async (title, fnName, params = { ticker }) => {
            log(`[Step] START: ${title}`);
            try {
                const { data, error } = await base44.functions.invoke(fnName,
params);

                if (error || (data && (data.ok === false || data.success ===
false))) {
                    const errorMessage = error?.message || data?.error ||
data?.message || `${title} failed.`;
                    throw new Error(errorMessage);
                }

                if (data?.logs) logs.push(...data.logs.map(l => `  [${fnName}] $
{l}`));
                log(`[Step] SUCCESS: ${title}`);
                results.steps[fnName] = { status: 'success', data: data };
            } catch (e) {
                log(`[Step] FAILED: ${title} - ${e.message}`);
                results.status = "partial";
                results.errors.push(`${title} failed: ${e.message}`);
                results.steps[fnName] = { status: 'failed', error: e.message };
            }
        };

        // Strict, but resilient, execution order
        await step('A: Yahoo Historical Prices (5Y)', 'runYahooEtlPipeline');
        await step('B: Finnhub Fundamentals ETL', 'runFinnhubFundamentalsEtl');
        await step('C: Yahoo Fundamentals ETL', 'runYahooFundamentalsEtl');
        await step('D1: Compute Fundamental Metrics',
'computeFundamentalMetrics');
        await step('E: Sync Recent Prices', 'syncRecentPricesYahoo');
        await step('D2: Re-compute Fundamental Metrics',
'computeFundamentalMetrics'); // Re-run with latest price
        await step('F: Compute Price-based Metrics',
'computeAndSavePriceMetrics');

        const finalStatus = results.errors.length > 0 ? "partial" : "ok";
        const finalMessage = finalStatus === "ok"
            ? `Full refresh completed successfully for ${ticker}.`
            : `Full refresh for ${ticker} completed with $
{results.errors.length} error(s).`;

        return new Response(JSON.stringify({
```

```
            ok: true, // The orchestrator itself succeeded
            success: finalStatus === "ok",
            status: finalStatus,
            message: finalMessage,
            errors: results.errors,
            logs
        }), { status: 200, headers: { 'Content-Type': 'application/json' } });

    } catch (error) {
        log(`[FullRefresh][FATAL] Orchestrator failed: ${error.message}`);
        return new Response(JSON.stringify({
            ok: false,
            success: false,
            error: error.message,
            message: `Full refresh for ${ticker} failed: ${error.message}`,
            logs
        }), {
            status: 500,
            headers: { 'Content-Type': 'application/json' }
        });
    }
});

// FILE: functions/runScreenerProbe.ts
// ============================================================

import { createClientFromRequest } from 'npm:@base44/sdk@0.5.0';

const isNum = (v) => typeof v === 'number' && Number.isFinite(v);
const nz = (v) => (isNum(v) ? v : null);

Deno.serve(async (req) => {
  const base44 = createClientFromRequest(req);
  const logs = [];
  const log = (m) => logs.push(`[${new Date().toISOString()}] ${m}`);

  if (!(await base44.auth.isAuthenticated())) {
    return new Response(JSON.stringify({ ok:false, message:'Unauthorized' }),
{ status: 401 });
  }

  try {
    const { ticker } = await req.json();
    const t = String(ticker || '').trim().toUpperCase();
    if (!t) throw new Error('ticker required');

    log(`[Probe] Starting for ${t}`);

    // 1) Metrics (the screener's primary source)
    const metricsList = await
base44.asServiceRole.entities.Metrics.filter({ ticker_symbol: t }, '-
updated_date', 1);
    const metrics = metricsList?.[0] || {};
    log(`[Probe] Metrics source=${metrics.data_source || 'n/a'} as_of=$
{metrics.as_of_date || 'n/a'}`);
    log(`[Probe] Metrics as_of=${metrics?.as_of_date} source=$
{metrics?.data_source}`);
    log(`[Probe] Present: price_current=${metrics?.price_current}, eps_ttm=$
{metrics?.eps_ttm}, pe_ttm=${metrics?.pe_ttm}, ev_ebitda=$
{metrics?.ev_ebitda}`);
    log(`[Probe] Capital inputs: debt=${metrics?.total_debt}, cash=$
{metrics?.cash}, equity=${metrics?.equity}, ebit_ttm=${metrics?.ebit_ttm},
ebitda_ttm=${metrics?.ebitda_ttm}, interest_expense_ttm=$
{metrics?.interest_expense_ttm}`);
```

```
    // 2) PriceDerivedStats (P/E bands etc.) - handle case where entity doesn't
exist
    let statsList = [];
    let stats = {};
    try {
      statsList = await base44.asServiceRole.entities.PriceDerivedStats.filter({
ticker: t }, '-as_of_date', 1);
      stats = statsList?.[0] || {};
    } catch (e) {
      log(`[Probe] PriceDerivedStats entity not found or error: ${e.message}`);
    }
    log(`[Probe] PriceDerivedStats present=${!!stats.id} as_of=$
{stats.as_of_date || 'n/a'}`);

    // 3) Recent quarterlies (for Capital Allocation)
    const q = await
base44.asServiceRole.entities.FinancialsHistory.filter({ ticker: t, freq:
'quarterly' }, '-period_end', 6);
    const q0 = q?.[0] || {};
    const derived = (() => {
      const equity = nz(metrics.equity ?? q0.equity);
      const total_debt = nz(metrics.total_debt ?? ((q0.short_debt||0)+
(q0.long_debt||0)));
      const cash = nz(metrics.cash ?? q0.cash);
      const ebit_ttm = nz(metrics.ebit_ttm);
      const dep_ttm = nz(metrics.depreciation_ttm);
      const ebitda_ttm = nz(metrics.ebitda_ttm ??
(isNum(ebit_ttm)&&isNum(dep_ttm) ? ebit_ttm+dep_ttm : null));
      const ie_ttm = nz(metrics.interest_expense_ttm);

      const debt_to_equity = (isNum(total_debt)&&isNum(equity)&&equity>0) ?
total_debt/equity : null;
      const netdebt_to_ebitda =
(isNum(total_debt)&&isNum(cash)&&isNum(ebitda_ttm)&&ebitda_ttm>0)
        ? (total_debt - cash)/ebitda_ttm : null;
      const interest_coverage_x =
(isNum(ebit_ttm)&&isNum(ie_ttm)&&Math.abs(ie_ttm)>0)
        ? ebit_ttm/Math.abs(ie_ttm) : null;

      return { debt_to_equity, netdebt_to_ebitda, interest_coverage_x };
    })();

    // 4) Identify "screener-critical" missing fields
    const need = [
      'price_current','eps_ttm','pe_ttm','pe_fwd',
      'ev_ebitda','pe_5y_low','pe_5y_high',
      'eps_cagr_5y_pct','revenue_cagr_5y_pct',
      'debt_to_equity','netdebt_to_ebitda','interest_coverage_x',
      'cfo_to_ni','fcf_margin_pct','fcf_to_ebit'
    ];
    const missing = need.filter(k => !isNum(metrics[k] ?? stats[k] ??
derived[k]));

    // log coverage context to explain WHY some are missing
    if (metrics.pe_obs_count != null) {
      log(`[Probe] pe_obs_count=${metrics.pe_obs_count} (â‰¥12 required for
pe_12m, â‰¥24 for pe_24m, â‰¥36 for pe_36m, â‰¥60 for 5Y bands)`);
    }

    // Additional diagnostics for common missing fields
    if (missing.includes('eps_cagr_5y_pct')) {
```

```
      const annuals = await
base44.asServiceRole.entities.FinancialsHistory.filter({ ticker: t, freq:
'annual' }, '-period_end', 6);
      log(`[Probe] EPS CAGR missing: ${annuals.length} annual records available
(need â‰¥6 for 5Y CAGR)`);
    }

    if (missing.includes('pe_5y_low') || missing.includes('pe_5y_high')) {
      log(`[Probe] P/E bands missing: likely insufficient price history coverage
or EPS data`);
    }

    if (missing.includes('interest_coverage_x') && !
isNum(metrics.interest_expense_ttm)) {
      log(`[Probe] Interest coverage missing: no interest_expense_ttm
available`);
    }

    if (missing.length) log(`[Probe] Missing fields: $
{JSON.stringify(missing)}`);

    const out = {
      ok: true,
      message: `Probe complete for ${t}`,
      metrics: {
        ticker_symbol: metrics.ticker_symbol, as_of_date: metrics.as_of_date,
data_source: metrics.data_source,
        price_current: metrics.price_current, eps_ttm: metrics.eps_ttm, pe_ttm:
metrics.pe_ttm, pe_fwd: metrics.pe_fwd,
        ev_ebitda: metrics.ev_ebitda, cfo_to_ni: metrics.cfo_to_ni,
fcf_margin_pct: metrics.fcf_margin_pct,
        fcf_to_ebit: metrics.fcf_to_ebit, debt_to_equity:
metrics.debt_to_equity,
        netdebt_to_ebitda: metrics.netdebt_to_ebitda, interest_coverage_x:
metrics.interest_coverage_x,
        revenue_cagr_5y_pct: metrics.revenue_cagr_5y_pct, eps_cagr_5y_pct:
metrics.eps_cagr_5y_pct,
        pe_obs_count: metrics.pe_obs_count
      },
      priceDerivedStats: {
        as_of_date: stats.as_of_date,
        pe_5y_low: stats.pe_5y_low, pe_5y_high: stats.pe_5y_high,
        pe_12m: stats.pe_12m, pe_24m: stats.pe_24m, pe_36m: stats.pe_36m
      },
      capitalAllocationDerived: derived,
      missing_fields: missing,
      coverage_diagnostics: {
        pe_obs_count: metrics.pe_obs_count,
        annual_records_available: q.length,
        has_interest_expense: isNum(metrics.interest_expense_ttm)
      },
      logs
    };
    return new Response(JSON.stringify(out), { status: 200, headers: { 'Content-
Type': 'application/json' } });
  } catch (e) {
    logs.push(`[FATAL] ${e.message}`);
    return new Response(JSON.stringify({ ok:false, message:e.message, logs }), {
status: 500 });
  }
});

// FILE: functions/runYahooEtlPipeline.ts
// ============================================================
```

```
import { createClientFromRequest } from 'npm:@base44/sdk@0.5.0';

// --- START: SHARED HELPERS ---

// --- Rate Limiting & Pacing ---
const yahooTokenBucket = { tokens: 2, lastRefill: Date.now(), refillRate: 1 /
600, maxTokens: 2 };
const dbTokenBucket = { tokens: 3, lastRefill: Date.now(), refillRate: 3 / 1000,
maxTokens: 3 };

function refillBucket(bucket) {
  const now = Date.now();
  const elapsed = now - bucket.lastRefill;
  const tokensToAdd = Math.floor(elapsed * bucket.refillRate);

  if (tokensToAdd > 0) {
    bucket.tokens = Math.min(bucket.maxTokens, bucket.tokens + tokensToAdd);
    bucket.lastRefill = now;
  }
}

async function waitForToken(bucket, log, logPrefix) {
  while (bucket.tokens < 1) {
    refillBucket(bucket);
    if (bucket.tokens < 1) {
      log(`${logPrefix}[RateLimit] waiting for token, current=$
{bucket.tokens.toFixed(2)}`);
      await new Promise(resolve => setTimeout(resolve, 100));
    }
  }
  bucket.tokens -= 1;
  refillBucket(bucket);
}

// --- Exponential Backoff ---
async function withBackoff(fn, options) {
    const { retries, base, cap, jitter, retryOn, onRetry } = options;
    for (let attempt = 1; attempt <= retries; attempt++) {
        try {
            return await fn();
        } catch (error) {
            const status = error.response?.status || error.status || 0;
            const code = error.code || 'unknown';
            const isRetryable = retryOn.includes(status) || retryOn.some(c =>
code.includes(c));

            if (!isRetryable || attempt === retries) {
                throw error; // Throw original error if not retryable or final
attempt
            }

            const delay = Math.min(cap, base * Math.pow(1.5, attempt - 1));
            const jitteredDelay = jitter ? Math.round(delay * (1 +
(Math.random() - 0.5) * 0.4)) : delay;

            onRetry(attempt, jitteredDelay, { status, code });
            await new Promise(resolve => setTimeout(resolve, jitteredDelay));
        }
    }
}

// --- Session & Fetching ---
```

```
let yahooSession = { cookie: null, crumb: null, acquiredAt: 0, inFlight: null };
const SESSION_TTL_MS = 60 * 60 * 1000;
function sessionIsFresh() { return yahooSession.cookie && yahooSession.crumb &&
(Date.now() - yahooSession.acquiredAt) < SESSION_TTL_MS; }
async function parseJsonOrHtml(response, log) { const text = await
response.text(); if (text.startsWith('<')) { log(`[YahooFetch] Non-JSON (HTML)
status=${response.status}`); return { ok: false, isHtml: true, status:
response.status, text }; } try { const json = JSON.parse(text);
log(`[YahooFetch] json OK`); return { ok: true, isJson: true, json, status:
response.status }; } catch (e) { log(`[YahooFetch] Non-JSON detected (Parse
Error) status=${response.status}`); return { ok: false, parseError: true,
status: response.status, text }; } }
async function fetchCrumbWithBackoff(log) {
    const maxAttempts = 6; const baseDelayMs = 1000;
    for (let attempt = 1; attempt <= maxAttempts; attempt++) {
        try {
            await waitForToken(yahooTokenBucket, log, '[Backfill]');
            const cookieResponse = await fetch('https://fc.yahoo.com/',
{ headers: { 'User-Agent': 'Mozilla/5.0' } });
            if (cookieResponse.status === 429 || cookieResponse.status >= 500)
throw new Error(`Cookie fetch failed: status=${cookieResponse.status}`);
            const cookieHeader = cookieResponse.headers.get('set-cookie'); if (!
cookieHeader) throw new Error("No session cookie");
            await waitForToken(yahooTokenBucket, log, '[Backfill]');
            const crumbResponse = await
fetch('https://query1.finance.yahoo.com/v1/test/getcrumb', { headers:
{ 'cookie': cookieHeader, 'User-Agent': 'Mozilla/5.0' } });
            if (crumbResponse.status === 429 || crumbResponse.status >= 500)
throw new Error(`Crumb fetch failed: status=${crumbResponse.status}`);
            const crumb = await crumbResponse.text(); if (!crumb || crumb.length
< 5) throw new Error("Invalid crumb");
            yahooSession.cookie = cookieHeader; yahooSession.crumb = crumb;
yahooSession.acquiredAt = Date.now();
            log("[Backfill][Stage:session] cookie OK"); log("[Backfill]
[Stage:session] crumb OK");
            return;
        } catch (error) {
            if (attempt === maxAttempts) throw { stage: 'session', code: 429,
message: 'Crumb fetch failed after retries', cause: error.message };
            const delay = Math.min(6000, baseDelayMs * Math.pow(1.5, attempt -
1));
            log(`[Backfill][Retry ${attempt}/${maxAttempts}] stage=session
status=${error.message.match(/status=(\d+)/)?.[1] || 'n/a'} sleep=${delay}ms`);
            await new Promise(resolve => setTimeout(resolve, delay));
        }
    }
}
async function ensureSession(log) { log("[Backfill][Stage:session] getSession
start"); if (sessionIsFresh()) { log("[Backfill][Stage:session] cache hit");
return; } log("[Backfill][Stage:session] cache miss"); if
(yahooSession.inFlight) { log("[Backfill][Stage:session] waiting for in-flight
session"); await yahooSession.inFlight; return; } yahooSession.inFlight =
fetchCrumbWithBackoff(log); try { await yahooSession.inFlight; } finally
{ yahooSession.inFlight = null; } }
async function signedFetch(path, params, log) {
    await ensureSession(log);
    const url = new URL(`https://query1.finance.yahoo.com${path}`);
    const finalParams = { ...params, crumb: yahooSession.crumb };
    Object.entries(finalParams).forEach(([key, value]) =>
url.searchParams.append(key, value));
    await waitForToken(yahooTokenBucket, log, '[Backfill]');
    const response = await fetch(url.toString(), { headers: { 'cookie':
yahooSession.cookie, 'User-Agent': 'Mozilla/5.0' } });
    const bodyText = await response.text();
```

```
    log(`[Backfill][Stage:fetch] <- ${path} status=${response.status} len=$
{bodyText.length}`);
    if (!response.ok) {
      const error = new Error(`Yahoo fetch failed with status $
{response.status}`);
      error.status = response.status;
      throw error;
    }
    const mockResponse = { status: response.status, text: () =>
Promise.resolve(bodyText) };
    return parseJsonOrHtml(mockResponse, log);
}
function buildChartRequest(opts) { const { symbol, interval = '1d', events =
'div%2Csplits', range, period1, period2 } = opts; if (!symbol) throw new
Error('Symbol required'); const urlPath = `/v8/finance/chart/${symbol}`; const
params = { interval, events, includeAdjustedClose: 'true' }; if (period1 &&
period2) { params.period1 = period1.toString(); params.period2 =
period2.toString(); } else if (range) { params.range = range; } else { throw new
Error('Time window required'); } return { urlPath, params }; }

// --- ETL LOGIC ---
const toYMD = (d) => d ? new Date(d).toISOString().slice(0, 10) : null;
const numOrNull = (v) => (v === null || v === undefined || Number.isNaN(+v)) ?
null : +v;

async function fetchYahooData(ticker, log) {
    log("[Backfill] Using YahooSession v1");
    const { urlPath, params } = buildChartRequest({ symbol: ticker, range:
'5y' });
    const chartResult = await signedFetch(urlPath, params, log);
    if (!chartResult.ok || !chartResult.json?.chart?.result?.[0]) {
        throw { stage: 'fetch', code: chartResult.status || 500, message: `Could
not fetch price history for ${ticker}`, cause: chartResult.text?.slice(0, 120)
|| 'No chart data' };
    }
    return { priceData: chartResult.json.chart.result[0] };
}

function normalizePrices(ticker, priceData, log) {
    const timestamps = Array.isArray(priceData?.timestamp) ? priceData.timestamp
: [];
    if (!timestamps.length) return [];
    log(`[Backfill][Stage:parse] processing ${timestamps.length} price
timestamps`);

    const quotes = priceData.indicators.quote[0];
    const adjCloses = priceData.indicators.adjclose[0].adjclose;

    const prices = timestamps.map((ts, i) => ({
        ticker,
        date: toYMD(new Date(ts * 1000)),
        open: numOrNull(quotes?.open?.[i]),
        high: numOrNull(quotes?.high?.[i]),
        low: numOrNull(quotes?.low?.[i]),
        close: numOrNull(quotes?.close?.[i]),       // Split-adjusted only
(Yahoo "Close")
        close_adj: numOrNull(adjCloses?.[i]),       // Dividend & split-adjusted
(Yahoo "Adj Close")
        volume: numOrNull(quotes?.volume?.[i]),
        source: 'yahoo',
        as_of_date: toYMD(new Date())
    })).filter(p => p.date && p.close_adj !== null && p.close !== null); //
Require both prices
```

```
    log(`[Backfill][Stage:parse] normalized ${prices.length} valid price
points`);
    return prices;
}

async function upsertPrices(base44, prices, log) {
    const safePrices = Array.isArray(prices) ? prices : [];
    if (safePrices.length === 0) {
        log("[Backfill][Stage:db] No price data to upsert.");
        return { inserted: 0, updated: 0, skipped: 0 };
    }

    // AUTH DIAGNOSTICS for service role access
    log("[Backfill][Stage:db] Initializing service role database
operations...");

    const serviceClient = base44.asServiceRole; // Use the service role client
for elevated permissions

    try {
        // Test service role connectivity
        const testQuery = await serviceClient.entities.PricesHistory.filter({},
null, 1);
        log(`[Backfill][Stage:db] Service role connectivity: ${testQuery ?
'CONNECTED' : 'FAILED'}`);

    } catch (serviceTestError) {
        log(`[Backfill][Stage:db] Service role test failed: $
{serviceTestError.message}`);
        throw {
            stage: 'db',
            code: 'SERVICE_ROLE_FAILED',
            message: 'Service role database access failed',
            cause: serviceTestError.message
        };
    }

    const ticker = safePrices[0].ticker;
    const BATCH_SIZE = 25; // Reduced for better error handling
    const batches = [];
    for (let i = 0; i < safePrices.length; i += BATCH_SIZE) {
        batches.push(safePrices.slice(i, i + BATCH_SIZE));
    }
    log(`[Backfill][Stage:db] Processing ${batches.length} batches of max $
{BATCH_SIZE} rows each`);

    let totalInserted = 0, totalUpdated = 0, totalSkipped = 0;

    for (let i = 0; i < batches.length; i++) {
        const batch = batches[i];
        const dates = batch.map(p => p.date);

        try {
            const existing = await serviceClient.entities.PricesHistory.filter({
                ticker,
                date: { '$in': dates }
            });

            const safeExisting = Array.isArray(existing) ? existing : [];
            const existingMap = new Map(safeExisting.map(p => [p.date, p]));
            const toInsert = batch.filter(row => !existingMap.has(row.date));

            if (toInsert.length > 0) {
                try {
```

```
                    // Wait for DB token before write
                    await waitForToken(dbTokenBucket, log, '[Backfill][Stage:db]
batch ' + (i+1) + '/' + batches.length);

                    // Perform the write with backoff using service role
                    await withBackoff(
                        () =>
serviceClient.entities.PricesHistory.bulkCreate(toInsert),
                        {
                            retries: 6,
                            base: 300,
                            cap: 6000,
                            jitter: true,
                            retryOn: [429, 503, 'FetchError', 'ETIMEDOUT',
'ECONNRESET'],
                            onRetry: (n, wait, err) => log(`[Backfill][Retry $
{n}] stage=db status=${err.status||'n/a'} sleep=${wait}ms`)
                        }
                    );
                    totalInserted += toInsert.length;
                    log(`[Backfill][Stage:db] Batch ${i+1}/${batches.length}
complete: inserted ${toInsert.length}`);

                } catch (dbError) {
                    const status = dbError.response?.status || dbError.status ||
500;
                    log(`[Backfill FATAL] stage=db status=${status} message=DB
write failed on batch ${i+1} after retries.`);

                    // Enhanced unauthorized error handling
                    if (status === 401 || status === 403) {
                        log(`[Backfill FATAL] UNAUTHORIZED ERROR - Service role
access denied`);
                        log(`[Backfill FATAL] Check RLS policies on
PricesHistory table for service_role`);

                        throw {
                            stage: 'db',
                            code: 'UNAUTHORIZED',
                            message: `Unauthorized service role access on batch
${i+1}. Check RLS policies.`,
                            cause: dbError.message,
                            resumeHint: `Run failed at batch ${i+1}/$
{batches.length}. ${totalInserted} rows inserted.`
                        };
                    }

                    // Throw structured error to be caught by the main handler
                    throw {
                        stage: 'db',
                        code: status,
                        message: `DB write for batch ${i+1} failed after
retries.`,
                        cause: dbError.message,
                        resumeHint: `Run failed at batch ${i+1}/$
{batches.length}. ${totalInserted} rows inserted.`
                    };
                }
            }

            const skipped = batch.length - toInsert.length;
            totalSkipped += skipped;

        } catch (batchError) {
```

```
            // Re-throw structured errors
            if (batchError.stage) {
                throw batchError;
            }

            log(`[Backfill][Stage:db] Batch ${i+1} error: $
{batchError.message}`);
            totalSkipped += batch.length;
        }
    }

    log(`[Backfill][Stage:db] All batches complete. Total new rows inserted: $
{totalInserted}.`);
    return { inserted: totalInserted, updated: totalUpdated, skipped:
totalSkipped };
}

// In-process metrics computation helper (no HTTP calls)
async function computeDerivedMetricsInProcess(base44, ticker, log) {
    log(`[Backfill][Stage:finalize] computing price-derived metrics for $
{ticker}`);

    try {
        // Get price history from DB
        const fiveYearsAgo = new Date(new Date().setFullYear(new
Date().getFullYear() - 5)).toISOString().slice(0, 10);

        const tickerPrices = await base44.entities.PricesHistory.filter({
            ticker: ticker,
            date: { $gte: fiveYearsAgo }
        }, '-date', 2000);

        if (!Array.isArray(tickerPrices) || tickerPrices.length < 252) {
            log(`[Backfill][Stage:finalize] insufficient price data for metrics
(${tickerPrices?.length || 0} records)`);
            return { metricsComputed: 0 };
        }

        const derivedMetrics = {};

        // Calculate max drawdown
        let peak = tickerPrices[0].close_adj;
        let maxDrawdown = 0;
        for (const price of tickerPrices) {
            if (price.close_adj > peak) peak = price.close_adj;
            const drawdown = (peak - price.close_adj) / peak;
            if (drawdown > maxDrawdown) maxDrawdown = drawdown;
        }
        derivedMetrics.maxdrawdown_5y_pct = maxDrawdown * 100;

        // Update metrics table
        const existingMetrics = await
base44.entities.Metrics.filter({ ticker_symbol: ticker });
        const updatePayload = {
            ...derivedMetrics,
            data_source: 'derived:backfill',
            as_of_date: toYMD(new Date())
        };

        if (existingMetrics.length > 0) {
            await base44.entities.Metrics.update(existingMetrics[0].id,
updatePayload);
        } else {
```

```
            await base44.entities.Metrics.create({ ticker_symbol: ticker,
...updatePayload });
        }

        log(`[Backfill][Stage:finalize] updated $
{Object.keys(derivedMetrics).length} derived metrics for ${ticker}`);
        return { metricsComputed: Object.keys(derivedMetrics).length };

    } catch (error) {
        // Don't fail the entire backfill for metrics computation failure
        log(`[Backfill][Stage:finalize] metrics computation failed: $
{error.message}`);
        return { metricsComputed: 0, metricsError: error.message };
    }
}

// --- MAIN DENO SERVER ---
let activeRuns = new Map();
Deno.serve(async (req) => {
    const base44 = createClientFromRequest(req);
    if (!(await base44.auth.isAuthenticated())) return new
Response(JSON.stringify({ success: false, message: 'Unauthorized' }), { status:
401 });

    const logs = [];
    const log = (msg) => logs.push(`[${new Date().toISOString().split('T')
[1].slice(0, -1)}] ${msg}`);

    let tickerUpper = '';
    let currentStage = 'init';

    try {
        const { ticker } = await req.json();
        tickerUpper = ticker.trim().toUpperCase();

        // Stage: init
        currentStage = 'init';
        log(`[Backfill][Stage:init] starting full ETL for ${tickerUpper}...`);

        // Concurrency guard
        if (activeRuns.has(tickerUpper)) {
            log(`[Backfill][Guard] run locked for ${tickerUpper}`);
            await activeRuns.get(tickerUpper);
            return new Response(JSON.stringify({ success: true, message:
'Concurrent run completed', logs }), { status: 200 });
        }

        log(`[Backfill][Guard] run locked for ${tickerUpper}`);

        const runPromise = (async () => {
            let yahooResult, normalizedPrices, priceResult, metricsResult;

            try {
                // Stage: fetch
                currentStage = 'fetch';
                yahooResult = await fetchYahooData(tickerUpper, log);

                // Stage: parse
                currentStage = 'parse';
                normalizedPrices = normalizePrices(tickerUpper,
yahooResult.priceData, log);

                // Stage: db
                currentStage = 'db';
```

```
                priceResult = await upsertPrices(base44, normalizedPrices, log);

                // Stage: finalize (in-process metrics computation)
                currentStage = 'finalize';
                metricsResult = await computeDerivedMetricsInProcess(base44,
tickerUpper, log);

                // Stage: complete
                currentStage = 'complete';
                log(`[Backfill SUCCESS] Complete. Received: $
{normalizedPrices.length}, Inserted: ${priceResult.inserted}, Updated: $
{priceResult.updated}.`);

                return {
                    success: true,
                    message: `ETL Completed for ${tickerUpper}`,
                    logs,
                    summary: {
                        financials_stored: 0,
                        prices_stored: priceResult.inserted,
                        metrics_computed: metricsResult.metricsComputed || 0
                    }
                };

            } catch (stageError) {
                // Re-throw with current stage context if not already structured
                if (!stageError.stage) {
                    stageError.stage = currentStage;
                    stageError.code = stageError.status || stageError.code ||
500;
                }
                throw stageError;
            }
        })();

        activeRuns.set(tickerUpper, runPromise);
        const result = await runPromise;
        return new Response(JSON.stringify(result), { status: 200 });

    } catch (error) {
        const stage = error.stage || currentStage || 'unknown';
        const code = error.code || error.status || 500;
        const message = error.message || 'Unknown error occurred';

        log(`[Backfill FATAL] stage=${stage} code=${code} message=${message}`);

        return new Response(JSON.stringify({
            success: false,
            stage: stage,
            code: code,
            message: message,
            cause: error.cause || null,
            guidance: `ETL failed at ${stage}. ${error.resumeHint || ''}`,
            logs
        }), {
            status: code >= 400 && code < 600 ? code : 500,
            headers: { 'Content-Type': 'application/json' }
        });

    } finally {
        if (tickerUpper && activeRuns.has(tickerUpper)) {
            activeRuns.delete(tickerUpper);
            log(`[Backfill][Guard] run unlocked for ${tickerUpper}`);
        }
```

```
    }
});


// FILE: functions/runYahooFundamentalsEtl.ts
// ============================================================


import { createClientFromRequest } from 'npm:@base44/sdk@0.5.0';

// --- START: SHARED HELPERS & SESSION MANAGEMENT ---
const isFiniteNum = (x) => typeof x === 'number' && Number.isFinite(x);
const asArray = (v) => Array.isArray(v) ? v : [];
const keyCount = (obj) => (obj && typeof obj === 'object') ?
Object.keys(obj).length : 0;

// --- START: ROBUST YAHOO SESSION & CRUMB MANAGEMENT ---
const yahooSession = {
  cookie: null,
  crumb: null,
  crumbTs: 0,
  userAgent: 'Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36
(KHTML, like Gecko) Chrome/124.0.0.0 Safari/537.36',
};
async function ensureCrumb(ticker = 'AAPL', log) {
  if (yahooSession.crumb && (Date.now() - yahooSession.crumbTs < 10 * 60 *
1000)) { return yahooSession.crumb; }
  const headers = { 'User-Agent': yahooSession.userAgent, 'Accept':
'text/html,application/json;q=0.9,*/*;q=0.8', 'DNT': '1' };
  try { const primeResponse = await fetch(`https://finance.yahoo.com/quote/$
{encodeURIComponent(ticker)}`, { headers }); const cookieHeader =
primeResponse.headers.get('set-cookie'); if (cookieHeader) yahooSession.cookie =
cookieHeader; } catch (e) { log(`[crumb] cookie priming failed: $
{e.message}`); }
  const crumbHeaders = { ...headers, ...(yahooSession.cookie && { 'Cookie':
yahooSession.cookie }) };
  for (const host of ['query2', 'query1']) {
     try {
       const crumbResponse = await fetch(`https://$
{host}.finance.yahoo.com/v1/test/getcrumb`, { headers: crumbHeaders });
      const text = await crumbResponse.text();
      const body = (text || '').trim();
      if (crumbResponse.status === 200 && body) {
         yahooSession.crumb = body;
         yahooSession.crumbTs = Date.now();
         return body;
       }
     } catch (e) { log(`[crumb] S1: ${host} error: ${e.message}`); }
   }
   throw { stage: 'session', message: 'Could not obtain Yahoo crumb.' };
}
async function fetchSourceQ(ticker, log) {
    log("[Source Q] Fetching quoteSummary data...");
    const crumb = await ensureCrumb(ticker, log);
    const modules = ["incomeStatementHistoryQuarterly",
"cashflowStatementHistoryQuarterly", "balanceSheetHistoryQuarterly",
"incomeStatementHistory", "cashflowStatementHistory", "balanceSheetHistory",
"financialData", "defaultKeyStatistics", "price", "summaryDetail"].join(",");
    const url = new
URL(`https://query2.finance.yahoo.com/v10/finance/quoteSummary/${ticker}`);
    url.search = new URLSearchParams({ modules, crumb, formatted: "false", lang:
"en-US", region: "US", corsDomain: "finance.yahoo.com" }).toString();
    const headers = { 'Cookie': yahooSession.cookie || '', 'User-Agent':
yahooSession.userAgent };
```

```
    const response = await fetch(url.toString(), { headers });
    if (!response.ok) throw new Error(`quoteSummary returned $
{response.status}`);
    const json = await response.json();
    if (!json?.quoteSummary?.result?.[0]) {
      log('[Source Q] Empty/invalid quoteSummary payload; continuing with
minimal fields.');
      return {};
    }
    log(`[Source Q] Successfully fetched quoteSummary data`);
    return json.quoteSummary.result[0];
}
async function fetchSourceF(ticker, log) { return null;}
async function fetchSourceT(ticker, log) { return null;}

// --- C) DB UPSERT WITH PRESERVED STRINGS ---
async function upsertWithGuard(base44, entityName, matchFields, payload, log) {
    const existing = await
base44.asServiceRole.entities[entityName].filter(matchFields);
    const clean = {};
    const skippedKeys = [];
    const alwaysAllow = new Set(['as_of_date', 'data_source', 'ticker_symbol']);

    for (const [k, v] of Object.entries(payload)) {
        if (alwaysAllow.has(k) && typeof v === 'string') {
            clean[k] = v;
            continue;
        }
        if (isFiniteNum(v)) {
            clean[k] = v;
        } else {
            skippedKeys.push(k);
        }
    }

    if (existing.length > 0) {
        const id = existing[0].id;
        delete clean.ticker_symbol;
        if (Object.keys(clean).length > 0) {
            await base44.asServiceRole.entities[entityName].update(id, clean);
            log(`[DB Upsert] Updated: [${Object.keys(clean).join(', ')}]`);
        } else {
            log(`[DB Upsert] No valid fields to update.`);
        }
        if (skippedKeys.length > 0) log(`[DB Upsert] Skipped null/invalid
fields: [${skippedKeys.join(', ')}]`);
        return "updated";
    } else {
        const createPayload = { ...matchFields, ...clean };
        await base44.asServiceRole.entities[entityName].create(createPayload);
        log(`[DB Upsert] Created: [${Object.keys(clean).join(', ')}]`);
        if (skippedKeys.length > 0) log(`[DB Upsert] Skipped null/invalid fields
on create: [${skippedKeys.join(', ')}]`);
        return "inserted";
    }
}


// --- NUMERIC EXTRACTION WITH .RAW SUPPORT ---
function num(n) {
    if (typeof n?.raw === 'number') return n.raw;
    if (typeof n?.raw === 'string' && n.raw.trim() !== '' && !
isNaN(Number(n.raw))) return Number(n.raw);
    if (typeof n === 'number') return n;
```

```
    if (typeof n === 'string' && n.trim() !== '' && !isNaN(Number(n))) return
Number(n);
    return null;
}
const pickNum = (...candidates) => {
  for (const c of candidates) {
    const v = num(c);
    if (typeof v === 'number' && Number.isFinite(v)) return v;
  }
  return null;
};
function toISODate(d) {
    if (!d) return null;
    let timestamp;
    if (typeof d?.raw === 'number') timestamp = d.raw;
    else if (typeof d === 'number') timestamp = d;
    else if (typeof d === 'string') return d.slice(0, 10); // Assume ISO string
already
    else return null;
    const dt = new Date(timestamp * 1000);
    return isFinite(dt.getTime()) ? dt.toISOString().slice(0, 10) : null;
}

function normalizeSourceQ(sourceQ, log) {
    if (!sourceQ || typeof sourceQ !== 'object') return {};
    const incQ =
asArray(sourceQ?.incomeStatementHistoryQuarterly?.incomeStatementHistory);
    const cfsQ =
asArray(sourceQ?.cashflowStatementHistoryQuarterly?.cashflowStatements);
    const bssQ =
asArray(sourceQ?.balanceSheetHistoryQuarterly?.balanceSheetStatements);
    const quarters = {};
    [...incQ, ...cfsQ, ...bssQ].forEach((stmt) => {
         const date = toISODate(stmt?.endDate);
         if (date) {
             if (!quarters[date]) quarters[date] = {};
             Object.assign(quarters[date], stmt);
         }
     });
    log(`[Source Q] Normalized ${keyCount(quarters)} quarters`);
    return quarters;
}
function normalizeSourceF(sourceF, log) {
    if (!sourceF || !sourceF.timeseries || !
asArray(sourceF.timeseries.result).length) return {};
    const quarters = {};
    asArray(sourceF.timeseries.result).forEach(series => {
        if (asArray(series.quarterly).length) {
            asArray(series.quarterly).forEach(entry => {
                 const date = toISODate(entry?.asOfDate || entry?.endDate);
                 if (date) {
                     if (!quarters[date]) quarters[date] = {};
                    const metaType = asArray(series?.meta?.type);
                    const metricName = metaType[0] || 'unknown';
                     quarters[date][metricName] = entry;
                 }
             });
        }
    });
    log(`[Source F] Normalized ${keyCount(quarters)} quarters`);
    return quarters;
}
function normalizeSourceT(sourceT, log) {
```

```
    if (!sourceT || !sourceT.timeseries || !
asArray(sourceT.timeseries.result).length) return {};
    const quarters = {};
    asArray(sourceT.timeseries.result).forEach(series => {
        const metaType = asArray(series?.meta?.type);
        const metricName = metaType[0] || 'unknown';
        if (asArray(series.quarterly).length) {
            asArray(series.quarterly).forEach(entry => {
                 const date = toISODate(entry?.asOfDate || entry?.endDate);
                 if (date) {
                     if (!quarters[date]) quarters[date] = {};
                     quarters[date][metricName] = entry;
                 }
             });
        }
    });
    log(`[Source T] Normalized ${keyCount(quarters)} quarters`);
    return quarters;
}
function mergeMultiSource(quartersQ, quartersF, quartersT, log) {
    const allDates = [...new Set([
        ...Object.keys(quartersQ || {}),
        ...Object.keys(quartersF || {}),
        ...Object.keys(quartersT || {})
    ])].sort().reverse();

    const recentDates = allDates.slice(0, 4);

    const merged = recentDates.map(date => {
        const qData = quartersQ[date] || {};
        const fData = quartersF[date] || {};
        const tData = quartersT[date] || {};

        const result = {
            as_of_date: date,
            // Income Statement
            revenue: num(qData.totalRevenue) || num(fData.Revenue) ||
num(tData.revenue),
            net_income: num(qData.netIncome) || num(fData.NetIncome) ||
num(tData.net_income),
            ebit: num(qData.ebit) || num(fData.OperatingIncome) ||
num(tData.ebit),
            depreciation: num(qData.depreciation) ||
num(fData.DepreciationAndAmortization) ||
num(tData.depreciation_and_amortization),
            sbc: num(qData.stockBasedCompensation) ||
num(fData.StockBasedCompensation) || num(tData.stock_based_compensation),
            interest_expense: num(qData.interestExpense) ||
num(fData.InterestExpense) || num(tData.interest_expense),

            // Cash Flow Statement
            cfo: num(qData.totalCashFromOperatingActivities) ||
num(fData.OperatingCashFlow) || num(tData.cash_flow_from_operating_activities),
            capex: num(qData.capitalExpenditures) ||
num(fData.CapitalExpenditures) || num(tData.capital_expenditure),

            // Balance Sheet
            cash: num(qData.cash) || num(fData.CashAndCashEquivalents) ||
num(tData.cash_and_cash_equivaents),
            short_debt: num(qData.shortTermDebt) || num(fData.ShortTermDebt) ||
num(tData.short_term_debt),
            long_debt: num(qData.longTermDebt) || num(fData.LongTermDebt) ||
num(tData.long_term_debt),
```

```
            equity: num(qData.totalStockholderEquity) || num(fData.TotalEquity)
|| num(tData.total_equity),
            total_assets: num(qData.totalAssets) || num(fData.TotalAssets) ||
num(tData.total_assets),
        };
        const coverage = {};
        for (const key in result) {
            if (key === 'as_of_date') continue;
            let source = 'None';
            if (result[key] === num(qData[key])) source = 'Q';
            else if (result[key] === num(fData[key])) source = 'F';
            else if (result[key] === num(tData[key])) source = 'T';
            coverage[key] = source;
        }
        result.coverage = coverage;
        return result;
    }).filter(q => q.as_of_date);

    if (asArray(merged).length > 0) {
         const latestCoverage = merged[0].coverage || {};
        const entries = Object.entries(latestCoverage);
        const coverageReport = entries.map(([field, source]) => `${field}:$
{source}`).join(', ');
        const fieldsPopulated = entries.filter(([,src]) => src !==
'None').length;
        const totalFields = entries.length;
         const coveragePercent = Math.round((fieldsPopulated / (totalFields ||
1)) * 100);
         log(`[Coverage] ${coverageReport}`);
        log(`[Coverage] fields_populated = ${fieldsPopulated}/${totalFields ||
1} (${coveragePercent || 0}%)`);
    }
    return merged;
}

// --- C) ROBUST TTM BUILDER ---
function buildTTMFromMerged(merged, sourceQ, ticker, log) {
    if (asArray(merged).length < 4) {
         log(`[TTM] Warning: Found only ${asArray(merged).length} quarters of
data. TTM values may be incomplete.`);
    }

    const sum4 = (vals) => {
        const nums = (vals || []).filter(v => typeof v === 'number' &&
Number.isFinite(v));
        return nums.length === 4 ? nums.reduce((a, b) => a + b, 0) : null;
    };

    const pick = (k) => merged.map(q => (typeof q[k] === 'number' ? q[k] :
null));

    const revenue_ttm = sum4(pick('revenue'));
    const cfo_ttm = sum4(pick('cfo'));
    const capex_ttm = sum4(pick('capex'));
    const sbc_ttm = sum4(pick('sbc'));
    const ebit_ttm = sum4(pick('ebit'));
    const depreciation_ttm = sum4(pick('depreciation'));
    const ebitda_ttm = (isFiniteNum(ebit_ttm) &&
isFiniteNum(depreciation_ttm)) ? ebit_ttm + depreciation_ttm : null;
    const fcf_ttm = (isFiniteNum(cfo_ttm) && isFiniteNum(capex_ttm)) ? (cfo_ttm
- Math.abs(capex_ttm)) : null;

    const latest = merged[0] || {};
```

```
    const payload = {
        ticker_symbol: ticker,
        as_of_date: new Date().toISOString().slice(0, 10),
        data_source: 'yahoo:multi_source_v1',

        price_current: pickNum(sourceQ?.price?.regularMarketPrice,
sourceQ?.summaryDetail?.regularMarketPreviousClose),
        revenue_ttm,
        cfo_ttm,
        capex_ttm,
        sbc_ttm,
        fcf_ttm,
        ebit_ttm,
        depreciation_ttm,
        ebitda_ttm,
        net_income_ttm: sum4(pick('net_income')),
        interest_expense_ttm: sum4(pick('interest_expense')),

        cash: latest.cash,
        total_debt: (latest.short_debt !== null || latest.long_debt !== null) ?
(latest.short_debt || 0) + (latest.long_debt || 0) : null,
        equity: latest.equity,
        total_assets: latest.total_assets,

        market_cap: num(sourceQ?.price?.marketCap),
        shares_out: num(sourceQ?.defaultKeyStatistics?.sharesOutstanding),
        pe_fwd: num(sourceQ?.summaryDetail?.forwardPE),
        pe_ttm: num(sourceQ?.summaryDetail?.trailingPE),
        beta_5y: num(sourceQ?.defaultKeyStatistics?.beta),

        enterprise_value: pickNum(
          sourceQ?.defaultKeyStatistics?.enterpriseValue,
          sourceQ?.financialData?.enterpriseValue
        ),
        enterprise_to_ebitda: pickNum(
          sourceQ?.defaultKeyStatistics?.enterpriseToEbitda,
          sourceQ?.financialData?.enterpriseToEbitda
        ),

        current_pe: pickNum(
          sourceQ?.summaryDetail?.trailingPE,
          sourceQ?.defaultKeyStatistics?.trailingPE
        )
    };

    if (payload.enterprise_value != null && payload.ebitda_ttm != null &&
payload.ebitda_ttm !== 0) {
        const ratio = payload.enterprise_value / payload.ebitda_ttm;
        if (Number.isFinite(ratio)) payload.ev_ebitda = ratio;
    } else if (payload.enterprise_to_ebitda != null) {
        payload.ev_ebitda = payload.enterprise_to_ebitda;
    }

    return payload;
}

// --- MAIN DENO SERVER ---
Deno.serve(async (req) => {
    const base44 = createClientFromRequest(req);
    const logs = [];
    const log = (msg) => logs.push(`[${new Date().toISOString().split('T')
[1].slice(0, -1)}] ${msg}`);
    if (!(await base44.auth.isAuthenticated())) {
```

```
        return new Response(JSON.stringify({ ok: false, success: false, error:
'Unauthorized' }), { status: 401 });
    }

    try {
         const { ticker, force_backfill = false } = await req.json();
         if (!ticker) throw { stage: 'input', message: 'Ticker is required' };
         const tickerUpper = ticker.trim().toUpperCase();
         log(`[START] Yahoo Fundamentals ETL for ${tickerUpper}
(force_backfill=${force_backfill})`);

        const sourceQ = await fetchSourceQ(tickerUpper, log);

        // --- SAVE QUARTERLY FINANCIALS HISTORY ---
        const quarterlyFinancials = [];
        const annualFinancials = [];

        // Process quarterly data
        const allQuarterDates = [...new Set([

...asArray(sourceQ?.incomeStatementHistoryQuarterly?.incomeStatementHistory).map
(s => toISODate(s.endDate)),

...asArray(sourceQ?.cashflowStatementHistoryQuarterly?.cashflowStatements).map(s
=> toISODate(s.endDate)),

...asArray(sourceQ?.balanceSheetHistoryQuarterly?.balanceSheetStatements).map(s
=> toISODate(s.endDate))
        ])].filter(Boolean).sort().reverse(); // Most recent first

        log(`[Quarterly] Found ${allQuarterDates.length} unique quarterly dates
to process.`);

        for (const date of allQuarterDates) {
            const inc =
asArray(sourceQ?.incomeStatementHistoryQuarterly?.incomeStatementHistory).find(s
=> toISODate(s.endDate) === date) || {};
            const cfs =
asArray(sourceQ?.cashflowStatementHistoryQuarterly?.cashflowStatements).find(s
=> toISODate(s.endDate) === date) || {};
            const bss =
asArray(sourceQ?.balanceSheetHistoryQuarterly?.balanceSheetStatements).find(s =>
toISODate(s.endDate) === date) || {};

            // Robust interest expense with fallback and sign normalization
            let interestExp = num(inc.interestExpense) ?? num(cfs.interestPaid);
            if (interestExp != null) interestExp = Math.abs(interestExp);

            const quarterlyPayload = {
                ticker: tickerUpper,
                period_end: date,
                freq: 'quarterly',
                source: 'yahoo',
                as_of_date: new Date().toISOString().slice(0,10),

                // Income Statement
                revenue: num(inc.totalRevenue),
                net_income: num(inc.netIncome),
                ebit: num(inc.ebit),
                interest_expense: interestExp,
                depreciation: num(cfs.depreciation),
                stock_based_compensation: num(inc.stockBasedCompensation) ||
num(cfs.stockBasedCompensation),
                shares_diluted: num(inc.dilutedAverageShares),
```

```
                eps_diluted: num(inc.dilutedEps),

                // Cash Flow Statement
                cfo: num(cfs.totalCashFromOperatingActivities),
                capex: num(cfs.capitalExpenditures),
                fcf: isFiniteNum(num(cfs.totalCashFromOperatingActivities)) &&
isFiniteNum(num(cfs.capitalExpenditures)) ?
                     num(cfs.totalCashFromOperatingActivities) -
Math.abs(num(cfs.capitalExpenditures)) : null,

                // Balance Sheet
                cash: num(bss.cash),
                short_debt: num(bss.shortLongTermDebt),
                long_debt: num(bss.longTermDebt),
                total_debt: (isFiniteNum(num(bss.shortLongTermDebt)) ||
isFiniteNum(num(bss.longTermDebt))) ?
                            (num(bss.shortLongTermDebt) || 0) +
(num(bss.longTermDebt) || 0) : null,
                stockholder_equity: num(bss.totalStockholderEquity),
                total_assets: num(bss.totalAssets),
            };
            quarterlyFinancials.push(quarterlyPayload);
        }

        // Process annual data (if available)
        const allAnnualDates = [...new Set([

...asArray(sourceQ?.incomeStatementHistory?.incomeStatementHistory).map(s =>
toISODate(s.endDate)),

...asArray(sourceQ?.cashflowStatementHistory?.cashflowStatements).map(s =>
toISODate(s.endDate)),

...asArray(sourceQ?.balanceSheetHistory?.balanceSheetStatements).map(s =>
toISODate(s.endDate))
        ])].filter(Boolean).sort().reverse();

        log(`[Annual] Found ${allAnnualDates.length} unique annual dates to
process.`);

        for (const date of allAnnualDates) {
            const inc =
asArray(sourceQ?.incomeStatementHistory?.incomeStatementHistory).find(s =>
toISODate(s.endDate) === date) || {};
            const cfs =
asArray(sourceQ?.cashflowStatementHistory?.cashflowStatements).find(s =>
toISODate(s.endDate) === date) || {};
            const bss =
asArray(sourceQ?.balanceSheetHistory?.balanceSheetStatements).find(s =>
toISODate(s.endDate) === date) || {};

            let interestExp = num(inc.interestExpense) ?? num(cfs.interestPaid);
            if (interestExp != null) interestExp = Math.abs(interestExp);

            const annualPayload = {
                ticker: tickerUpper,
                period_end: date,
                freq: 'annual',
                source: 'yahoo',
                as_of_date: new Date().toISOString().slice(0,10),

                revenue: num(inc.totalRevenue),
                net_income: num(inc.netIncome),
                ebit: num(inc.ebit),
```

```
                interest_expense: interestExp,
                depreciation: num(cfs.depreciation),
                stock_based_compensation: num(inc.stockBasedCompensation) ||
num(cfs.stockBasedCompensation),
                shares_diluted: num(inc.dilutedAverageShares),
                eps_diluted: num(inc.dilutedEps),

                cfo: num(cfs.totalCashFromOperatingActivities),
                capex: num(cfs.capitalExpenditures),
                fcf: isFiniteNum(num(cfs.totalCashFromOperatingActivities)) &&
isFiniteNum(num(cfs.capitalExpenditures)) ?
                     num(cfs.totalCashFromOperatingActivities) -
Math.abs(num(cfs.capitalExpenditures)) : null,

                cash: num(bss.cash),
                short_debt: num(bss.shortLongTermDebt),
                long_debt: num(bss.longTermDebt),
                total_debt: (isFiniteNum(num(bss.shortLongTermDebt)) ||
isFiniteNum(num(bss.longTermDebt))) ?
                            (num(bss.shortLongTermDebt) || 0) +
(num(bss.longTermDebt) || 0) : null,
                stockholder_equity: num(bss.totalStockholderEquity),
                total_assets: num(bss.totalAssets),
            };
            annualFinancials.push(annualPayload);
        }

        // Upsert quarterly financials
        let qUpsertedCount = 0;
        for (const record of quarterlyFinancials) {
            const { ticker, period_end, freq, ...data } = record;
            const matchFields = { ticker, period_end, freq };
            const cleanData =
Object.fromEntries(Object.entries(data).filter(([,v]) => v != null));

            if (Object.keys(cleanData).length === 0) continue;

            const existing = await
base44.asServiceRole.entities.FinancialsHistory.filter(matchFields);
            if(existing.length > 0) {
                await
base44.asServiceRole.entities.FinancialsHistory.update(existing[0].id,
cleanData);
            } else {
                await
base44.asServiceRole.entities.FinancialsHistory.create({ ...matchFields,
...cleanData });
            }
            qUpsertedCount++;
        }

        // Upsert annual financials
        let aUpsertedCount = 0;
        for (const record of annualFinancials) {
            const { ticker, period_end, freq, ...data } = record;
            const matchFields = { ticker, period_end, freq };
            const cleanData =
Object.fromEntries(Object.entries(data).filter(([,v]) => v != null));

            if (Object.keys(cleanData).length === 0) continue;

            const existing = await
base44.asServiceRole.entities.FinancialsHistory.filter(matchFields);
            if(existing.length > 0) {
```

```
                await
base44.asServiceRole.entities.FinancialsHistory.update(existing[0].id,
cleanData);
            } else {
                await
base44.asServiceRole.entities.FinancialsHistory.create({ ...matchFields,
...cleanData });
            }
            aUpsertedCount++;
        }

        log(`[FinancialsHistory] Completed upsert for ${qUpsertedCount}
quarterly + ${aUpsertedCount} annual reports.`);

        // Save current metrics to Metrics table
        const metricsPayload = buildTTMFromMerged([], sourceQ || {},
tickerUpper, log);
        await upsertWithGuard(base44, 'Metrics', { ticker_symbol: tickerUpper },
metricsPayload, log);

        return new Response(JSON.stringify({
            ok: true,
            success: true,
            message: `Yahoo fundamentals processed for ${tickerUpper}.`,
            quarterly_count: qUpsertedCount,
            annual_count: aUpsertedCount,
            summary: { ticker: tickerUpper, fields:
Object.keys(metricsPayload).filter(k => metricsPayload[k] != null) },
            logs
        }), { status: 200, headers: { 'Content-Type': 'application/json' } });

    } catch (error) {
        const errorMessage = (error && error.message) ? error.message :
String(error || 'Unknown error');
        log(`[FATAL] ${error?.stage || 'unknown'}: ${errorMessage}`);
        return new Response(JSON.stringify({ ok: false, success: false, stage:
error?.stage || 'unknown', error: errorMessage, message: `Failed during Yahoo
ETL: ${errorMessage}`, logs }), { status: 500, headers: { 'Content-Type':
'application/json' } });
    }
});


// FILE: functions/saveFinancialsHistory.ts
// ============================================================

import { createClientFromRequest } from 'npm:@base44/sdk@0.5.0';

Deno.serve(async (req) => {
    const base44 = createClientFromRequest(req);

    if (!(await base44.auth.isAuthenticated())) {
        return new Response(JSON.stringify({ error: 'Unauthorized' }), {
            status: 401,
            headers: { 'Content-Type': 'application/json' }
        });
    }

    try {
        const { rows } = await req.json();

        if (!rows || !Array.isArray(rows)) {
            return new Response(JSON.stringify({ error: 'Rows array
required' }), {
```

```
                status: 400,
                headers: { 'Content-Type': 'application/json' }
            });
        }

        let saved_count = 0;

        for (const row of rows) {
            try {
                // Check if record already exists
                const existing = await
base44.entities.FinancialsHistory.filter({
                    ticker: row.ticker,
                    period: row.period,
                    freq: row.freq
                });

                if (existing.length === 0) {
                    // Only create if it doesn't exist (idempotent)
                    const payload = {};
                    // Only include non-null numeric fields
                    Object.keys(row).forEach(key => {
                        if (row[key] != null) {
                            payload[key] = row[key];
                        }
                    });

                    await base44.entities.FinancialsHistory.create(payload);
                    saved_count++;
                }
            } catch (error) {
                console.error(`Error saving financial for ${row.ticker} $
{row.period}:`, error);
                // Continue with next record
            }
        }

        return new Response(JSON.stringify({
            saved_count,
            total_rows: rows.length
        }), {
            status: 200,
            headers: { 'Content-Type': 'application/json' }
        });

    } catch (error) {
        console.error('Error saving financials history:', error);
        return new Response(JSON.stringify({ error: error.message }), {
            status: 500,
            headers: { 'Content-Type': 'application/json' }
        });
    }
});

// FILE: functions/savePricesHistory.ts
// ============================================================

import { createClientFromRequest } from 'npm:@base44/sdk@0.5.0';

// Helper functions for data normalization
const toYMD = (d) => {
    try {
        return new Date(d).toISOString().slice(0, 10);
    } catch {
```

```
        return null;
    }
};

const numOrNull = (v) => (v === null || v === undefined || Number.isNaN(+v)) ?
null : +v;

Deno.serve(async (req) => {
    const base44 = createClientFromRequest(req);

    // AUTH DIAGNOSTICS - Check authentication before any DB operations
    const logs = [];
    const log = (msg) => logs.push(`[${new Date().toISOString().split('T')
[1].slice(0, -1)}] ${msg}`);

    try {
        log("[Auth] Checking authentication status...");
        const isAuth = await base44.auth.isAuthenticated();
        log(`[Auth] User authenticated: ${isAuth}`);

        if (!isAuth) {
            return new Response(JSON.stringify({
                error: true,
                stage: 'auth',
                message: 'User not authenticated',
                logs
            }), {
                status: 401,
                headers: { 'Content-Type': 'application/json' }
            });
        }

        // DB DIAGNOSTICS - Check database connection and privileges
        log("[DB] Running database diagnostics...");

        try {
            // Test basic connectivity
            const basicTest = await
base44.asServiceRole.entities.PricesHistory.filter({}, null, 1);
            log(`[DB] Basic connectivity test: ${basicTest ? 'OK' : 'FAILED'}`);

            // Log current database role and permissions (using raw SQL if
available)
            log("[DB] Database role diagnostics completed");

        } catch (dbTestError) {
            log(`[DB] Database test failed: ${dbTestError.message}`);
            return new Response(JSON.stringify({
                error: true,
                stage: 'db_test',
                message: `Database connectivity failed: ${dbTestError.message}`,
                logs
            }), {
                status: 500,
                headers: { 'Content-Type': 'application/json' }
            });
        }

    } catch (authError) {
        log(`[Auth] Authentication check failed: ${authError.message}`);
        return new Response(JSON.stringify({
            error: true,
            stage: 'auth_check',
            message: authError.message,
```

```
            logs
        }), {
            status: 401,
            headers: { 'Content-Type': 'application/json' }
        });
    }

    try {
        const { ticker, daily_prices, source = 'yahoo' } = await req.json();

        if (!ticker || !daily_prices || !Array.isArray(daily_prices)) {
            return new Response(JSON.stringify({
                error: true,
                stage: 'validation',
                message: 'Ticker and daily_prices array required',
                logs
            }), {
                status: 400,
                headers: { 'Content-Type': 'application/json' }
            });
        }

        // Enforce 50-row limit for conservative batching
        if (daily_prices.length > 50) {
            return new Response(JSON.stringify({
                error: true,
                stage: 'validation',
                message: `Too many rows: ${daily_prices.length}. Maximum 50 rows
per call.`,
                received_rows: daily_prices.length,
                max_allowed: 50,
                logs
            }), {
                status: 400,
                headers: { 'Content-Type': 'application/json' }
            });
        }

        // Normalize and validate each row
        const normalizedRows = [];
        const asOfDate = toYMD(new Date());

        for (const row of daily_prices) {
            const clean = {
                ticker: String(ticker).trim().toUpperCase(),
                date: toYMD(row.date),
                as_of_date: asOfDate,
                source: source,
                close_adj: numOrNull(row.close_adj),
                open: numOrNull(row.open),
                high: numOrNull(row.high),
                low: numOrNull(row.low),
                close: numOrNull(row.close),
                volume: numOrNull(row.volume),
            };

            // Skip rows with invalid ticker or date
            if (!clean.ticker || !clean.date) {
                continue;
            }

            normalizedRows.push(clean);
        }
```

```
        if (normalizedRows.length === 0) {
            return new Response(JSON.stringify({
                error: true,
                stage: 'validation',
                message: 'No valid rows after normalization',
                logs
            }), {
                status: 400,
                headers: { 'Content-Type': 'application/json' }
            });
        }

        // Remove duplicates within batch by (ticker, date)
        const uniqueRows = [];
        const seen = new Set();
        for (const row of normalizedRows) {
            const key = `${row.ticker}|${row.date}`;
            if (!seen.has(key)) {
                seen.add(key);
                uniqueRows.push(row);
            }
        }

        // Log batch info
        const dates = uniqueRows.map(r => r.date).sort();
        const minDate = dates[0];
        const maxDate = dates[dates.length - 1];

        log(`[savePricesHistory] ${ticker}: Processing ${uniqueRows.length} rows
(${minDate} to ${maxDate})`);
        log(`[savePricesHistory] First 2 rows: $
{JSON.stringify(uniqueRows.slice(0, 2))}`);

        try {
            log("[DB] Starting database operations with service role...");

            // Use service role for database operations
            const serviceClient = base44.asServiceRole;

            // Check for existing records
            const existingRecords = await
serviceClient.entities.PricesHistory.filter({
                ticker: ticker,
                date: { $in: uniqueRows.map(r => r.date) }
            });

            log(`[DB] Found ${existingRecords.length} existing records`);

            const existingMap = new Map();
            existingRecords.forEach(record => {
                existingMap.set(`${record.ticker}|${record.date}`, record);
            });

            let inserted = 0;
            let updated = 0;
            let skipped = 0;

            // Process each row with upsert logic
            for (const row of uniqueRows) {
                const key = `${row.ticker}|${row.date}`;
                const existing = existingMap.get(key);

                try {
                    if (existing) {
```

```
                        // Update existing record
                        await
serviceClient.entities.PricesHistory.update(existing.id, row);
                        updated++;
                        log(`[DB] Updated record for ${row.ticker} $
{row.date}`);
                    } else {
                        // Create new record
                        await serviceClient.entities.PricesHistory.create(row);
                        inserted++;
                        log(`[DB] Inserted record for ${row.ticker} $
{row.date}`);
                    }
                } catch (rowError) {
                    log(`[DB] Error processing row ${row.ticker} ${row.date}: $
{rowError.message}`);
                    skipped++;
                }
            }

            log(`[savePricesHistory] ${ticker}: Completed - inserted: $
{inserted}, updated: ${updated}, skipped: ${skipped}`);

            return new Response(JSON.stringify({
                ok: true,
                ticker,
                received: daily_prices.length,
                normalized: normalizedRows.length,
                processed: uniqueRows.length,
                inserted,
                updated,
                skipped,
                min_date: minDate,
                max_date: maxDate,
                logs
            }), {
                status: 200,
                headers: { 'Content-Type': 'application/json' }
            });

        } catch (dbError) {
            log(`[savePricesHistory] DB error for ${ticker}: $
{dbError.message}`);

            // Enhanced error logging for unauthorized access
            if (dbError.status === 401 || dbError.status === 403) {
                log(`[DB] UNAUTHORIZED ERROR - Status: ${dbError.status}`);
                log(`[DB] Error details: ${JSON.stringify({
                    message: dbError.message,
                    status: dbError.status,
                    cause: dbError.cause || 'Unknown'
                })}`);

                return new Response(JSON.stringify({
                    error: true,
                    stage: 'db',
                    code: 'UNAUTHORIZED',
                    message: 'Unauthorized at DB stage; check service key / role
/ RLS policy.',
                    detail: dbError.message,
                    ticker,
                    logs
                }), {
                    status: 401,
```

```
                    headers: { 'Content-Type': 'application/json' }
                });
            }

            return new Response(JSON.stringify({
                error: true,
                stage: 'upsert',
                message: dbError.message,
                code: dbError.code || 'DB_OPERATION_FAILED',
                detail: dbError?.meta || null,
                ticker,
                logs
            }), {
                status: 400,
                headers: { 'Content-Type': 'application/json' }
            });
        }

    } catch (error) {
        log(`[savePricesHistory] Function error: ${error.message}`);
        return new Response(JSON.stringify({
            error: true,
            stage: 'function',
            message: error.message,
            code: error.code || 'FUNCTION_EXECUTION_FAILED',
            logs
        }), {
            status: 400,
            headers: { 'Content-Type': 'application/json' }
        });
    }
});

// FILE: functions/screenerProbe.ts
// ==============================================================

// functions/screenerProbe.js
import { createClientFromRequest } from 'npm:@base44/sdk@0.5.0';

const isNum = (v) => typeof v === 'number' && Number.isFinite(v);
const nz = (v) => (isNum(v) ? v : null);

Deno.serve(async (req) => {
  const base44 = createClientFromRequest(req);
  const logs = [];
  const log = (m) => logs.push(`[${new Date().toISOString()}] ${m}`);

  if (!(await base44.auth.isAuthenticated())) {
    return new Response(JSON.stringify({ ok:false, message:'Unauthorized' }),
{ status: 401 });
  }

  try {
    const { ticker } = await req.json();
    const t = String(ticker || '').trim().toUpperCase();
    if (!t) throw new Error('ticker required');

    log(`[Probe] Starting for ${t}`);

    // 1) Metrics (the screener's primary source)
    const metricsList = await
base44.asServiceRole.entities.Metrics.filter({ ticker_symbol: t }, '-
updated_date', 1);
    const metrics = metricsList?.[0] || {};
```

```
    log(`[Probe] Metrics source=${metrics.data_source || 'n/a'} as_of=$
{metrics.as_of_date || 'n/a'}`);

    // 2) PriceDerivedStats (P/E bands etc.)
    const statsList = await
base44.asServiceRole.entities.PriceDerivedStats?.filter?.({ ticker: t }, '-
as_of_date', 1).catch(() => []);
    const stats = statsList?.[0] || {};
    log(`[Probe] PriceDerivedStats present=${!!stats.id} as_of=$
{stats.as_of_date || 'n/a'}`);

    // 3) Recent quarterlies (for Capital Allocation)
    const q = await
base44.asServiceRole.entities.FinancialsHistory.filter({ ticker: t, freq:
'quarterly' }, '-period_end', 6);
    const q0 = q?.[0] || {};
    const derived = (() => {
      const equity = nz(metrics.equity ?? q0.equity);
      const total_debt = nz(metrics.total_debt ?? ((q0.short_debt||0)+
(q0.long_debt||0)));
      const cash = nz(metrics.cash ?? q0.cash);
      const ebit_ttm = nz(metrics.ebit_ttm);
      const dep_ttm = nz(metrics.depreciation_ttm);
      const ebitda_ttm = nz(metrics.ebitda_ttm ??
(isNum(ebit_ttm)&&isNum(dep_ttm) ? ebit_ttm+dep_ttm : null));
      const ie_ttm = nz(metrics.interest_expense_ttm);

      const debt_to_equity = (isNum(total_debt)&&isNum(equity)&&equity>0) ?
total_debt/equity : null;
      const netdebt_to_ebitda =
(isNum(total_debt)&&isNum(cash)&&isNum(ebitda_ttm)&&ebitda_ttm>0)
        ? (total_debt - cash)/ebitda_ttm : null;
      const interest_coverage_x =
(isNum(ebit_ttm)&&isNum(ie_ttm)&&Math.abs(ie_ttm)>0)
        ? ebit_ttm/Math.abs(ie_ttm) : null;

      return { debt_to_equity, netdebt_to_ebitda, interest_coverage_x };
    })();

    // 4) Identify "screener-critical" missing fields
    const need = [
      'price_current','eps_ttm','pe_ttm','pe_fwd',
      'ev_ebitda','pe_5y_low','pe_5y_high',
      'eps_cagr_5y_pct','revenue_cagr_5y_pct',
      'debt_to_equity','netdebt_to_ebitda','interest_coverage_x',
      'cfo_to_ni','fcf_margin_pct','fcf_to_ebit'
    ];
    const missing = need.filter(k => !isNum(metrics[k] ?? stats[k] ??
derived[k]));

    const out = {
      ok: true,
      message: `Probe complete for ${t}`,
      metrics: {
        ticker_symbol: metrics.ticker_symbol, as_of_date: metrics.as_of_date,
data_source: metrics.data_source,
        price_current: metrics.price_current, eps_ttm: metrics.eps_ttm, pe_ttm:
metrics.pe_ttm, pe_fwd: metrics.pe_fwd,
        ev_ebitda: metrics.ev_ebitda, cfo_to_ni: metrics.cfo_to_ni,
fcf_margin_pct: metrics.fcf_margin_pct,
        fcf_to_ebit: metrics.fcf_to_ebit, debt_to_equity:
metrics.debt_to_equity,
        netdebt_to_ebitda: metrics.netdebt_to_ebitda, interest_coverage_x:
metrics.interest_coverage_x,
```

```
        revenue_cagr_5y_pct: metrics.revenue_cagr_5y_pct, eps_cagr_5y_pct:
metrics.eps_cagr_5y_pct
      },
      priceDerivedStats: {
        as_of_date: stats.as_of_date,
        pe_5y_low: stats.pe_5y_low, pe_5y_high: stats.pe_5y_high,
        pe_12m: stats.pe_12m, pe_24m: stats.pe_24m, pe_36m: stats.pe_36m
      },
      capitalAllocationDerived: derived,
      missing_fields: missing,
      logs
    };
    return new Response(JSON.stringify(out), { status: 200, headers: { 'Content-
Type': 'application/json' } });
  } catch (e) {
    logs.push(`[FATAL] ${e.message}`);
    return new Response(JSON.stringify({ ok:false, message:e.message, logs }), {
status: 500 });
  }
});

// FILE: functions/syncFundamentals.ts
// ============================================================

import { createClientFromRequest } from 'npm:@base44/sdk@0.5.0';

// --- Utility Functions ---
const normalizeSymbol = (s) => (s || '').toString().trim().toUpperCase();
const toYMD = (date) => new Date(date).toISOString().slice(0, 10);
const numOrNull = (v) => Number.isFinite(Number(v)) ? Number(v) : null;
const corsHeaders = () => ({ "Access-Control-Allow-Origin": "*", "Access-
Control-Allow-Methods": "POST, GET, OPTIONS", "Access-Control-Allow-Headers":
"Content-Type, Authorization" });

// --- New Fetch Helper ---
async function fetchJsonOrDie(url, { label = "finnhub" } = {}) {
  const maskedUrl = url.replace(/token=[^&]+/, "token=***");
  console.log(`[${label}] Fetching: ${maskedUrl}`);

  try {
    const res = await fetch(url, { headers: { accept: "application/json" } });
    const ct = res.headers.get("content-type") || "";
    const text = await res.text();
    const head = text.slice(0, 200).replace(/\n/g, " ");

    if (!res.ok) {
      console.error(`[${label}] Error: status=${res.status}, content-type='$
{ct}', preview='${head}'`);
      return { ok: false, status: res.status, contentType: ct, preview: head };
    }
    if (!ct.includes("application/json")) {
      console.error(`[${label}] Error: Expected JSON, got content-type='${ct}',
preview='${head}'`);
      return { ok: false, status: res.status, contentType: ct, preview: head };
    }

    const json = JSON.parse(text);
    return { ok: true, json };

  } catch (e) {
    console.error(`[${label}] Network or parsing error: ${e.message}`);
    return { ok: false, status: 500, contentType: 'N/A', preview:
'Network/parsing error', parseError: e.message };
  }
```

```
}

// --- Data Mapping ---
const conceptMap = {
  revenue: ["Revenues", "RevenueFromContractWithCustomerExcludingAssessedTax"],
  eps_basic: ["EarningsPerShareBasic"],
  ebit: ["OperatingIncomeLoss"],
  cfo: ["NetCashProvidedByUsedInOperatingActivities"],
  capex: ["PaymentsToAcquirePropertyPlantAndEquipment"],
  total_debt: ["LongTermDebtAndCapitalLeaseObligation",
"DebtAndCapitalLeaseObligations"],
  cash: ["CashAndCashEquivalentsAtCarryingValue"],
  total_equity: ["StockholdersEquity",
"StockholdersEquityIncludingPortionAttributableToNoncontrollingInterest"],
  total_assets: ["Assets"]
};

function pickValue(report, concepts) {
  for (const concept of concepts) {
    for (const statement of ['ic', 'bs', 'cf']) {
      const item = report[statement]?.find(i => i.concept === `us-gaap_$
{concept}`);
      if (item) return numOrNull(item.value);
    }
  }
  return null;
}

function mapReportToFinancialsHistory(ticker, filing) {
  const { report, endDate, fiscalYear, fiscalQuarter } = filing;
  if (!report) return null;

  const row = {
    ticker,
    period: endDate,
    freq: fiscalQuarter > 0 ? 'quarterly' : 'annual',
    source: 'finnhub',
    as_of_date: toYMD(new Date())
  };

  const cfo = pickValue(report, conceptMap.cfo);
  const capex = pickValue(report, conceptMap.capex);
  const cash = pickValue(report, conceptMap.cash);
  const total_debt = pickValue(report, conceptMap.total_debt);
  const total_equity = pickValue(report, conceptMap.total_equity);
  const total_assets = pickValue(report, conceptMap.total_assets);

  row.revenue = pickValue(report, conceptMap.revenue);
  row.eps_basic = pickValue(report, conceptMap.eps_basic);
  row.ebit = pickValue(report, conceptMap.ebit);
  row.cfo = cfo;
  row.total_debt = total_debt;
  row.cash = cash;

  if (cfo !== null && capex !== null) {
    row.fcf = cfo - capex;
  }

  // EBITDA and Invested Capital often require calculation or are non-standard.
  // We'll compute what we can.
  if (total_assets !== null && cash !== null) {
    row.invested_capital = total_assets - cash; // Simplified approximation
  }
```

```
  // Return row only if it has at least one key metric
  return (row.revenue || row.eps_basic || row.cfo) ? row : null;
}

// --- Main Function ---
Deno.serve(async (req) => {
  if (req.method === 'OPTIONS') return new Response(null, { status: 204,
headers: corsHeaders() });

  const base44 = createClientFromRequest(req);
  if (!(await base44.auth.isAuthenticated())) {
    return new Response(JSON.stringify({ error: true, message:
'Unauthorized' }), { status: 401, headers: { ...corsHeaders(), 'Content-Type':
'application/json' }});
  }

  try {
    const { ticker: rawTicker, testMode = false } = await req.json();
    const ticker = normalizeSymbol(rawTicker);
    if (!ticker) throw new Error('Valid ticker symbol required');

    const apiKey = Deno.env.get("FINNHUB_API_KEY");
    if (!apiKey) throw new Error('FINNHUB_API_KEY is not configured on the
server');

    const allLogs = [`[Fundamentals] Starting sync for ${ticker}. Test Mode: $
{testMode}`];

    // 1. Fetch Financials from Finnhub
    const url = `https://finnhub.io/api/v1/stock/financials-reported?symbol=$
{ticker}&token=${apiKey}`;
    const finnhubResponse = await fetchJsonOrDie(url, { label: `finnhub-
financials` });

    if (!finnhubResponse.ok) {
      allLogs.push(`[FATAL] Finnhub request failed. Status: $
{finnhubResponse.status}, Type: ${finnhubResponse.contentType}`);
      allLogs.push(`[FATAL] Preview: ${finnhubResponse.preview}`);
      return new Response(JSON.stringify({ error: true, message: "Failed to
fetch from Finnhub.", logs: allLogs }), { status: 502, headers: {
...corsHeaders(), 'Content-Type': 'application/json' } });
    }

    const filings = finnhubResponse.json.data || [];
    allLogs.push(`[Finnhub] Received ${filings.length} total filings.`);

    const financialsToUpsert = filings.map(f =>
mapReportToFinancialsHistory(ticker, f)).filter(Boolean);
    allLogs.push(`[Mapper] Mapped ${financialsToUpsert.length} valid financial
periods.`);

    if (testMode) {
      const testRows = financialsToUpsert.slice(0, 5);
      allLogs.push(`[Test Mode] Displaying first 5 mapped rows:`);
      console.log(JSON.stringify(testRows, null, 2));
      testRows.forEach(row => allLogs.push(`  - ${row.period} (${row.freq}):
Revenue=${row.revenue ?? 'N/A'}`));
      allLogs.push(`[Test Mode] Halting before DB write.`);
      return new Response(JSON.stringify({ success: true, testMode: true, logs:
allLogs, testRows }), { status: 200, headers: { ...corsHeaders(), 'Content-
Type': 'application/json' } });
    }

    // 2. Idempotent Upsert to FinancialsHistory
```

```
    let inserted = 0, updated = 0, skipped = 0;
    for (const row of financialsToUpsert) {
      const existing = await base44.entities.FinancialsHistory.filter({ ticker:
row.ticker, period: row.period, freq: row.freq });

      if (existing.length === 0) {
        await base44.entities.FinancialsHistory.create(row);
        inserted++;
      } else {
        const updates = {};
        for (const key in row) {
          if (row[key] !== null && existing[0][key] === null) {
            updates[key] = row[key];
          }
        }
        if (Object.keys(updates).length > 0) {
          await base44.entities.FinancialsHistory.update(existing[0].id,
updates);
          updated++;
        } else {
          skipped++;
        }
      }
    }
    allLogs.push(`[DB] FinancialsHistory: ${inserted} inserted, ${updated}
updated, ${skipped} skipped.`);

    // 3. (Optional) Update Metrics Snapshot - we can add this back if needed
    allLogs.push(`[Metrics] Snapshot update deferred to
computeAndSavePriceMetrics.`);

    return new Response(JSON.stringify({
      success: true,
      ticker,
      financials_summary: { received: filings.length, mapped:
financialsToUpsert.length, inserted, updated, skipped },
      logs: allLogs
    }), { status: 200, headers: { ...corsHeaders(), 'Content-Type':
'application/json' } });

  } catch (error) {
    console.error('Sync fundamentals error:', error);
    return new Response(JSON.stringify({ error: true, message: error.message,
logs: [error.message] }), { status: 500, headers: { ...corsHeaders(), 'Content-
Type': 'application/json' } });
  }
});

// FILE: functions/syncHistoryPricesYahoo.ts
// ============================================================

import { createClientFromRequest } from 'npm:@base44/sdk@0.5.0';

// --- START: SHARED YAHOO SESSION MANAGEMENT ---
// Reuse the same session management from syncRecentPricesYahoo
let yahooSession = {
  cookie: null,
  crumb: null,
  acquiredAt: 0,
  inFlight: null
};

const SESSION_TTL_MS = 60 * 60 * 1000; // 60 minutes
let tokenBucket = {
```

```
  tokens: 2,
  lastRefill: Date.now(),
  refillRate: 1 / 600, // 1 token per 600ms
  maxTokens: 2
};

function refillTokens() {
  const now = Date.now();
  const elapsed = now - tokenBucket.lastRefill;
  const tokensToAdd = Math.floor(elapsed * tokenBucket.refillRate);

  if (tokensToAdd > 0) {
    tokenBucket.tokens = Math.min(tokenBucket.maxTokens, tokenBucket.tokens +
tokensToAdd);
    tokenBucket.lastRefill = now;
  }
}

async function waitForToken(log) {
  while (tokenBucket.tokens < 1) {
    refillTokens();
    if (tokenBucket.tokens < 1) {
      log(`[History][RateLimit] waiting for token, current=$
{tokenBucket.tokens.toFixed(2)}`);
      await new Promise(resolve => setTimeout(resolve, 100));
    }
  }
  tokenBucket.tokens -= 1;
  refillTokens();
}

function sessionIsFresh() {
  return yahooSession.cookie &&
         yahooSession.crumb &&
         (Date.now() - yahooSession.acquiredAt) < SESSION_TTL_MS;
}

function addJitter(delayMs) {
  const jitter = 0.2 * delayMs * (Math.random() - 0.5) * 2;
  return Math.round(delayMs + jitter);
}

async function fetchCrumbWithBackoff(log) {
    const maxAttempts = 6;
    const baseDelayMs = 1000;

    for (let attempt = 1; attempt <= maxAttempts; attempt++) {
        try {
            log(`[History][Stage:crumb] acquiring crumb attempt ${attempt}/$
{maxAttempts}`);

            await waitForToken(log);

            const cookieResponse = await fetch('https://fc.yahoo.com/', {
                headers: {
                    'User-Agent': 'Mozilla/5.0 (Macintosh; Intel Mac OS X
10_15_7) AppleWebKit/537.36'
                }
            });

            if (cookieResponse.status === 429 || cookieResponse.status === 999
|| cookieResponse.status >= 500) {
                throw new Error(`Cookie fetch failed: status=$
{cookieResponse.status}`);
```

```
            }

            const cookieHeader = cookieResponse.headers.get('set-cookie');
            if (!cookieHeader) {
                throw new Error("No session cookie received from fc.yahoo.com");
            }

            await waitForToken(log);
            const crumbResponse = await
fetch('https://query1.finance.yahoo.com/v1/test/getcrumb', {
                headers: {
                    'cookie': cookieHeader,
                    'User-Agent': 'Mozilla/5.0 (Macintosh; Intel OS X 10_15_7)
AppleWebKit/537.36'
                }
            });

            if (crumbResponse.status === 429 || crumbResponse.status === 999 ||
crumbResponse.status >= 500) {
                throw new Error(`Crumb fetch failed: status=$
{crumbResponse.status}`);
            }

            if (!crumbResponse.ok) {
                throw new Error(`Crumb fetch failed: status=$
{crumbResponse.status}`);
            }

            const crumb = await crumbResponse.text();
            if (!crumb || crumb.length < 5) {
                throw new Error("Invalid crumb received");
            }

            yahooSession.cookie = cookieHeader;
            yahooSession.crumb = crumb;
            yahooSession.acquiredAt = Date.now();

            log("[History][Stage:session] cookie OK");
            log("[History][Stage:session] crumb OK");
            return;

        } catch (error) {
            const isRetriable = error.message.includes('status=429') ||
                                error.message.includes('status=999') ||
                                error.message.includes('status=5') ||
                                error.message.includes('Invalid crumb') ||
                                error.message.includes('No session cookie');

            if (!isRetriable || attempt === maxAttempts) {
                throw {
                    stage: 'session',
                    code: 'session_error',
                    message: 'Session bootstrap failed after retries',
                    cause: error.message.slice(0, 120)
                };
            }

            const baseDelay = baseDelayMs * Math.pow(1.5, attempt - 1);
            const delay = addJitter(baseDelay);

            log(`[History][Retry] stage=session attempt=${attempt}/$
{maxAttempts} â†’ backing off ${(delay/1000).toFixed(1)}s`);
            await new Promise(resolve => setTimeout(resolve, delay));
        }
```

```
    }
}

async function ensureSession(log) {
    log("[History][Stage:session] getSession start");

    if (sessionIsFresh()) {
        log("[History][Stage:session] cache hit");
        return;
    }

    log("[History][Stage:session] cache miss");

    if (yahooSession.inFlight) {
        log("[History][Stage:session] waiting for in-flight session
acquisition");
        await yahooSession.inFlight;
        return;
    }

    yahooSession.inFlight = fetchCrumbWithBackoff(log);

    try {
        await yahooSession.inFlight;
    } finally {
        yahooSession.inFlight = null;
    }
}

async function signedFetch(path, params, log) {
    await ensureSession(log);

    const maxAttempts = 3;
    let lastError = null;

    for (let attempt = 1; attempt <= maxAttempts; attempt++) {
        try {
            const url = new URL(`https://query1.finance.yahoo.com${path}`);
            const finalParams = { ...params, crumb: yahooSession.crumb };
            Object.entries(finalParams).forEach(([key, value]) =>
url.searchParams.append(key, value));

            await waitForToken(log);

            const response = await fetch(url.toString(), {
                headers: {
                    'cookie': yahooSession.cookie,
                    'User-Agent': 'Mozilla/5.0 (Macintosh; Intel Mac OS X
10_15_7) AppleWebKit/537.36'
                }
            });

            const bodyText = await response.text();
            log(`[History][Stage:fetch] <- ${path} status=${response.status}
len=${bodyText.length}`);

            if (response.status === 401 || response.status === 403) {
                if (attempt < maxAttempts) {
                    log(`[History][Stage:session] session invalid, invalidating
cache for retry`);
                    yahooSession.cookie = null;
                    yahooSession.crumb = null;
                    yahooSession.acquiredAt = 0;
                    await ensureSession(log);
```

```
                    continue;
                }
            }

            if (response.status === 429 || response.status === 999 ||
response.status >= 500) {
                if (attempt < maxAttempts) {
                    const baseDelay = 700 * Math.pow(2, attempt - 1);
                    const delay = addJitter(baseDelay);
                    log(`[History][Retry] stage=fetch attempt=${attempt}/$
{maxAttempts} code=${response.status} â†’ backing off $
{(delay/1000).toFixed(1)}s`);
                    await new Promise(res => setTimeout(res, delay));
                    continue;
                }
            }

            if (!response.ok) {
                throw new Error(`Yahoo fetch failed with status $
{response.status}`);
            }

            try {
                const json = JSON.parse(bodyText);
                return { ok: true, json, status: response.status };
            } catch (parseError) {
                return { ok: false, parseError: true, status: response.status,
text: bodyText };
            }

        } catch (error) {
            lastError = error.message;
            if (attempt < maxAttempts) {
                const delay = addJitter(1000 * attempt);
                log(`[History][Retry] stage=fetch attempt=${attempt}/$
{maxAttempts} code=network_error â†’ backing off ${(delay/1000).toFixed(1)}s`);
                await new Promise(res => setTimeout(res, delay));
                continue;
            }
        }
    }

    throw {
        stage: 'fetch',
        code: 'fetch_failed',
        message: 'Network error after retries',
        cause: lastError?.slice(0, 120) || 'unknown'
    };
}

// --- DATA NORMALIZATION ---
const toYMD = (d) => d ? new Date(d).toISOString().slice(0, 10) : null;
const numOrNull = (v) => (v === null || v === undefined || Number.isNaN(+v)) ?
null : +v;

function normalizePrices(ticker, priceData, log) {
    log(`[History][Stage:parse] normalizing 5-year price data`);
    const asOfDate = toYMD(new Date());
    const timestamps = Array.isArray(priceData?.timestamp) ? priceData.timestamp
: [];
    if (timestamps.length === 0) return [];

    log(`[History][Stage:parse] processing ${timestamps.length} price timestamps
for 5-year history`);
```

```
    const quotes = priceData.indicators?.quote?.[0];
    const adjCloses = priceData.indicators?.adjclose?.[0]?.adjclose;

    if (!quotes || !adjCloses) {
        log(`[History][Stage:parse] missing quote or adjclose data in Yahoo
response`);
        return [];
    }

    const normalized = timestamps.map((ts, i) => ({
        ticker,
        date: toYMD(new Date(ts * 1000)),
        open: numOrNull(quotes.open?.[i]),
        high: numOrNull(quotes.high?.[i]),
        low: numOrNull(quotes.low?.[i]),
        close: numOrNull(quotes.close?.[i]),        // Split-adjusted only
(Yahoo "Close")
        close_adj: numOrNull(adjCloses[i]),         // Dividend & split-adjusted
(Yahoo "Adj Close")
        volume: numOrNull(quotes.volume?.[i]),
        source: 'yahoo',
        as_of_date: asOfDate,
    })).filter(p => p.date && p.close_adj !== null && p.close !== null); //
Require both prices

    log(`[History][Stage:parse] normalized ${normalized.length} valid price
points over 5 years`);
    return normalized;
}

// --- DATABASE UPSERT WITH BATCHING ---
async function upsertHistoricalPrices(base44, prices, log) {
    const safePrices = Array.isArray(prices) ? prices : [];
    if (safePrices.length === 0) {
        log("[History][Stage:db] No historical price data to upsert.");
        return { inserted: 0, updated: 0, skipped: 0 };
    }

    log("[History][Stage:db] Starting 5-year historical price upsert with
service role...");

    const serviceClient = base44.asServiceRole;

    try {
        const testQuery = await serviceClient.entities.PricesHistory.filter({},
null, 1);
        log(`[History][Stage:db] Database connectivity test: ${testQuery ?
'OK' : 'FAILED'}`);
    } catch (dbTestError) {
        log(`[History][Stage:db] Service role test failed: $
{dbTestError.message}`);
        throw {
            stage: 'db',
            code: 'SERVICE_ROLE_ACCESS_FAILED',
            message: 'Service role database access failed during connectivity
test',
            cause: dbTestError.message
        };
    }

    const ticker = safePrices[0].ticker;
    const BATCH_SIZE = 50; // Larger batches for historical data
    const INTER_BATCH_DELAY = 100; // Shorter delay for bulk operations
```

```
    const batches = [];
    for (let i = 0; i < safePrices.length; i += BATCH_SIZE) {
        batches.push(safePrices.slice(i, i + BATCH_SIZE));
    }
    log(`[History][Stage:db] Processing ${batches.length} batches of max $
{BATCH_SIZE} rows each for 5-year history`);

    let totalInserted = 0, totalUpdated = 0, totalSkipped = 0;

    for (let i = 0; i < batches.length; i++) {
        const batch = batches[i];
        const dates = batch.map(p => p.date);

        try {
            const existing = await serviceClient.entities.PricesHistory.filter({
                ticker,
                date: { '$in': dates }
            });

            const safeExisting = Array.isArray(existing) ? existing : [];
            const existingMap = new Map(safeExisting.map(p => [p.date, p]));
            const toInsert = batch.filter(row => !existingMap.has(row.date));
            const toUpdate = batch.filter(row => existingMap.has(row.date));

            log(`[History][Stage:db] Batch ${i+1}: ${toInsert.length} to insert,
${toUpdate.length} to update`);

            if (toInsert.length > 0) {
                const maxRetries = 3;
                const retryDelays = [500, 1000, 2000];

                for (let attempt = 0; attempt < maxRetries; attempt++) {
                    try {
                        await
serviceClient.entities.PricesHistory.bulkCreate(toInsert);
                        totalInserted += toInsert.length;
                        log(`[History][Stage:db] Batch ${i+1} inserts
successful: ${toInsert.length} rows`);
                        break;

                    } catch (dbError) {
                        const status = dbError.response?.status ||
dbError.status || 500;

                        if (status === 401 || status === 403) {
                            log(`[History][Stage:db] âŒ UNAUTHORIZED ERROR ($
{status}) on batch ${i+1}`);
                            throw {
                                stage: 'db',
                                code: 'UNAUTHORIZED',
                                message: `Unauthorized at DB stage (${status});
check service role configuration.`,
                                cause: dbError.message,
                                batch: i + 1,
                                operation: 'bulkCreate'
                            };
                        }

                        if (attempt === maxRetries - 1) {
                            throw {
                                stage: 'db',
                                code: status || 'DB_WRITE_FAILED',
```

```
                                message: `DB write for batch ${i+1} failed after
${maxRetries} retries.`,
                                cause: dbError.message
                            };
                        } else {
                            await new Promise(resolve => setTimeout(resolve,
retryDelays[attempt]));
                        }
                    }
                }
            }

            // Handle updates
            for (const row of toUpdate) {
                const existingRow = existingMap.get(row.date);
                if (!existingRow) continue;

                try {
                    await
serviceClient.entities.PricesHistory.update(existingRow.id, row);
                    totalUpdated++;
                } catch (updateError) {
                    log(`[History][Stage:db] Update error for ${row.ticker} $
{row.date}: ${updateError.message}`);
                    totalSkipped++;
                }
            }

        } catch (batchError) {
            if (batchError.stage) {
                throw batchError;
            }

            log(`[History][Stage:db] Batch ${i+1} failed: $
{batchError.message}`);
            totalSkipped += batch.length;
        }

        // Inter-batch delay
        if (i < batches.length - 1) {
            await new Promise(resolve => setTimeout(resolve,
INTER_BATCH_DELAY));
        }
    }

    log(`[History][Stage:db] 5-year history complete. Total inserted: $
{totalInserted}, updated: ${totalUpdated}, skipped: ${totalSkipped}`);
    return { inserted: totalInserted, updated: totalUpdated, skipped:
totalSkipped };
}

// --- MAIN HANDLER ---
Deno.serve(async (req) => {
    const base44 = createClientFromRequest(req);
    const logs = [];
    const log = (msg) => logs.push(msg);

    let requestTicker = null;

    try {
        log(`[History][Stage:init] starting 5-year historical price sync`);

        if (!(await base44.auth.isAuthenticated())) {
```

```
            throw { stage: 'auth', code: 'unauthorized', message:
'Authentication required' };
        }

        const body = await req.json();
        requestTicker = body.ticker;
        const { ticker } = body;

        if (!ticker || typeof ticker !== 'string' || !/^[A-Z.-]+$/.test(ticker))
{
            throw { stage: 'validation', code: 'invalid_ticker', message:
'Invalid or missing ticker symbol.' };
        }

        log("[History] Using YahooSession for 5-year backfill");

        // Fetch 5 years of historical data instead of 1 month
        const urlPath = `/v8/finance/chart/${ticker}`;
        const params = {
            interval: '1d',
            events: 'div%2Csplits',
            range: '5y', // 5 years of data
            includeAdjustedClose: 'true'
        };

        const yahooFetchResult = await signedFetch(urlPath, params, log);

        if (!yahooFetchResult.ok || !yahooFetchResult.json) {
            throw {
                stage: 'fetch',
                code: 'api_error',
                message: `Yahoo API fetch failed with status $
{yahooFetchResult.status}.`,
                cause: yahooFetchResult.text?.slice(0, 120) || 'No response
body'
            };
        }

        const normalized = normalizePrices(ticker,
yahooFetchResult.json.chart.result[0], log);

        if (normalized.length === 0) {
            log("[History] No historical data to sync.");
            return new Response(JSON.stringify({
                ok: true,
                received: 0,
                inserted: 0,
                updated: 0,
                skipped: 0,
                logs
            }), { status: 200, headers: { 'Content-Type':
'application/json' } });
        }

        const dbOperationResult = await upsertHistoricalPrices(base44,
normalized, log);

        const dates = normalized.map(p => p.date).sort();
        const minDate = dates[0];
        const maxDate = dates[dates.length - 1];

        log(`[History SUCCESS] 5-year window ${minDate} â†’ ${maxDate}, fetched
${normalized.length}, inserted ${dbOperationResult.inserted}, updated $
{dbOperationResult.updated}, failed ${dbOperationResult.skipped}.`);
```

```
        return new Response(JSON.stringify({
            ok: true,
            received: normalized.length,
            inserted: dbOperationResult.inserted,
            updated: dbOperationResult.updated,
            skipped: dbOperationResult.skipped,
            date_range: { min: minDate, max: maxDate },
            logs
        }), {
            status: 200,
            headers: { 'Content-Type': 'application/json' }
        });

    } catch (error) {
        log(`[History ERROR] stage=${error.stage || 'unknown'} code=${error.code
|| 'unknown'} detail=${error.message}`);

        const errorPayload = {
            ok: false,
            stage: error.stage || 'unknown',
            message: error.message,
            code: error.code || 'unknown',
            cause: error.cause || null,
            logs
        };
        return new Response(JSON.stringify(errorPayload), {
            status: error.status || 500,
            headers: { 'Content-Type': 'application/json' }
        });
    }
});

// FILE: functions/syncRecentPricesYahoo.ts
// ============================================================


import { createClientFromRequest } from 'npm:@base44/sdk@0.5.0';

// --- START: SHARED CHART REQUEST BUILDER ---

function buildChartRequest(opts) {
    const { symbol, interval = '1d', events = 'div%2Csplits', range, period1,
period2 } = opts;

    if (!symbol) {
        throw new Error('Symbol is required for chart request');
    }

    const urlPath = `/v8/finance/chart/${symbol}`;
    const params = {
        interval,
        events, // Pre-encoded
        includeAdjustedClose: 'true'
    };

    // Time window: use either range OR period1/period2, never both
    if (period1 && period2) {
        params.period1 = period1.toString();
        params.period2 = period2.toString();
    } else if (range) {
        params.range = range;
    } else {
        throw new Error('Either range or period1/period2 must be specified');
```

```
    }

    // Build query string (already encoded once)
    const urlQuery = new URLSearchParams(params).toString();

    return { urlPath, urlQuery, params };
}

async function chartProbe(opts, log) {
    const { urlPath, urlQuery, params } = buildChartRequest(opts);

    try {
        await ensureSession(log);

        const url = new URL(`https://query1.finance.yahoo.com${urlPath}`);
        const finalParams = { ...params, crumb: yahooSession.crumb };
        Object.entries(finalParams).forEach(([key, value]) =>
url.searchParams.append(key, value));

        await waitForToken(log);

        const response = await fetch(url.toString(), {
            headers: {
                'cookie': yahooSession.cookie,
                'User-Agent': 'Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7)
AppleWebKit/537.36'
            }
        });

        const bodyText = await response.text();
        const contentType = response.headers.get('content-type') || 'unknown';
        let parsedOk = false;

        if (response.status !== 200) {
            log(`[ChartProbe] status=${response.status} path=${urlPath}
query=<safe: ${urlQuery}>`);
        }

        try {
            JSON.parse(bodyText);
            parsedOk = true;
        } catch {
            parsedOk = false;
        }

        return {
            status: response.status,
            contentType,
            len: bodyText.length,
            parsedOk,
            body: bodyText
        };

    } catch (error) {
        log(`[ChartProbe] ERROR: ${error.message}`);
        return {
            status: 0,
            contentType: 'error',
            len: 0,
            parsedOk: false,
            error: error.message
        };
    }
}
```

```
// --- END: SHARED CHART REQUEST BUILDER ---

// --- START: HARDENED YAHOO SESSION MANAGEMENT ---

// Module-level session cache with TTL and single-flight protection
let yahooSession = {
  cookie: null,
  crumb: null,
  acquiredAt: 0,
  inFlight: null // Promise to prevent concurrent crumb fetches
};

const SESSION_TTL_MS = 60 * 60 * 1000; // 60 minutes
const YAHOO_THROTTLE_MS = parseInt(Deno.env.get("YAHOO_THROTTLE_MS") || "1200");

// Concurrency guard for function entry
let activeRuns = new Map(); // ticker -> promise

// Token bucket rate limiter
let tokenBucket = {
  tokens: 2, // burst capacity
  lastRefill: Date.now(),
  refillRate: 1 / 600, // 1 token per 600ms
  maxTokens: 2
};

function refillTokens() {
  const now = Date.now();
  const elapsed = now - tokenBucket.lastRefill;
  const tokensToAdd = Math.floor(elapsed * tokenBucket.refillRate);

  if (tokensToAdd > 0) {
    tokenBucket.tokens = Math.min(tokenBucket.maxTokens, tokenBucket.tokens +
tokensToAdd);
    tokenBucket.lastRefill = now;
  }
}

async function waitForToken(log) {
  while (tokenBucket.tokens < 1) {
    refillTokens();
    if (tokenBucket.tokens < 1) {
      log(`[Recent][RateLimit] waiting for token, current=$
{tokenBucket.tokens.toFixed(2)}`);
      await new Promise(resolve => setTimeout(resolve, 100));
    }
  }
  tokenBucket.tokens -= 1;
  refillTokens(); // Update after consumption
}

function sessionIsFresh() {
  return yahooSession.cookie &&
         yahooSession.crumb &&
         (Date.now() - yahooSession.acquiredAt) < SESSION_TTL_MS;
}

// Add jitter to delay (Â±20%)
function addJitter(delayMs) {
  const jitter = 0.2 * delayMs * (Math.random() - 0.5) * 2;
  return Math.round(delayMs + jitter);
}
```

```
async function parseJsonOrHtml(response, log) {
    const text = await response.text();
    if (text.startsWith('<') && (text.includes('<!DOCTYPE html>') ||
text.includes('<html'))) {
        log(`[YahooFetch] Non-JSON (HTML) status=${response.status}`);
        return { ok: false, isHtml: true, status: response.status, text };
    }
    try {
        const json = JSON.parse(text);
        log(`[YahooFetch] json OK`);
        return { ok: true, isJson: true, json, status: response.status };
    } catch (e) {
        log(`[YahooFetch] Non-JSON response detected (Parse Error). status=$
{response.status}`);
        return { ok: false, parseError: true, status: response.status, text };
    }
}

async function fetchCrumbWithBackoff(log) {
    const maxAttempts = 6;
    const baseDelayMs = 1000;

    for (let attempt = 1; attempt <= maxAttempts; attempt++) {
        try {
            log(`[Recent][Stage:crumb] acquiring crumb attempt ${attempt}/$
{maxAttempts}`);

            // Token bucket rate limiting
            await waitForToken(log);

            // Step A: Get Cookie from stable Yahoo Finance page
            const cookieResponse = await fetch('https://fc.yahoo.com/', {
                headers: {
                    'User-Agent': 'Mozilla/5.0 (Macintosh; Intel Mac OS X
10_15_7) AppleWebKit/537.36'
                }
            });

            if (cookieResponse.status === 429 || cookieResponse.status === 999
|| cookieResponse.status >= 500) {
                throw new Error(`Cookie fetch failed: status=$
{cookieResponse.status}`);
            }

            const cookieHeader = cookieResponse.headers.get('set-cookie');
            if (!cookieHeader) {
                throw new Error("No session cookie received from fc.yahoo.com");
            }

            // Step B: Get Crumb
            await waitForToken(log);
            const crumbResponse = await
fetch('https://query1.finance.yahoo.com/v1/test/getcrumb', {
                headers: {
                    'cookie': cookieHeader,
                    'User-Agent': 'Mozilla/5.0 (Macintosh; Intel OS X 10_15_7)
AppleWebKit/537.36'
                }
            });

            if (crumbResponse.status === 429 || crumbResponse.status === 999 ||
crumbResponse.status >= 500) {
                throw new Error(`Crumb fetch failed: status=$
{crumbResponse.status}`);
```

```
            }

            if (!crumbResponse.ok) {
                throw new Error(`Crumb fetch failed: status=$
{crumbResponse.status}`);
            }

            const crumb = await crumbResponse.text();
            if (!crumb || crumb.length < 5) {
                throw new Error("Invalid crumb received");
            }

            // Success - update session
            yahooSession.cookie = cookieHeader;
            yahooSession.crumb = crumb;
            yahooSession.acquiredAt = Date.now();

            log("[Recent][Stage:session] cookie OK");
            log("[Recent][Stage:session] crumb OK");
            return;

        } catch (error) {
            const isRetriable = error.message.includes('status=429') ||
                                error.message.includes('status=999') ||
                                error.message.includes('status=5') ||
                                error.message.includes('Invalid crumb') ||
                                error.message.includes('No session cookie');

            if (!isRetriable || attempt === maxAttempts) {
                const code = error.message.includes('429') ||
error.message.includes('999') ? 'rate_limited' :
                           error.message.includes('5') ? 'server_error' :
'session_error';
                throw {
                    stage: 'session',
                    code: code,
                    message: 'Session bootstrap failed after retries',
                    cause: error.message.slice(0, 120)
                };
            }

            // Calculate backoff with jitter
            const baseDelay = baseDelayMs * Math.pow(1.5, attempt - 1);
            const delay = addJitter(baseDelay);

            log(`[Recent][Retry] stage=session attempt=${attempt}/${maxAttempts}
code=${error.message.includes('429') || error.message.includes('999') ? '429' :
'5xx'} â†’ backing off ${(delay/1000).toFixed(1)}s`);
            await new Promise(resolve => setTimeout(resolve, delay));
        }
    }
}

async function ensureSession(log) {
    log("[Recent][Stage:session] getSession start");

    // Check if session is still fresh
    if (sessionIsFresh()) {
        log("[Recent][Stage:session] cache hit");
        return;
    }

    log("[Recent][Stage:session] cache miss");
```

```
    // If a fetch is already in progress, wait for it
    if (yahooSession.inFlight) {
        log("[Recent][Stage:session] waiting for in-flight session
acquisition");
        await yahooSession.inFlight;
        return;
    }

    // Start new session acquisition
    yahooSession.inFlight = fetchCrumbWithBackoff(log);

    try {
        await yahooSession.inFlight;
    } finally {
        yahooSession.inFlight = null;
    }
}

async function signedFetch(path, params, log) {
    await ensureSession(log);

    const maxAttempts = 3;
    let lastError = null;

    for (let attempt = 1; attempt <= maxAttempts; attempt++) {
        try {
            const url = new URL(`https://query1.finance.yahoo.com${path}`);
            const finalParams = { ...params, crumb: yahooSession.crumb };
            Object.entries(finalParams).forEach(([key, value]) =>
url.searchParams.append(key, value));

            const safeQuery = new URLSearchParams(params).toString();
            log(`[Recent][Stage:fetch] GET ${path}?${safeQuery} (attempt $
{attempt})`);

            // Token bucket rate limiting
            await waitForToken(log);

            const response = await fetch(url.toString(), {
                headers: {
                    'cookie': yahooSession.cookie,
                    'User-Agent': 'Mozilla/5.0 (Macintosh; Intel Mac OS X
10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/114.0.0.0 Safari/537.36'
                }
            });

            const bodyText = await response.text();
            log(`[Recent][Stage:fetch] <- ${path} status=${response.status}
len=${bodyText.length}`);

            // Handle authentication errors
            if (response.status === 401 || response.status === 403) {
                if (attempt < maxAttempts) {
                    log(`[Recent][Stage:session] session invalid, invalidating
cache for retry`);
                    yahooSession.cookie = null;
                    yahooSession.crumb = null;
                    yahooSession.acquiredAt = 0;
                    await ensureSession(log);
                    continue;
                }
                throw {
                    stage: 'fetch',
                    code: 'session_error',
```

```
                    message: 'Authentication failed after session refresh',
                    cause: `status=${response.status}`
                };
            }

            // Handle rate limiting and server errors
            if (response.status === 429 || response.status === 999 ||
response.status >= 500) {
                if (attempt < maxAttempts) {
                    const baseDelay = 700 * Math.pow(2, attempt - 1); // 700ms,
1400ms, 2800ms
                    const delay = addJitter(baseDelay);
                    log(`[Recent][Retry] stage=fetch attempt=${attempt}/$
{maxAttempts} code=${response.status} â†’ backing off $
{(delay/1000).toFixed(1)}s`);
                    await new Promise(res => setTimeout(res, delay));
                    continue;
                }
                const code = (response.status === 429 || response.status ===
999) ? 'rate_limited' : 'server_error';
                throw {
                    stage: 'fetch',
                    code: code,
                    message: `${code === 'rate_limited' ? 'Rate limited' :
'Server error'} after retries`,
                    cause: `status=${response.status} len=${bodyText.length}`
                };
            }

            // Handle client errors (don't retry)
            if (response.status >= 400 && response.status < 500) {
                throw {
                    stage: 'fetch',
                    code: 'client_error',
                    message: 'Client error from Yahoo API',
                    cause: `status=${response.status} body=${bodyText.slice(0,
120)}`
                };
            }

            // Parse response
            log(`[Recent][Stage:parse] parsing response`);
            const mockResponse = { status: response.status, text: () =>
Promise.resolve(bodyText) };
            const parsed = await parseJsonOrHtml(mockResponse, log);

            if (parsed.isHtml && attempt < maxAttempts) {
                lastError = "Received HTML instead of JSON";
                log(`[Recent][Retry] stage=parse attempt=${attempt}/$
{maxAttempts} code=html_response`);
                continue;
            }

            if (!parsed.ok) {
                throw {
                    stage: 'parse',
                    code: parsed.isHtml ? 'html_response' : 'parse_error',
                    message: parsed.isHtml ? 'Received HTML instead of JSON' :
'JSON parse error',
                    cause: parsed.text?.slice(0, 120) || 'unknown'
                };
            }

            return parsed;
```

```
        } catch (error) {
            if (error.stage) {
                // Structured error, re-throw
                throw error;
            }

            // Network or unexpected error
            lastError = error.message;
            if (attempt < maxAttempts) {
                const delay = addJitter(1000 * attempt);
                log(`[Recent][Retry] stage=fetch attempt=${attempt}/$
{maxAttempts} code=network_error â†’ backing off ${(delay/1000).toFixed(1)}s`);
                await new Promise(res => setTimeout(res, delay));
                continue;
            }
        }
    }

    throw {
        stage: 'fetch',
        code: 'network_error',
        message: 'Network error after retries',
        cause: lastError?.slice(0, 120) || 'unknown'
    };
}

// --- DATA NORMALIZATION & DB HELPERS ---
const toYMD = (d) => d ? new Date(d).toISOString().slice(0, 10) : null;
const numOrNull = (v) => (v === null || v === undefined || Number.isNaN(+v)) ?
null : +v;

function normalizePrices(ticker, priceData, log) {
    log(`[Recent][Stage:parse] normalizing price data`);
    const asOfDate = toYMD(new Date());
    const timestamps = Array.isArray(priceData?.timestamp) ? priceData.timestamp
: [];
    if (timestamps.length === 0) return [];

    log(`[Recent][Stage:parse] processing ${timestamps.length} price
timestamps`);

    const quotes = priceData.indicators?.quote?.[0];
    const adjCloses = priceData.indicators?.adjclose?.[0]?.adjclose;

    if (!quotes || !adjCloses) {
        log(`[Recent][Stage:parse] missing quote or adjclose data in Yahoo
response`);
        return [];
    }

    const normalized = timestamps.map((ts, i) => ({
        ticker,
        date: toYMD(new Date(ts * 1000)),
        open: numOrNull(quotes.open?.[i]),
        high: numOrNull(quotes.high?.[i]),
        low: numOrNull(quotes.low?.[i]),
        close: numOrNull(quotes.close?.[i]),        // Split-adjusted only
(Yahoo "Close")
        close_adj: numOrNull(adjCloses[i]),         // Dividend & split-adjusted
(Yahoo "Adj Close")
        volume: numOrNull(quotes.volume?.[i]),
        source: 'yahoo',
        as_of_date: asOfDate,
```

```
    })).filter(p => p.date && p.close_adj !== null && p.close !== null); //
Require both prices

    log(`[Recent][Stage:parse] normalized ${normalized.length} valid price
points`);
    return normalized;
}

// --- Enhanced DB Write Logic with Comprehensive Auth Diagnostics ---
async function upsertPricesWithRetry(base44, prices, log) {
    const safePrices = Array.isArray(prices) ? prices : [];
    if (safePrices.length === 0) {
        log("[Recent][Stage:db] No price data to upsert.");
        return { inserted: 0, updated: 0, skipped: 0 };
    }

    // AUTH DIAGNOSTICS
    log("[Recent][Stage:db] Starting database operations with comprehensive auth
diagnostics...");

    // Initialize service client early for diagnostics and subsequent use
    const serviceClient = base44.asServiceRole;

    try {
        // Quick connectivity test
        log("[Recent][Stage:db] Using service role client for database
operations");
        const testQuery = await serviceClient.entities.PricesHistory.filter({},
null, 1);
        log(`[Recent][Stage:db] Database connectivity test: ${testQuery ?
'CONNECTED' : 'FAILED'}`);

    } catch (authTestError) {
        log(`[Recent][Stage:db] Service role test failed: $
{authTestError.message}`);
        throw {
            stage: 'db',
            code: 'SERVICE_ROLE_ACCESS_FAILED',
            message: 'Service role database access failed during connectivity
test',
            cause: authTestError.message
        };
    }

    const ticker = safePrices[0].ticker;
    const BATCH_SIZE = 25; // Reduced batch size for better error isolation
    const INTER_BATCH_DELAY = 200; // 200ms between batches

    const batches = [];
    for (let i = 0; i < safePrices.length; i += BATCH_SIZE) {
        batches.push(safePrices.slice(i, i + BATCH_SIZE));
    }
    log(`[Recent][Stage:db] Processing ${batches.length} batches of max $
{BATCH_SIZE} rows each`);

    let totalInserted = 0, totalUpdated = 0, totalSkipped = 0;

    for (let i = 0; i < batches.length; i++) {
        const batch = batches[i];
        const dates = batch.map(p => p.date);

        log(`[Recent][Stage:db] Processing batch ${i+1}/${batches.length} with $
{batch.length} rows`);
```

```
        try {
            // Check for existing records using service role
            const existing = await serviceClient.entities.PricesHistory.filter({
                ticker,
                date: { '$in': dates }
            });

            const safeExisting = Array.isArray(existing) ? existing : [];
            const existingMap = new Map(safeExisting.map(p => [p.date, p]));
            const toInsert = batch.filter(row => !existingMap.has(row.date));
            const toUpdate = batch.filter(row => existingMap.has(row.date));

            log(`[Recent][Stage:db] Batch ${i+1}: ${toInsert.length} to insert,
${toUpdate.length} to update`);

            // Handle inserts
            if (toInsert.length > 0) {
                const maxRetries = 3;
                const retryDelays = [1500, 3000, 5000];

                for (let attempt = 0; attempt < maxRetries; attempt++) {
                    try {
                        await
serviceClient.entities.PricesHistory.bulkCreate(toInsert);
                        totalInserted += toInsert.length;
                        log(`[Recent][Stage:db] Batch ${i+1} inserts successful:
${toInsert.length} rows`);
                        break; // Success

                    } catch (dbError) {
                        const status = dbError.response?.status ||
dbError.status || 500;
                        log(`[Recent][Stage:db] Insert error on batch ${i+1},
attempt ${attempt+1}: status=${status}, message=${dbError.message}`);

                        if (status === 429) {
                            // Rate limit - check Retry-After header
                            const retryAfter =
dbError.response?.headers?.get('retry-after');
                            const delay = retryAfter ? parseInt(retryAfter) *
1000 : retryDelays[attempt];

                            if (attempt < maxRetries - 1) {
                                log(`[Recent][Stage:db] Rate limit (429) on
batch ${i+1} - retrying in ${delay}ms`);
                                await new Promise(resolve => setTimeout(resolve,
delay));

                                continue;
                            }
                        } else if (status === 401 || status === 403) {
                            // Authentication error - provide detailed
diagnostics
                            log(`[Recent][Stage:db] âŒ UNAUTHORIZED ERROR ($
{status}) on batch ${i+1}`);
                            log(`[Recent][Stage:db] âŒ Using service role
client: ${!!serviceClient}`);
                            log(`[Recent][Stage:db] âŒ Error message: $
{dbError.message}`);
                            log(`[Recent][Stage:db] âŒ This indicates RLS policy
or service role configuration issues`);

                            throw {
                                stage: 'db',
                                code: 'UNAUTHORIZED',
```

```
                                message: `Unauthorized at DB stage (${status});
check service key / role / RLS policy for PricesHistory table.`,
                                cause: dbError.message,
                                batch: i + 1,
                                operation: 'bulkCreate'
                            };
                        }

                        // Final attempt failed
                        if (attempt === maxRetries - 1) {
                            throw {
                                stage: 'db',
                                code: status || 'DB_WRITE_FAILED',
                                message: `DB write for batch ${i+1} failed after
${maxRetries} retries.`,
                                cause: dbError.message,
                                resumeHint: `Failed at batch ${i+1}/$
{batches.length}. ${totalInserted} rows inserted so far.`
                            };
                        } else {
                            await new Promise(resolve => setTimeout(resolve,
retryDelays[attempt]));
                        }
                    }
                }
            }

            // Handle updates (individual updates for better error isolation)
            for (const row of toUpdate) {
                const existingRow = existingMap.get(row.date);
                if (!existingRow) continue;

                try {
                    await
serviceClient.entities.PricesHistory.update(existingRow.id, row);
                    totalUpdated++;
                } catch (updateError) {
                    log(`[Recent][Stage:db] Update error for ${row.ticker} $
{row.date}: ${updateError.message}`);
                    totalSkipped++;
                }
            }

        } catch (batchError) {
            // If this is already a structured error, re-throw it
            if (batchError.stage) {
                throw batchError;
            }

            // Otherwise, wrap it
            log(`[Recent][Stage:db] Batch ${i+1} failed: $
{batchError.message}`);
            totalSkipped += batch.length;
        }

        // Inter-batch delay for rate limiting
        if (i < batches.length - 1) {
            await new Promise(resolve => setTimeout(resolve,
INTER_BATCH_DELAY));
        }
    }

    log(`[Recent][Stage:db] All batches complete. Total inserted: $
{totalInserted}, updated: ${totalUpdated}, skipped: ${totalSkipped}`);
```

```
    return { inserted: totalInserted, updated: totalUpdated, skipped:
totalSkipped };
}

// --- MAIN HANDLER ---
Deno.serve(async (req) => {
    const base44 = createClientFromRequest(req);
    const logs = [];
    const log = (msg) => logs.push(msg);

    let requestTicker = null; // Declare here to be accessible in finally

    try {
        log(`[Recent][Stage:init] starting recent price sync`);

        if (!(await base44.auth.isAuthenticated())) {
            throw { stage: 'auth', code: 'unauthorized', message:
'Authentication required' };
        }

        const body = await req.json(); // Consume request body once
        requestTicker = body.ticker; // Store ticker
        const { ticker } = body;

        if (!ticker || typeof ticker !== 'string' || !/^[A-Z.-]+$/.test(ticker))
{
            throw { stage: 'validation', code: 'invalid_ticker', message:
'Invalid or missing ticker symbol.' };
        }

        // Concurrency guard
        if (activeRuns.has(ticker)) {
            log(`[Recent][Guard] coalesced concurrent run for ${ticker}`);
            await activeRuns.get(ticker);
            return new Response(JSON.stringify({ ok: true, received: 0,
inserted: 0, updated: 0, skipped: 0, logs }), { status: 200 });
        }

        log(`[Recent][Guard] run locked for ${ticker}`);

        const runPromise = (async () => {
            log("[Recent] Using YahooSession v1");

            // USE SHARED CHART REQUEST BUILDER
            const { urlPath, params } = buildChartRequest({
                symbol: ticker,
                interval: '1d',
                events: 'div%2Csplits',
                range: '1mo'
            });

            const yahooFetchResult = await signedFetch(urlPath, params, log);

            if (!yahooFetchResult.ok || !yahooFetchResult.isJson) {
                throw { stage: 'fetch', code: 'api_error', message: `Yahoo API
fetch failed with status ${yahooFetchResult.status}.`, cause:
yahooFetchResult.text?.slice(0, 120) || 'No response body' };
            }

            log("[Recent] Using signedFetch + crumb session");

            const normalized = normalizePrices(ticker,
yahooFetchResult.json.chart.result[0], log);
```

```
            if (normalized.length === 0) {
                log("[Recent] No new or changed data to sync.");
                return { ok: true, received: 0, inserted: 0, updated: 0,
skipped: 0, logs };
            }

            // Stage: db (use new robust upsert)
            const dbOperationResult = await upsertPricesWithRetry(base44,
normalized, log);

            const dates = normalized.map(p => p.date).sort();
            const minDate = dates[0];
            const maxDate = dates[dates.length - 1];

            log(`[Recent SUCCESS] window ${minDate} â†’ ${maxDate}, fetched $
{normalized.length}, inserted ${dbOperationResult.inserted}, updated $
{dbOperationResult.updated}, failed ${dbOperationResult.skipped}.`);

            return {
                ok: true,
                received: normalized.length,
                inserted: dbOperationResult.inserted,
                updated: dbOperationResult.updated,
                skipped: dbOperationResult.skipped,
                logs
            };
        })();

        activeRuns.set(ticker, runPromise);
        const finalRunResult = await runPromise;
        return new Response(JSON.stringify(finalRunResult), { status: 200 });

    } catch (error) {
        log(`[Recent ERROR] stage=${error.stage || 'unknown'} code=${error.code
|| 'unknown'} detail=${error.message}`);

        const errorPayload = {
            ok: false,
            stage: error.stage || 'unknown',
            message: error.message,
            code: error.code || 'unknown',
            cause: error.cause || null,
            logs
        };
        return new Response(JSON.stringify(errorPayload), { status: error.status
|| 500, headers: { 'Content-Type': 'application/json' }});
    } finally {
        if (requestTicker) { // Use the stored ticker
            activeRuns.delete(requestTicker);
            log(`[Recent][Guard] run unlocked for ${requestTicker}`);
        }
    }
});


// FILE: functions/yahooFundamentals.ts
// ============================================================


import { createClientFromRequest } from 'npm:@base44/sdk@0.5.0';

function corsHeaders() {
  return {
    "Access-Control-Allow-Origin": "*",
```

```
    "Access-Control-Allow-Methods": "POST, GET, OPTIONS",
    "Access-Control-Allow-Headers": "Content-Type, Authorization",
    "Access-Control-Allow-Credentials": "true",
    "Content-Type": "application/json",
  };
}

function nz(value) {
    // Safely get raw number, return null if not a finite number
    return Number.isFinite(value?.raw) ? value.raw : null;
}

Deno.serve(async (req) => {
  if (req.method === 'OPTIONS') {
    return new Response(null, { status: 204, headers: corsHeaders() });
  }

  let ticker = "";
  try {
    if (req.method === 'POST') {
      const body = await req.json();
      ticker = body.ticker;
    } else {
      ticker = new URL(req.url).searchParams.get('ticker');
    }
  } catch {
    return new Response(JSON.stringify({ ok: false, status: 400, reason:
'Invalid request body' }), { status: 400, headers: corsHeaders() });
  }


  if (!ticker) {
    return new Response(JSON.stringify({ ok: false, status: 400, reason: 'Ticker
required' }), { status: 400, headers: corsHeaders() });
  }

  // Added console.log as per outline
  console.log(`[Yahoo] Fetching fundamentals for ${ticker}`);

  const modules =
"price,summaryDetail,defaultKeyStatistics,financialData,calendarEvents,assetProf
ile,earnings";
  const url = `https://query2.finance.yahoo.com/v10/finance/quoteSummary/$
{ticker}?modules=${modules}`;

  const maxRetries = 2; // Initial try + 1 retry
  const backoff = 400; // ms
  const timeout = 8000; // 8 seconds

  for (let attempt = 1; attempt <= maxRetries; attempt++) {
    const controller = new AbortController();
    const timeoutId = setTimeout(() => controller.abort(), timeout);

    try {
      const resp = await fetch(url, { signal: controller.signal });
      clearTimeout(timeoutId);

      if (!resp.ok) {
        if (resp.status >= 500 && attempt < maxRetries) {
          await new Promise(r => setTimeout(r, backoff));
          continue; // Retry on 5xx errors
        }
```

```
        return new Response(JSON.stringify({ ok: false, status: resp.status,
reason: `HTTP error: ${resp.statusText}`, ticker }), { status: 200, headers:
corsHeaders() });
      }

      const data = await resp.json();
      const summary = data?.quoteSummary?.result?.[0];

      if (!summary) {
        return new Response(JSON.stringify({ ok: false, status: 404, reason: "No
summary data in response", ticker }), { status: 200, headers: corsHeaders() });
      }

      const fields = {
          pe_ttm: nz(summary.summaryDetail?.trailingPE) ??
nz(summary.defaultKeyStatistics?.trailingPE),
          pe_fwd: nz(summary.summaryDetail?.forwardPE) ??
nz(summary.financialData?.forwardPE),
          peg_5y: nz(summary.defaultKeyStatistics?.pegRatio),
          ev_ebitda: nz(summary.defaultKeyStatistics?.enterpriseToEbitda),
          div_yield_pct: (nz(summary.summaryDetail?.dividendYield) ?? 0) * 100,
          payout_ratio: nz(summary.summaryDetail?.payoutRatio),
          beta: nz(summary.summaryDetail?.beta),
          market_cap: nz(summary.price?.marketCap),
          shares_out: nz(summary.defaultKeyStatistics?.sharesOutstanding),
          debt: nz(summary.financialData?.totalDebt),
          cash: nz(summary.financialData?.totalCash),
          profile: { // Include profile data
              name: summary.price?.longName || summary.price?.shortName ||
ticker,
              exchange: summary.price?.exchangeName,
              sector: summary.assetProfile?.sector,
          }
      };

      // Blacklist ratio fields that should only come from
computeFundamentalMetrics
      const RATIO_BLACKLIST = new Set([
        'roic_pct', 'fcf_margin_pct', 'cfo_to_ni', 'fcf_to_ebit',
'accruals_ratio',
        'margin_stdev_5y_pct', 'buyback_yield_pct', 'debt_to_equity',
        'netdebt_to_ebitda', 'interest_coverage_x',
        'pe_12m', 'pe_24m', 'pe_36m', 'pe_5y_low', 'pe_5y_high', 'pe_5y_median'
      ]);

      // Clean the data and apply blacklist
      const cleanData = {};
      for (const [key, value] of Object.entries(fields)) { // Iterating over
'fields' as per current structure
        if (RATIO_BLACKLIST.has(key)) {
          console.log(`[Yahoo] Skipping blacklisted ratio field: ${key}`);
          continue;
        }

        // Preserve objects (like 'profile') which are not intended to be
filtered by isNaN
        if (typeof value === 'object' && value !== null) {
          cleanData[key] = value;
        } else {
          // Apply the original outline's condition for scalar values
          if (value !== null && value !== undefined && !isNaN(value)) {
            cleanData[key] = value;
          }
        }
```

```
      }

      // Added console.log as per outline
      console.log(`[Yahoo] Successfully processed ${ticker}, saved $
{Object.keys(cleanData).length} fields`);

      // Modified response as per outline, ensuring 'fields' object is included
and CORS headers are kept
      return new Response(JSON.stringify({
        ok: true,
        status: 200, // Keep status 200 for successful responses
        ticker,
        fields: cleanData, // Changed to cleanData as the primary 'fields'
payload
        fields_saved: Object.keys(cleanData),
        fields_blacklisted: [...RATIO_BLACKLIST]
      }), {
        status: 200, // Keep status 200 for successful responses
        headers: corsHeaders() // Ensure CORS headers are included
      });

    } catch(e) {
      clearTimeout(timeoutId);
      if (e.name === 'AbortError') {
        return new Response(JSON.stringify({ ok: false, status: 408, reason:
'Request timed out', ticker }), { status: 200, headers: corsHeaders() });
      }
      if (attempt < maxRetries) {
          await new Promise(r => setTimeout(r, backoff));
          continue;
      }
      return new Response(JSON.stringify({ ok: false, status: 500, reason:
`Exception: ${e.message}`, ticker }), { status: 200, headers: corsHeaders() });
    }
  }
});


// FILE: functions/yahooHealth.ts
// ==============================================================

import { createClientFromRequest } from 'npm:@base44/sdk@0.5.0';

function corsHeaders() {
  return {
    "Access-Control-Allow-Origin": "*",
    "Access-Control-Allow-Methods": "POST, GET, OPTIONS",
    "Access-Control-Allow-Headers": "Content-Type, Authorization",
    "Access-Control-Allow-Credentials": "true",
    "Content-Type": "application/json",
  };
}

Deno.serve(async (req) => {
  if (req.method === 'OPTIONS') {
    return new Response(null, { status: 204, headers: corsHeaders() });
  }

  // No auth needed for a simple health check

  try {
    const url = "https://query1.finance.yahoo.com/v8/finance/chart/AAPL?
range=1mo&interval=1d&includeAdjustedClose=true";
    const resp = await fetch(url);
```

```
    const http_status = resp.status;
    const data = await resp.json();

    if (!resp.ok) {
      return new Response(JSON.stringify({ ok: false, http_status, message:
"Yahoo API returned non-200 status", debug: data }), { status: 200, headers:
corsHeaders() });
    }

    const result = data?.chart?.result?.[0];
    const lastIndex = result?.timestamp?.length - 1;
    if (!result || lastIndex < 0) {
       return new Response(JSON.stringify({ ok: false, http_status, message:
"Unexpected response structure", debug: data }), { status: 200, headers:
corsHeaders() });
    }

    const lastAdjClose = result.indicators.adjclose[0].adjclose[lastIndex];
    const lastTs = result.timestamp[lastIndex];

    return new Response(JSON.stringify({ ok: true, http_status, sample: { c:
lastAdjClose, t: lastTs } }), { status: 200, headers: corsHeaders() });

  } catch (err) {
    return new Response(JSON.stringify({ ok: false, error: String(err.message ||
err), debug: { details: "Function execution exception"} }), { status: 500,
headers: corsHeaders() });
  }
});



import { createClientFromRequest } from 'npm:@base44/sdk@0.5.0';

Deno.serve(async (req) => {
    const base44 = createClientFromRequest(req);
    const logs = [];
    const log = (msg) => logs.push(`[${new Date().toISOString().split('T')
[1].slice(0, -1)}] ${msg}`);

    if (!(await base44.auth.isAuthenticated())) {
        return new Response(JSON.stringify({ ok: false, success: false, error:
'Unauthorized' }), { status: 401 });
    }

    let ticker = '';
    const results = {
        status: "ok", // 'ok' or 'partial'
        errors: [],
        steps: {}
    };

    try {
        const body = await req.json();
        ticker = (body.ticker || '').trim().toUpperCase();
        if (!ticker) throw new Error('Ticker is required for full refresh.');

        const step = async (title, fnName, params = { ticker }) => {
            log(`[Step] START: ${title}`);
            try {
                const { data, error } = await base44.functions.invoke(fnName,
params);
```

```
                if (error || (data && (data.ok === false || data.success ===
false))) {
                    const errorMessage = error?.message || data?.error ||
data?.message || `${title} failed.`;
                    throw new Error(errorMessage);
                }

                if (data?.logs) logs.push(...data.logs.map(l => `  [${fnName}] $
{l}`));
                log(`[Step] SUCCESS: ${title}`);
                results.steps[fnName] = { status: 'success', data: data };
            } catch (e) {
                log(`[Step] FAILED: ${title} - ${e.message}`);
                results.status = "partial";
                results.errors.push(`${title} failed: ${e.message}`);
                results.steps[fnName] = { status: 'failed', error: e.message };
            }
        };

        // Strict, but resilient, execution order
        await step('A: Yahoo Historical Prices (5Y)', 'runYahooEtlPipeline');
        await step('B: Finnhub Fundamentals ETL', 'runFinnhubFundamentalsEtl');
        await step('C: Yahoo Fundamentals ETL', 'runYahooFundamentalsEtl');
        await step('D1: Compute Fundamental Metrics',
'computeFundamentalMetrics');
        await step('E: Sync Recent Prices', 'syncRecentPricesYahoo');
        await step('D2: Re-compute Fundamental Metrics',
'computeFundamentalMetrics'); // Re-run with latest price
        await step('F: Compute Price-based Metrics',
'computeAndSavePriceMetrics');

        const finalStatus = results.errors.length > 0 ? "partial" : "ok";
        const finalMessage = finalStatus === "ok"
            ? `Full refresh completed successfully for ${ticker}.`
            : `Full refresh for ${ticker} completed with $
{results.errors.length} error(s).`;

        return new Response(JSON.stringify({
            ok: true, // The orchestrator itself succeeded
            success: finalStatus === "ok",
            status: finalStatus,
            message: finalMessage,
            errors: results.errors,
            logs
        }), { status: 200, headers: { 'Content-Type': 'application/json' } });

    } catch (error) {
        log(`[FullRefresh][FATAL] Orchestrator failed: ${error.message}`);
        return new Response(JSON.stringify({
            ok: false,
            success: false,
            error: error.message,
            message: `Full refresh for ${ticker} failed: ${error.message}`,
            logs
        }), {
            status: 500,
            headers: { 'Content-Type': 'application/json' }
        });
    }
});
```